



FOUR SEASONS
HOTEL
BALTIMORE

ENERGY AUDIT REPORT

November 2021

EEG			
EEG – ENERGY EFFICIENCY INTERNATIONAL LTD		Document Type:	Energy Audit Report
Energy Audit Report for the Four Seasons Hotel, Baltimore		Contract Ref:	
		Revision no.:	A
Document Control			
Revision / Date	Issued by:	Reviewed by:	Approved by:
A-for Client Review	Joe El Khoury	Ronald Diab	Ronald Diab
B-for Client Approval			
Disclaimer			
<i>This report has been prepared by EEG with all reasonable skill, care and diligence within the terms of the contract with the client, incorporating our General Terms and Conditions of Business and taking account of the resources devoted to it by agreement with the client. The information contained in this report is, to the best of our knowledge, correct at the time of printing. The interpretations and recommendations are based on our experience, using reasonable professional skill and judgment, and based upon the information that was available to us. This report is confidential to the client and we accept no responsibility whatsoever to third parties to whom this report, or any part thereof, is made known. Any such party relies on the report at their own risk.</i>			

Table of Contents

I. Table of Contents	3
II. List of Tables	7
III. List of Figures	8
IV. 1. Introduction	13
1.1. Energy Efficiency Rationale	13
1.2. Energy Audit Goals and Tasks	14
1.2.1. End Goals	14
1.2.2. Methodology	14
V. 2. Facility General Description	15
2.1. Introduction	15
2.2. Building Envelope	17
2.3. Occupancy	20
2.4. Building Automation and Monitoring	21
2.5. Baltimore Climatic Conditions	22
2.6. Utilities: Energy Sources & Water	24
VI. 3. Utilities Accounting	26
3.1. Electricity	26
3.1.1. FSB Electricity Baseline Breakdown	26
3.1.2. FSB Overall Electricity Consumption and Cost	28
3.1.3. Electricity Baseline – Regression Analysis	29
3.2. District Cooling	31
3.2.1. CW Total Consumption and Cost	31
3.2.2. CW Regression Analysis	32
3.3. District Heating	34
3.3.1. HW Total Consumption and Cost	34
3.3.2. HW Regression Analysis	35
3.4. Natural Gas	37
3.4.1. FSB Overall Gas Consumption and Cost	37
3.5. Water	38
3.5.1. Water Consumption and Cost	38
3.6. Utilities Accounting Summary	39
3.7. Energy and Carbon Footprint	40

3.7.1.	Energy Footprint	40
3.7.2.	FSB Carbon Emissions	41
VII.	4. Electrical Energy Load Analysis & Energy Balance	42
4.1.	Electrical Load Categories	42
4.1.1.	Lighting	43
4.1.2.	HVAC Fans	45
4.1.3.	Motors	52
4.1.4.	Kitchen and Refrigeration	58
4.1.5.	Miscellaneous	60
4.2.	Electrical Energy Balance	62
4.3.	Thermal Energy	64
4.3.1.	District Cooling	64
4.3.2.	District Heating	67
VIII.	5. Temperature and Relative Humidity Logging	73
IX.	6. Energy Efficiency Measures	78
6.1.	EEM – 01: [LCNC] Review Cooling and Heating Capacity Charges in Contracts 	79
6.1.1.	Rationale	79
6.1.2.	District Cooling	79
6.1.3.	District Heating	80
6.1.4.	Energy Savings	81
6.2.	EEM – 02: [LCNC] Lighting Control 	82
6.2.1.	Rationale	82
6.2.2.	Turn off all the lighting in functions rooms during vacancies (no events)	82
6.2.3.	Turn off lighting fixtures during daylight	83
6.3.	EEM – 03: [CAPEX] Motion Detectors Installation 	84
6.3.1.	Rationale	84
6.3.2.	Efficiency Action and Energy Savings	85
6.4.	EEM – 04: [CAPEX] Lighting Retrofit 	86
6.4.1.	Rationale	86
6.4.2.	Efficiency Action and Energy Savings	86
6.5.	EEM – 05: [CAPEX] Improve Thermal Insulation of the Hot Water Network 	88
6.5.1.	Rationale	88
6.5.2.	Efficiency Action and Energy Savings	88
6.6.	EEM – 06: [LCNC] HVAC Control 	91

6.6.1.	Rationale.....	91
6.6.2.	Reset setpoints	91
6.6.3.	Aggressive Scheduling control	92
6.6.4.	Proactive Control of VFDs.....	94
6.6.5.	Remove unnecessary reheat	95
6.6.6.	Re-commission/calibrate sensors	95
6.6.7.	Kitchen Hoods and Extractor Fans	96
6.6.8.	GRMS programing during unrented room	97
6.6.9.	Proactive Delta P settings at secondary chilled water pumps	97
6.6.10.	Energy Savings	98
6.7.	EEM – 07: [LCNC] Primary Hot Water Pumps Operation 	99
6.7.1.	Rationale.....	99
6.7.2.	Efficiency Action and Energy Savings	100
6.8.	EEM – 08: [CAPEX] Installation of VFDs on Fans and Pumps 	101
6.8.1.	Rationale.....	101
6.8.2.	VFDs on AHUs	101
6.8.3.	VFDs on Pumps	102
6.8.4.	Energy Savings	102
6.9.	EEM – 09: [CAPEX] Pool Cover for residences' indoor pool 	103
6.9.1.	Rationale.....	103
6.9.2.	Installation of pool cover	103
6.9.3.	Pool pumps duty cycling and Energy Saving	103
6.10.	EEM – 10: [LCNC] Outdoor Pool freeze protection control 	105
6.10.1.	Rationale.....	105
6.10.2.	Efficiency Action and Energy Savings	105
6.11.	EEM – 11: [LCNC] Building Envelope – Heat gain mitigation for the first 21 Floors  107	
6.11.1.	Rationale.....	107
6.11.2.	Efficiency Action and Energy Savings	107
6.12.	EEM – 12: [CAPEX] Operation of the Blending Mixing Valve on Chilled Water Line  109	
6.12.1.	Rationale.....	109
6.12.2.	What is a blending mixing valve?.....	109
6.12.3.	Recommended Action	110
6.13.	EEM – 13: [CAPEX] Recovery Wheels for FAHUs 	112
6.13.1.	Rationale.....	112

6.13.2.	Recommended action and energy efficiency	113
6.14.	EEM – 14: [CAPEX] Solar Water Heaters Installation 	114
6.14.1.	Rationale.....	114
6.14.2.	Efficiency Action and Energy Savings	114
6.15.	EEM – 15: [CAPEX] Photovoltaic Electricity Generation 	118
6.15.1.	Rationale.....	118
6.15.2.	Efficiency Action and Energy Savings	118
6.16.	EEM – 16: [CAPEX] Re-Commissioning and Water balancing : Chilled and Hot Water Loops 	121
6.16.1.	Rationale.....	121
6.16.2.	Action Plan for TAB and Re-commissioning	121
X. 6.	Energy Efficiency Summary Figures	125

List of Tables

Table 1: Monthly Occupancy Rate in FSB through the Baseline Period	20
Table 2: Utilities Annual Costs	25
Table 3: FSB Electricity Consumption Breakdown	26
Table 4: FSB Monthly Electricity Consumption and Cost.....	28
Table 5: Electricity Projected Consumption, Residuals and variation	30
Table 6: FSB Monthly Chilled Water Consumption and Cost.....	31
Table 7: CW Projected Consumption, Residuals and Variation	32
Table 8: FSB Monthly Hot Water Consumption and Cost.....	34
Table 9: HW Projected Consumption, Residuals and Variation.....	36
Table 10: Baseline Gas Consumption and Cost.....	37
Table 11: Monthly Water Consumption and Cost	38
Table 12: Total Utilities Accounting Breakdown.....	39
Table 13: Total Utilities Energy Breakdown.....	40
Table 14: FSB Carbon Emissions Monthly Breakdown.....	41
Table 15: List of AHUs in the FSB Hotel side	46
Table 16: HVAC Fans Breakdown.....	51
Table 17: Motors Breakdown	57
Table 18: Electrical Load & Energy Consumption Breakdown by Category.....	62
Table 19: Hot Water Consumption Breakdown.....	70
Table 20: Energy Efficiency Measures Color Codes	78
Table 21: Energy Efficiency Measures	78
Table 22: VFDs on AHUs.....	102
Table 23: Solar Panels Key Figures.....	115
Table 24: Solar Thermal Energy Figures from Installing Solar Panels.....	116
Table 25: SWH Savings	117
Table 26: PV Plant Parameters.....	118
Table 27: PV Electricity Production Calculation.....	118
Table 28: PV Energy Savings	119
Table 29: Water Balancing Procedure	122
Table 30: EEMs Summary.....	125
Table 31: EEMs Summary.....	126

List of Figures

Figure 1: Four Seasons Baltimore – facades views	15
Figure 2: Pool Bar (Left) and Bygone Restaurant (Right)	15
Figure 3: Up - Fitness Room (Left) ; Outdoor Pool (Right) Down – Function areas	16
Figure 4: Four Seasons Baltimore Glazing Facade	17
Figure 5: Main façade thermal image: Lower Floors (Left), Upper Floors (Right)	18
Figure 6: Thermal images from internal surfaces: Residences lounge (Left); Guest room in the 8 th floor (Right).....	18
Figure 7: Guest Rooms Curtain	19
Figure 8: Guest Rooms Thermal Images	19
Figure 9: Occupancy Rate Variations	20
Figure 10: BMS Snapshot	21
Figure 11: Inncom GRMS	21
Figure 12: Climate in Baltimore (source: © WeatherSpark.com).....	22
Figure 13: Temperature variation in Baltimore	22
Figure 14: Humidity variation in Baltimore.....	23
Figure 15: Degree Days Variations in Baltimore	23
Figure 16: Annual Utilities Costs Breakdown.....	25
Figure 17: FSB Electricity Breakdown.....	27
Figure 18: FSB Electricity Breakdown.....	27
Figure 19: Electricity Consumption Monthly Variations	28
Figure 20: Electrical Regression and Actual Energy Consumption	30
Figure 21: Chilled Water Cost Monthly Variations	32
Figure 22: CW Regression and Actual Energy Consumption.....	33
Figure 23: Chilled Water Cost Monthly Variations	35
Figure 24: HW Regression and Actual Energy Consumption	36
Figure 25: Gas Consumption Variations.....	37
Figure 26: Water Monthly Variations	38
Figure 27: Total Utilities Breakdown.....	39
Figure 28: Total Utilities Energy Breakdown.....	40
Figure 29: FSB Monthly Carbon Emissions.....	41
Figure 30: Lighting Fixtures in FSB	43
Figure 31: Lighting Monthly Variations.....	44
Figure 32: AHU 17 (Left) and AHUs 18-19 (Right).....	45
Figure 33: AHU 4 - Lobby BMS Snapshot	45
Figure 34: AHU scheduling.....	46
Figure 35: AHU-2 Supply Air Temperature Trend	47
Figure 36: AHU-4 VFD Feedback Trend.....	47
Figure 37: AHU-16 VFD Trend	48
Figure 38: AHU-10 VFD Trend	48
Figure 39: AHUs Thermal Images.....	49
Figure 40: Residence Fresh AHU with Energy Recovery Unit (ERU)	49
Figure 41: Banquet Ecology Unit (Left) and Guest Rooms Toilets Extract Fan (Right)	50

Figure 42: Exhaust Fans List	50
Figure 43: GRMS controlling guest room thermostat	51
Figure 44: HVAC Fans Monthly Variations	51
Figure 45: Primary Chilled Water Pumps (Left) with VFD Reading (Right)	52
Figure 46: Primary Pump Chilled Water Trend	52
Figure 47: Primary Hot Water Pumps Nameplate (Left) with VFD Reading (Right).....	53
Figure 48: Secondary Hot Water Pump VFD Feedback.....	53
Figure 49: Return Hot Water Pumps.....	54
Figure 50: AHUs Hot Water Circulation Pumps	54
Figure 51: Domestic Water Booster Pumps.....	54
Figure 52: Hot Water Pumps.....	55
Figure 53: Chilled Water Pumps	55
Figure 54: Vitality Pool Blowers	55
Figure 55: Main Pool Filtration Pump	56
Figure 56: Podium Secondary Pumps	56
Figure 57: Motors Monthly Variations.....	57
Figure 58: Cold Rooms Compressor Racks (Up) and Kitchen Equipment (Down)	58
Figure 59: General kitchens view	59
Figure 60: Kitchen and Refrigeration Monthly Variations	59
Figure 61: Miscellaneous Equipment.....	60
Figure 62: Laundry Equipment.....	60
Figure 63: Miscellaneous Monthly Variations	61
Figure 64: Electrical Load & Energy Breakdown	62
Figure 65: Energy Consumption by Category.....	63
Figure 67: District Cooling.....	64
Figure 66: District Cooling Heat Exchanger (Left) and Nameplate (Right).....	65
Figure 71: Thermal Images of the Chilled Water Network	65
Figure 68: Chilled Water Supply and Return Temperature with Delta T	65
Figure 69: Differential Pressure Tracking.....	66
Figure 70: CW Pumps 3 and 4 VFD Feedback	66
Figure 72: District Heating Heat Exchangers (Left) and Nameplate (Right).....	67
Figure 73: B1 (Left) and 30 th Floor (Right) Calorifiers	67
Figure 74: Hot Water Network.....	68
Figure 75: Primary pumps Hot Water Supply and Return Temperature with Delta T.....	68
Figure 76: Secondary Hot Water Pumps VFD Feedback	69
Figure 77: Backup Steam Boilers.....	69
Figure 78: Pools BMS Snapshot	71
Figure 79: Residences Indoor Pool (Left) and Thermal Image (Right)	71
Figure 80: Hot Water Network Thermal Images.....	72
Figure 81: Lobby Temperature and Relative Humidity Variations.....	73
Figure 82: Grand Ballroom Temperature and Relative Humidity Variations.....	74
Figure 83: SPA Entrance Temperature and Relative Humidity Variations	74
Figure 84: 28th Floor indoor Pool Temperature and Relative Humidity Variations	75
Figure 85: Kitchen Corridor Temperature and Relative Humidity Variations.....	75

Figure 86: Elevator lobby Temperature and Relative Humidity Variations	76
Figure 87: Unrented Room Temperature and Relative Humidity Variations	76
Figure 88: Rented Room (68°F) Temperature and Relative Humidity Variations	77
Figure 89: Chilled Water Demand Breakdown	79
Figure 90: Peak Demand vs. Maximum Capacity	80
Figure 91: Hot Water Demand Breakdown	80
Figure 92: District Cooling Peak Demand vs. Maximum Capacity	81
Figure 93: Lighting Control Targeted Areas	82
Figure 94: Areas where Lighting Fixtures were kept on during the day	83
Figure 95: Motion detectors Targeted areas	84
Figure 96: Motion Detector 360° Coverage	85
Figure 97: CFL recessed light (Left) CFL round fixtures (Right)	86
Figure 98: Thermal Images of Hot Water Pipes System	88
Figure 99: Example of AHU Temperature reset	91
Figure 100: AHU 1 Schedule	92
Figure 101: AHU 4 BMS Snapshot (Left) and VFD (Right)	94
Figure 102: AHU 22 with reheat coil open	95
Figure 103: AHU 22 with reheat coil open	95
Figure 104: Dishwasher (Left) and Kitchen Hood (Right)	96
Figure 105: Chilled Water Delta P	97
Figure 106: Primary Hot Water Pumps	99
Figure 107: Hot Water Delta T	99
Figure 108: Power and Speed Variations	101
Figure 109: Existing Pool (Left), Floating Pool Cover Example (Right)	103
Figure 110: FSB Outdoor Pool	105
Figure 111: FSB Building Envelope	107
Figure 111: Curtains in guest rooms and their heat gain mitigation impact	108
Figure 112: Chilled Water System supply: BMS control screen (Up), MEP drawing (Down)	110
Figure 112: Fresh AHUs with separate supply and exhaust	112
Figure 112: AHU 23 (Left); TE 6 (Right)	113
Figure 113: FSB Roof	114
Figure 114: Average Monthly Irradiance – Baltimore	115
Figure 115: Flat Panel solar thermal panels	116
Figure 116: PV project savings	120

Abbreviations

A	Amperage
AC	Air Conditioning
Amps	Amperage
BMS	Building Management System
CHW	Chilled Water System
CF	Compact Fluorescent
CO ₂	Carbon Dioxide
COP	Coefficient of Performance ($KW_{cooling}/KW_{electrical}$)
EEM	Energy Efficiency Measures
ESCO	Energy Services Company
Fluo	Fluorescent
GHG	Green House Gases
Hal	Halogen
HVAC	Heating Ventilation Air Conditioning
IRR	Internal Rate of Return
KVA	kilo Volt Ampere
KW	Kilowatt
KWh	Kilowatt-hour
LED	Light Emitting Diode
m	Meter
m ²	Square meter
M3	Cubic meter
MDB	Main Distribution Board
MH	Metal Halide
MJ	Mega Joules
°C	Degree Celsius
°K	Degree Kelvin
°F	Degree Fahrenheit
PF	Power Factor
RT	Refrigerated Tons
TR	Transformer
UPS	Uninterruptible Power Supply
V	Volt
VAV	Variable Air Volume
VFD	Variable Frequency Drive
W	Watt
W m ² /°K	Watt Square meter per degree Kelvin

Conversion Factors

from	to	multiply by
meter	Feet	3.2808399
meter	Inches	39.370079
kg	Pounds	2.2046226
tons	Kg	1000
liter	Gallons (US beer)	0.264179
liter	Cubic feet	0.035315
kWh	MJ	3.6
kWh	BTU	3412
W/m ²	BTU/ft ²	0.317
kJ/kg K	BTU/lb °F	0.2388
W (heat flow rate)	BTU/h	3.412
W/m ² k	BTU/ft ² h °F	0.1761
kJ/kg	BTU/lb	0.4299
MJ/m ³	BTU/ft ³	26.84
Mm Hg (mercury)	Pa	133.332
Ft of water	kPa	2.98898
m ³ /kg	ft ³ /lb	16.02
m/s	ft/s	3.281

1. Introduction

Following an agreement with the management, Energy Efficiency Group **EEG** has undertaken an Ashrae Level 2 energy audit of the Four Seasons Hotel Baltimore, MD (abbreviated by **FSB**).

The energy audit was carried out by a combination of thorough field surveys and extensive energy analysis off site. A close cooperation was made with the facility's engineering department throughout the audit who showed great support and provided full assistance.

The present report is divided in two main sections. The first section includes a detailed description of the utilities costs (energy baseline) and the operation of all existing equipment (energy balance). The second section of the report comprises a list of suggested energy efficiency measures, where each measure is expanded to its technical background and financial feasibility.

1.1. Energy Efficiency Rationale

Simply put, Energy Efficiency is using less energy to accomplish the same amount of work needed. Ever since energy emerged as a top priority for policymakers, it has become a critical economic issue. In other words, unsustainable energy supply has real implications. Efficient buildings offer considerable opportunities to save money while reducing greenhouse gas emissions and taking leadership in sustainable development.

Within all sectors, energy costs may be a small percentage of turnovers, but reducing those costs can directly impact the profitability as they go straight to the bottom line, making businesses more competitive, especially with the rising of energy prices. The implementation of energy efficiency and sustainability measures increases levels of occupants' comfort as well as improving staff's general morale.

Energy audits include a review of all energy systems such as cooling, lighting, heating, ventilation, and building envelope. An analysis then serves the purpose of identifying where and how a facility uses energy and developing the set of Energy Efficiency Measures [EEMs] that could optimize the overall energy costs and which can range from small energy projects to major capital investments on operations and maintenance.

The audit focuses on a first phase on Actions with little or no cost which are improvements in current operations and practices that involve little or no cost of execution [LCNC]. Then it develops a road map for additional Energy efficiency measures that would cover all loads and infrastructure and will have different levels of CAPEX (Capital Expenditures) and Prioritization.

1.2. Energy Audit Goals and Tasks

EEG's energy audit is in fact considered an Investment Grade Energy audit which exceeds the Ashrae Level II Procedure for Commercial Building Energy Auditing,

1.2.1. End Goals

The goals of an energy audit are to:

- ✓ Identify the types and costs of energy/utilities use.
- ✓ Understand how the energy is being used – and possibly wasted.
- ✓ Determine and analyze various EEM alternatives such as improved operational techniques and/or new equipment that could substantially reduce energy costs.
- ✓ Perform an economic analysis on those alternatives and determine which ones are cost effective for the business or industry involved.

1.2.2. Methodology

The energy audit's methodology includes the following set of tasks:

- ✓ Gathering all historical utilities billing database.
- ✓ Gathering all facility layout, description, load data, operational hours.
- ✓ Undertake an in-depth field survey and auditing covering all sections and utilities end uses of the facility, completing the load inventory, analyzing equipment status & operational performances...
- ✓ Perform all needed real time measurements and data logging.
- ✓ Build an Energy Simulation of the facility using an Energy Analysis software. This modeling will allow us to have a detailed load and cost breakdown (Energy balance) along an indepth study on the consumption of the facility.
- ✓ Investigate potential Energy Efficiency Measures (EEMs) from the low cost/no cost ones to those with low CAPEX and quick returns and then those with higher CAPEX and longer returns.
- ✓ Develop the chosen EEMs with their impact on the energy consumption and related financials.
- ✓ Prepare a comprehensive report that would be divided into 2 main parts and the appendixes:
 - In depth analysis of the present situation including utilities analysis, energy baseline and energy balance, detailed systems description...
 - EEMs: Technical analysis, savings development, financial analysis.
 - Appendixes: Full load inventory list and any supporting technical documentation.

2. Facility General Description

2.1. Introduction

The Four Seasons Hotel Baltimore is a luxury hotel that first started operation in November 2011. It initially consisted of 21 floors and 256 rooms, but in 2016 10 residential floors were added with 62 different new private apartments, out of which only 17 residences are used today. The total built up area of the hotel, including the new floors, is around 550,000ft².

Figure 1: Four Seasons Baltimore – facades views



FSB offers a wide range of rooms with three main types: The ordinary guest rooms, suites, and accessible rooms. Each type has also many sub-types where rooms vary in size, views, and design.

The property has four F&B outlets which operations are all outsourced with most of the related utilities back charged : the rooftop Bygone, Maximon Restaurants, the Loch Bar and Azumi. In addition, there is a large retail facility (Anthropology) located in the Ground Floor and which utilities are also back charged.

Figure 2: Pool Bar (Left) and Bygone Restaurant (Right)



FSB also has a fitness center, a SPA with 11 treatment rooms, one indoor, one outdoor pool with all year maintained temperature, along with a total of 1,913m² of meeting rooms and function areas.

Figure 3: Up - Fitness Room (Left) ; Outdoor Pool (Right) | Down – Function areas



As previously indicated, restaurants' kitchens operations are outsourced and FSB has one main kitchen serving the in-room dining and the banqueting services.

Each private apartment in the upper floors has its own submeter covering every utility's supply and is being charged accordingly; however, all the common areas and services of the residences floors are directly managed by FSB and part of the core OPEX. These include an indoor pool, a fitness room and a large lounge.

2.2. Building Envelope

A building envelope is a major component of a building energy performance and the physical separator between conditioned and unconditioned environment. Its function is mainly to resist heat, cold, and noise transfer. The envelope includes all elements of the outer shell to maintain a dry, heated, or cooled indoor environment and facilitate its climate control. It is considered inefficient when it allows high rates of heat transfer between interior spaces and the outdoor environment. Eliminating heat gain and heat loss completely is not possible, but they can be minimized to improve efficiency.

When a building is poorly insulated, there is plenty of heat conduction through the roof, walls and floor slabs. This translates into heat gain during the summer, and heat loss during the winter. The insulation level is normally measured with the R-Value, which describes heat flow resistance per unit of area. For example, a building with R-20 insulation loses 75% less heat than an identical building with R-5 insulation. Air leaks increase heat transfer in a different way, but the effect on energy consumption is the same. The loss of cool air increases air conditioning costs in summer, while the loss of warm air increases heating costs in winter.

FSB building's facades consist of an all glazing surface, top to bottom, with an original intent of a high thermal efficiency. As can be seen in the below picture, some floors include balconies that provide shading.

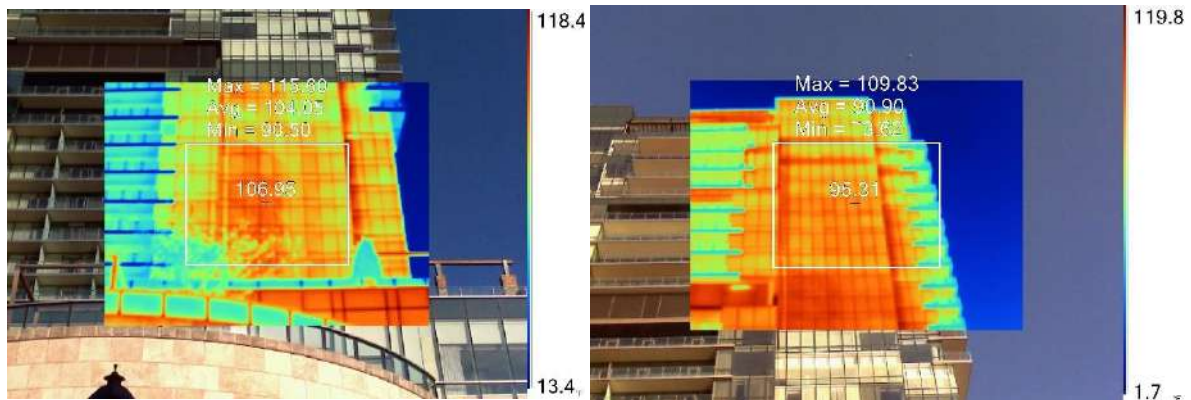
Figure 4: Four Seasons Baltimore Glazing Facade



The thermal images in the following page were taken during the field audit, in September 2021. The image on the left represents the building façade of the guest rooms' floors (i.e.: Lower than the 21st floor), whereas the image on the right represents the building façade of the residences' floors (i.e.: last 10 floors of the building), which was built few years after the hotel's opening. As can be clearly seen, the heat gains through the original hotel's envelope are much greater than

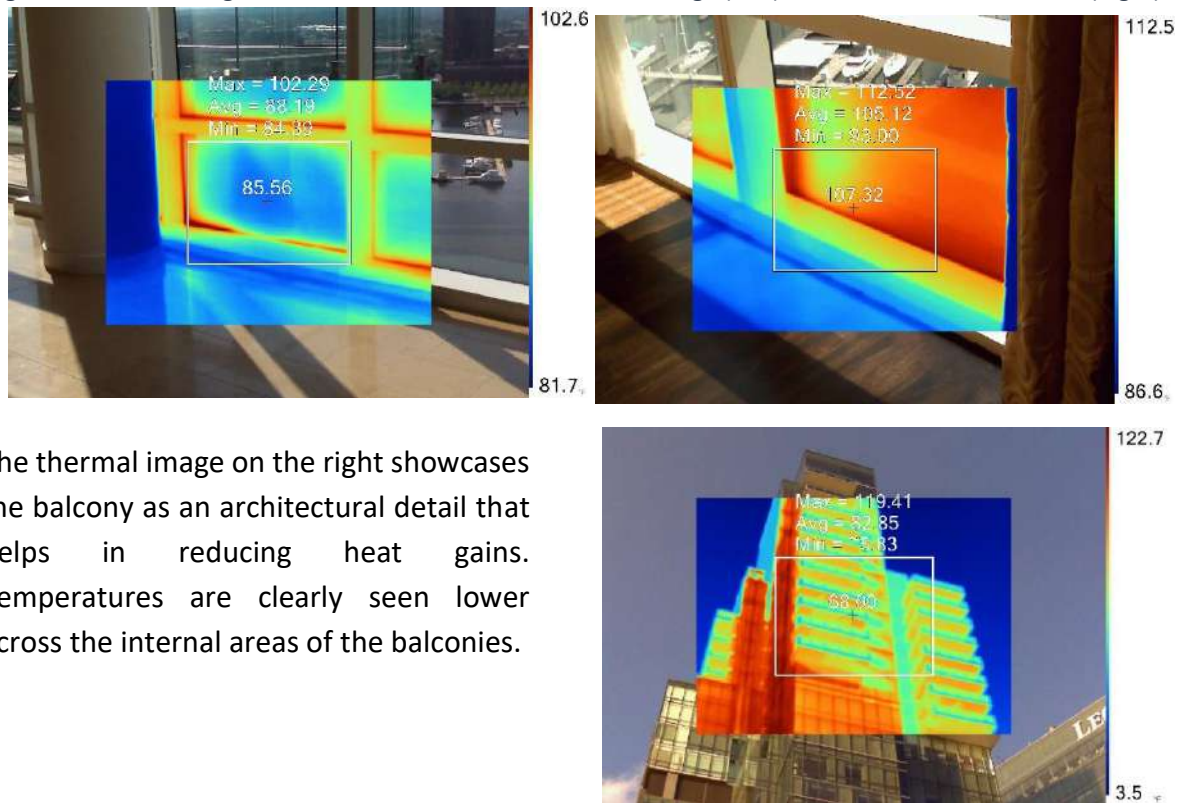
those of the residences' envelope,. The higher temperature at the glazed surface in the lower floors indicate a lower reflection of the solar radiations and therefore higher heat gains. This has also been verified from the inside

Figure 5: Main façade thermal image: Lower Floors (Left), Upper Floors (Right)



As indicated, thermal images from the internal glazing showed the substantially higher heat gains in the lower floors (guest rooms) as temperature topped 107°F on the glazing whereas it was only 85°F in the lounge of the residences lounge. The two sets of thermal images clearly indicate that the glazing of the additional residences floors added in 2017 has a much better thermal efficiency compared to the originally built floors.

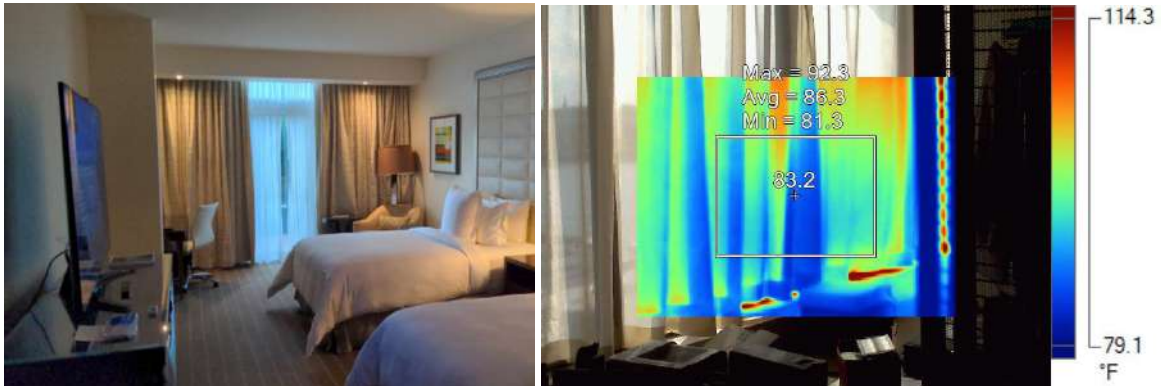
Figure 6: Thermal images from internal surfaces: Residences lounge (Left); Guest room in the 8th floor (Right)



The thermal image on the right showcases the balcony as an architectural detail that helps in reducing heat gains. Temperatures are clearly seen lower across the internal areas of the balconies.

While the glazing in the hotel's floors was seen to have a lower thermal efficiency compared to the upper newer floors, curtains in guest rooms do alleviate the problem of heat gains as seen in the below thermal image.

Figure 7: Guest Rooms Curtain



Restaurants in the Ground Floor have large air infiltration due to the frequent terrace access which do affect the cooling demand in these specific areas. On the other hand, the main hotel access is through a revolving door which is the optimal choice in terms of thermal efficiency.

Figure 8: Guest Rooms Thermal Images



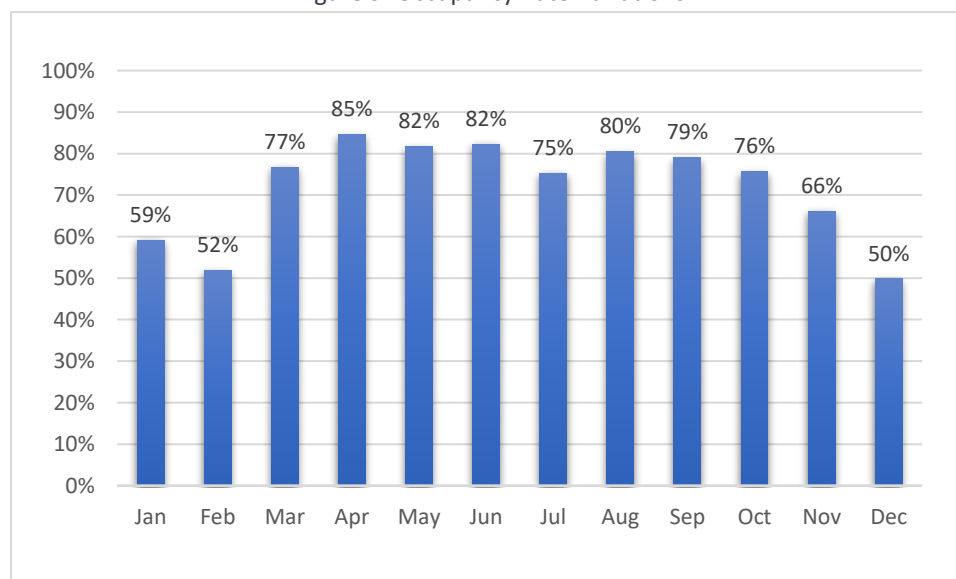
2.3. Occupancy

The following table depicts the occupancy rate variation of the hotel during the baseline year which was chosen to be 2019 as it was an ordinary and typical year compared to 2020 that was affected by the pandemic. As can be seen, the occupancy was fluctuating during the baseline period between a minimum of 50% in December and a maximum of 85% in April. The average occupancy rate throughout the baseline year is 72%.

Table 1: Monthly Occupancy Rate in FSB through the Baseline Period

Month	Occupancy	Availability	Occupancy Rate
Unit	Room Nights	Room Nights	%
Jan	4,693	7,936	59%
Feb	3,723	7,168	52%
Mar	6,088	7,936	77%
Apr	6,494	7,680	85%
May	6,484	7,936	82%
Jun	6,318	7,680	82%
Jul	5,971	7,936	75%
Aug	6,384	7,936	80%
Sep	6,065	7,680	79%
Oct	6,001	7,936	76%
Nov	5,075	7,680	66%
Dec	3,951	7,936	50%
TOTAL	67,247	93,440	72%
Maximum	6,494 [Apr]	7,936 [Jan]	85% [Apr]
Average	5,604	7,787	72%
Minimum	3,723 [Feb]	7,168 [Feb]	50% [Dec]

Figure 9: Occupancy Rate Variations



2.4. Building Automation and Monitoring

FSB has a Johnson Controls BMS system that is connected to most existing mechanical and electrical infrastructure and assets in the hotel and residences, notably all AHUs, extractor fans, pumps, and lifts, as well as the Energy Transfer Systems of both District Cooling and District Heating. The system was expanded to include the residences floors which have a separate graphical interface. The platform was seen to be comprehensive with full visibility and trending.

Figure 10: BMS Snapshot

ACSs	Supply Fan Status	Supply Fan Command	Heating Valve Output	Cooling Valve Output	Pre-Filter Status	Low Temperature Alarm	Discharge Air Temperature
AHU-1	On	On	0 %	61 %	N/A	Normal	54.1 deg F
AHU-2	On	On	0.0 %	61.1 %	N/A	Normal	53.2 deg F
AHU-3	On	On	0 %	63 %	N/A	Normal	55.1 deg F
AHU-4	On	On	0 %	71 %	N/A	Normal	55.0 deg F
ACS - 4	On	On	0.0 %	94.6 %	Normal	Normal	59.7 deg F
ACS - 6	On	On	0.0 %	9.9 %	Normal	Normal	57.5 deg F
ACS - 7	On	On	0 %	44 %	Normal	Normal	52.0 deg F
ACS - 8	On	On	0 %	26 %	Normal	Normal	57.0 deg F
ACS - 9	On	On	0 %	0 %	Alarm	Normal	56.4 deg F
ACS - 10	On	On	0 %	21 %	Normal	Normal	59.6 deg F
ACS - 11	On	On	0 %	26 %	Normal	Normal	55.3 deg F
ACS - 12	On	On	0.0 %	100.0 %	Normal	Normal	58.3 deg F

FSB has a number of meters (electricity, gas, water and BTU) , specially used for back charging outsourced restaurants operations and retail areas. Some are accessed by the BMS while the others are manually accessed. Furthermore, FSB has an Inncom GRMS that controls all the rooms' loads (thermostat and lighting) based on the set renting scenarios which was seen to properly operating .

Figure 11: Inncom GRMS

Default		Red		Mstr		Violet		Orange		DND		---		DND	
501-506	ROOM 501 L/R	ROOM 501 B/R	ROOM 502 L/R	ROOM 502 B/R	ROOM 503 L/R	COMMON DOOR 503		ROOM 504	ROOM 505						
	Dec	Dec	Ren Dec	Ren Dec DND		Ren Dec		Dec	Dec						
506-50612	ROOM 506	ROOM 507	ROOM 508	ROOM 509	ROOM 510 L/R	ROOM 511	ROOM 512 L/R	COMMON DOOR 512							
	Ren Dec DND			Ren Dec	Ren Dec	Dec		Ren Dec							
513-518	ROOM 513 L/R	ROOM 514	ROOM 515 L/R	ROOM 516	ROOM 517	ROOM 518	ROOM 519 L/R								
	Dec	Dec	Dec DND		Dec		Ren Dec								
10618-10624	ROOM 519 B/R	ROOM 520	ROOM 521	ROOM 524 L/R	ROOM 524 B/R										
	Ren Dec DND		Dec	Dec	Dec										
501-605	ROOM 501 L/R	ROOM 501 B/R	ROOM 502 L/R	ROOM 502 B/R 13-15	ROOM 503 L/R	ROOM 503 C	ROOM 504	ROOM 505							
		Ren Dec	Ren Dec	Ren Dec	Ren Dec	Ren Dec	Ren Dec	Dec							

2.5. Baltimore Climatic Conditions

Baltimore, Md has a humid subtropical climate that is mild with no dry season, constantly moist (year-round rainfall). Summers are hot and muggy with thunderstorms. Winters are mild with precipitation from mid-latitude cyclones.

The hot season lasts for 3.5 months, from May 30 to September 15, with an average daily high temperature above 79°F. The hottest month of the year in Baltimore is July, with an average high of 88°F and low of 73°F.

The cold season lasts for 3.2 months, from December 1 to March 8, with an average daily high temperature below 52°F. The coldest month of the year in Baltimore is January, with an average low of 31°F and high of 43°F.

Baltimore experiences extreme seasonal variation in the perceived humidity. The muggier period of the year lasts for 4.0 months, from May 26 to September 27, during which time the comfort level is muggy, oppressive, or miserable at least 15% of the time. The month with the most muggy days in Baltimore is July, with 16.8 days that are muggy or worse. The month with the fewest muggy days in Baltimore is February, with 0.0 days that are muggy or worse.

Figure 12: Climate in Baltimore (source: © WeatherSpark.com)

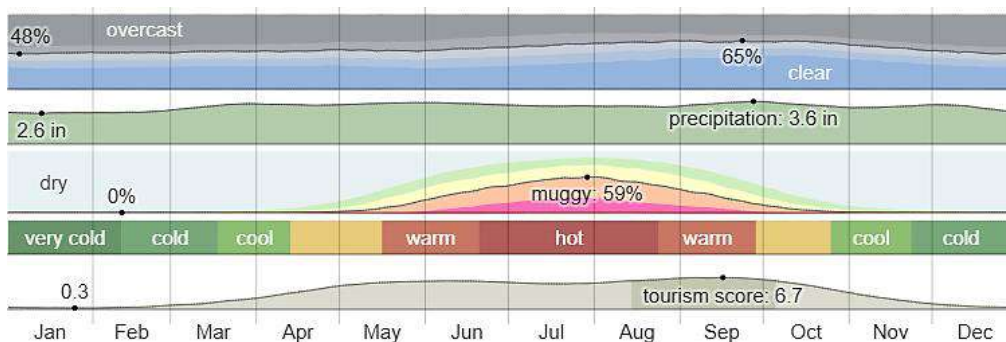


Figure 13: Temperature variation in Baltimore

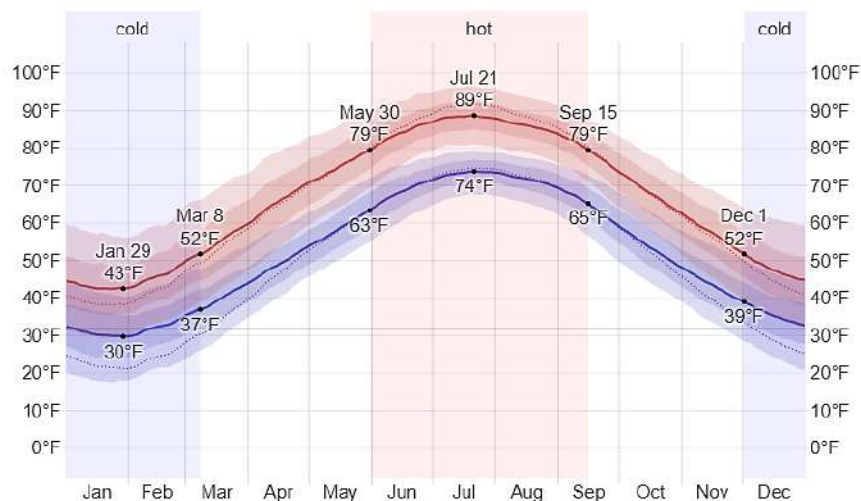
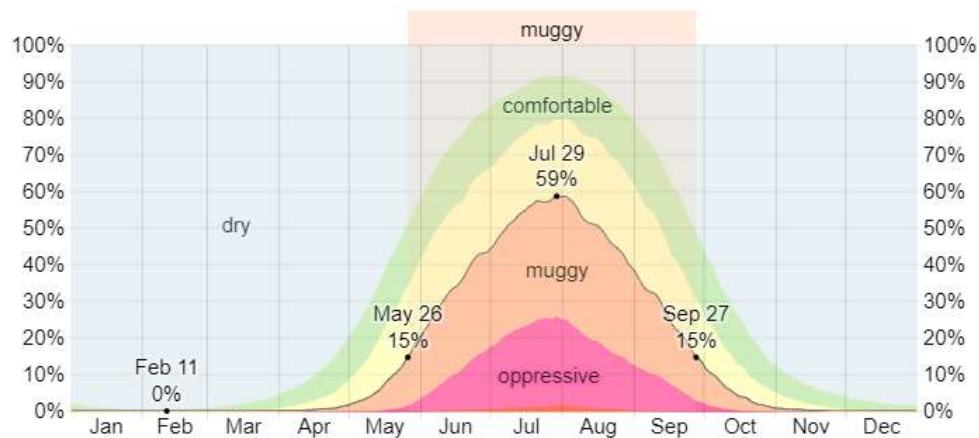


Figure 14: Humidity variation in Baltimore



Degree Days:

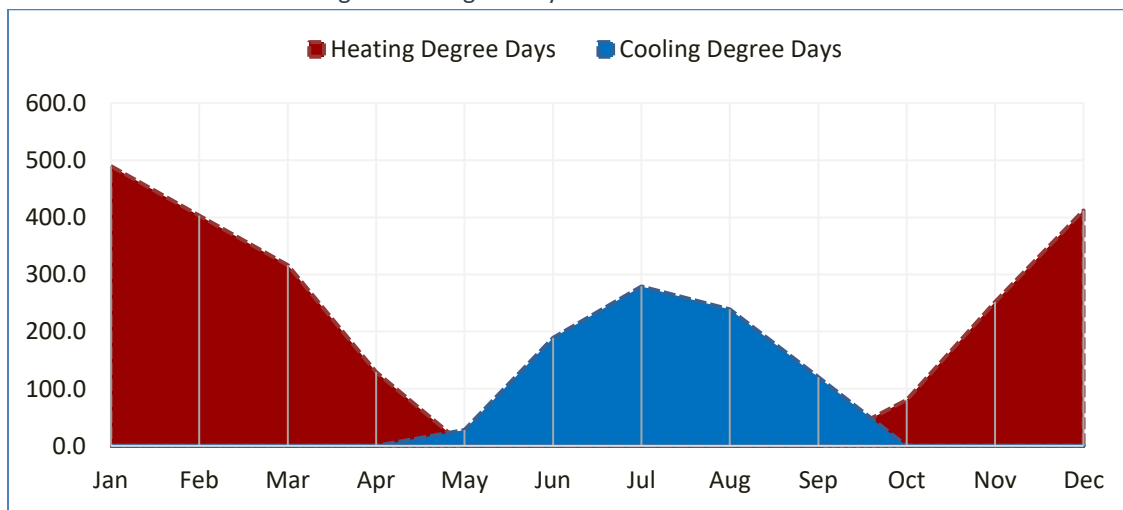
Degree days are essentially a simplified representation of outside air-temperature data. They are widely used in the energy industry for calculations relating to the effect of outside air temperature on building energy consumption.

"Heating degree days", or "HDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was lower than a specific "base temperature" (or "balance point"). They are used for calculations relating to the energy consumption required to heat buildings.

"Cooling degree days", or "CDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was higher than a specific base temperature. They are used for calculations relating to the energy consumption required to cool buildings.

The below chart provides the CDD and HDD in Baltimore across the baseline year with the highest CDD recorded in the baseline year in July (279) and the highest HDD in January (490)

Figure 15: Degree Days Variations in Baltimore



2.6. Utilities: Energy Sources & Water

FSB's utility sources are electricity, natural gas, water, district cooling, and district heating.

Electricity

Electricity is provided by the public electricity grid and distributed to the hotel through 2x2000KVA main transformers located in the 3rd floor's main electrical room, with a third transformer on the 18th floor serving the Bygone Restaurant and the residences.

Electricity charges are at \$384,396 representing **23%** of the total energy & water related utilities expenditures.



District Cooling

Chilled Water is supplied from a District Cooling Plant and delivered to the hotel through two heat exchangers. The subscribed cooling capacity in the contract is 1150 Refrigeration Tons. The sets of primary and secondary chilled water pumps enable the distribution of chilled water throughout all the hotel and residences.

Chilled Water energy annual cost for the baseline year is \$515,802 representing **31%** of the total utilities expenditures.



District Heating

Hot Water is supplied from a District Heating Plant and delivered to FSB through two heat exchangers. The subscribed heating capacity of is 18.2 mmBTU/hr. The supplied hot water is used for space heating, pools and the production of domestic hot water that is stored in various calorifiers in B1 and the 30th floor.

Hot Water energy annual cost for the baseline year is \$446,978 representing **27%** of the total utilities expenditures.



Gas

Natural gas is used in FSB for all the Kitchens' operation. Gas charges are at \$72,971 representing **4%** of the total utilities expenditures.

Water

Fresh water is supplied to FSB by public water supplier. The total water charges are at \$266,553 per year representing **16%** of the total utilities cost.

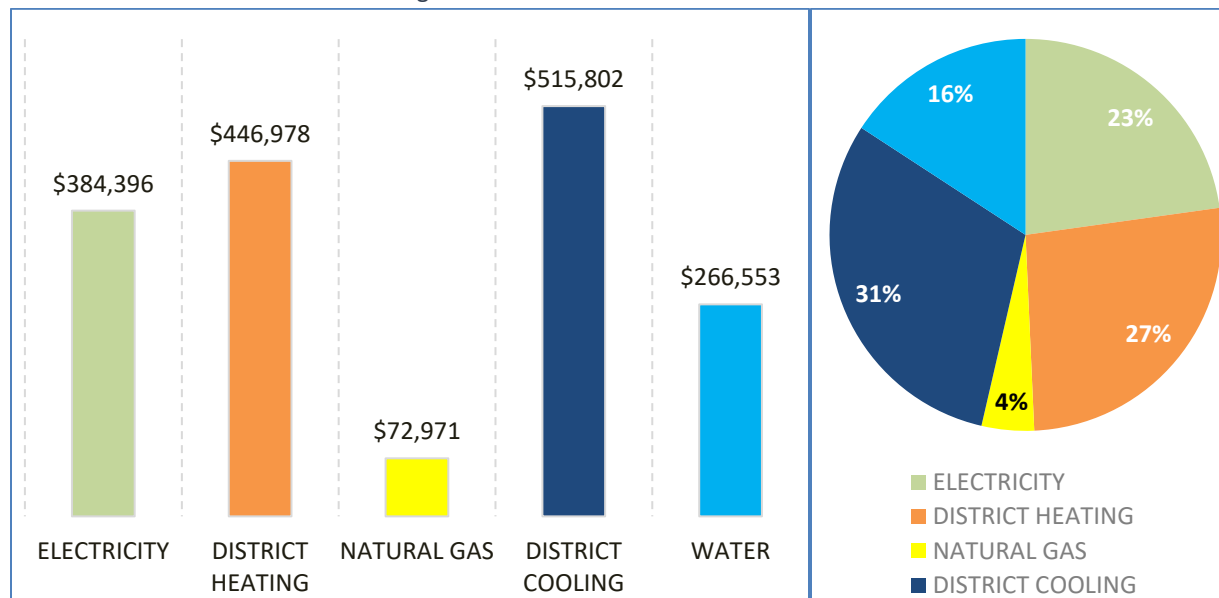
Total Utilities

The below table provides the cost of each utility supply as compiled for the baseline year of 2019.

Table 2: Utilities Annual Costs

UTILITY	ELECTRICITY	DISTRICT HEATING	NATURAL GAS	DISTRICT COOLING	WATER	ALL UTILITIES
TOTAL COST	\$384,396	\$446,978	\$72,971	\$515,802	\$266,553	\$1,686,700
MAXIMUM	\$36,030 (May)	\$65,052 (Jan)	8,525 (Mar)	\$62,106 (Jul)	\$29,376 (Jan)	\$162,456 (Jan)
AVERAGE	\$32,033	\$37,248	\$6,081	\$42,983	\$22,213	\$140,558
MINIMUM	\$26,966 (Apr)	\$27,656 (Aug)	3,389 (Nov)	\$31,433 (Jan)	\$16,586 (Dec)	\$126,744 (Nov)
PERCENTAGE	23%	27%	4%	31%	16%	100%

Figure 16: Annual Utilities Costs Breakdown



3. Utilities Accounting

The following sections detail the accounting numbers for the four utilities over the **baseline period from January 2019 until December 2019**. The annual consumption and related cost baseline are the aggregated consumption and associated costs for FSB throughout the baseline period. This energy baseline serves as a benchmark consumption to assess the current energy performance along with the calculation of all the proposed Energy Efficiency Measures. The present energy audit and the related energy simulation tackles the full facility space, excluding all the restaurants, residences, and retail shops.

3.1. Electricity

3.1.1. FSB Electricity Baseline Breakdown

The present energy audit covers all the tower excluding the outsourced restaurants, the retail (Anthropology) and the residences units (the common residential services are considered part of the Hotel analysis). The below table provide the full electricity data for the chosen baseline year of 2019 and whereby the last column 'Core Hotel' includes the parts considered in the energy analysis.

Table 3: FSB Electricity Consumption Breakdown

Month	Maximon	Loch Bar & Azumi	Bygone	Anthropology	Residences	Core Hotel	Total Hotel
Jan	91,410	46,530	16,120	10,931	35,461	321,756	522,208
Feb	51,621	22,966	14,000	10,089	28,983	291,819	419,478
Mar	52,575	23,351	16,120	14,115	31,146	390,629	527,936
Apr	64,238	30,246	15,600	13,383	28,018	302,987	454,472
May	54,304	20,140	17,980	11,434	28,546	404,833	537,237
Jun	68,240	31,245	18,600	13,641	30,648	375,971	538,345
Jul	63,424	20,341	26,363	13,572	32,874	383,263	539,837
Aug	47,707	18,200	27,550	10,283	33,058	383,262	508,510
Sep	55,566	17,648	23,692	12,856	28,720	375,772	514,254
Oct	65,832	34,639	16,022	13,226	26,404	367,878	524,001
Nov	61,272	26,846	16,995	12,857	26,830	348,482	493,282
Dec	52,098	24,032	15,349	14,202	28,911	372,403	506,995
Total	728,287	316,184	224,391	150,589	359,599	4,319,055	6,086,555
Maximum	91,410 (Jan)	46,530 (Jan)	27,550 (Aug)	14,202 (Dec)	35,461 (Jan)	404,833 (May)	539,837 (Jul)
Average	60,691	26,349	18,699	12,549	29,967	359,921	507,213
Minimum	47,707 (Aug)	17,648 (Sep)	14,000 (Feb)	10,089 (Feb)	26,404 (Oct)	291,819 (Feb)	419,478 (Feb)
Percentage	12%	5%	4%	2%	6%	71%	100%

As can be seen in the below chart, the 'Core Hotel' part that is considered in the present energy analysis represent 71% of the total electricity consumption. Maximon restaurant comes second with 12% of the total while apartments' units represent only 6% of the total.

Figure 17: FSB Electricity Breakdown

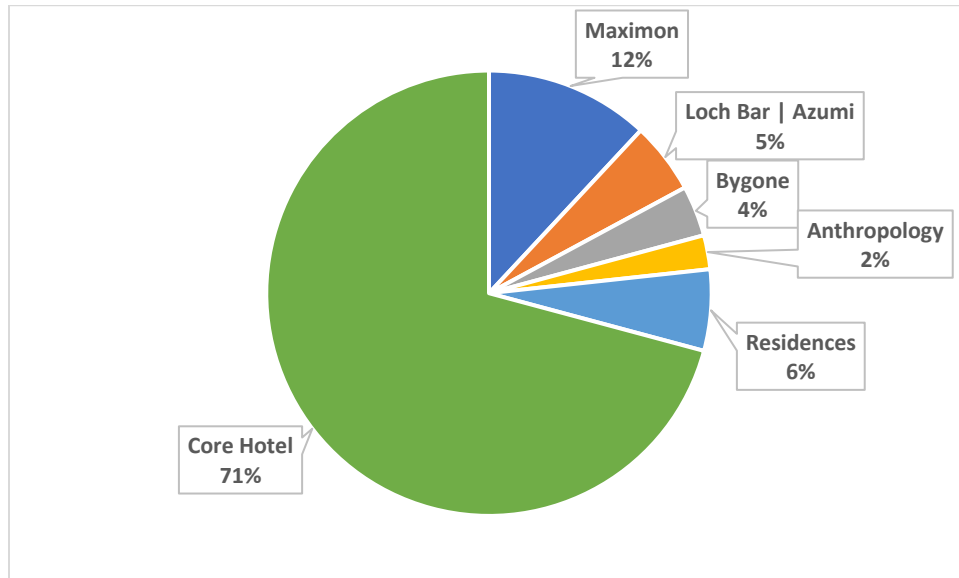
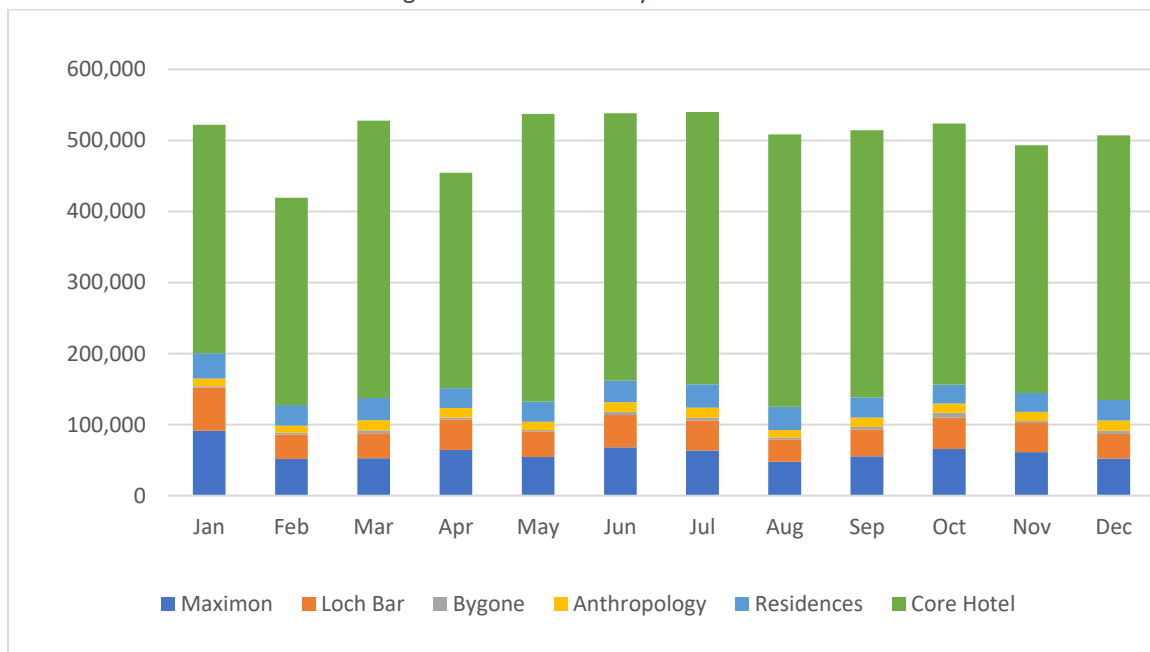


Figure 18: FSB Electricity Breakdown



3.1.2. FSB Overall Electricity Consumption and Cost

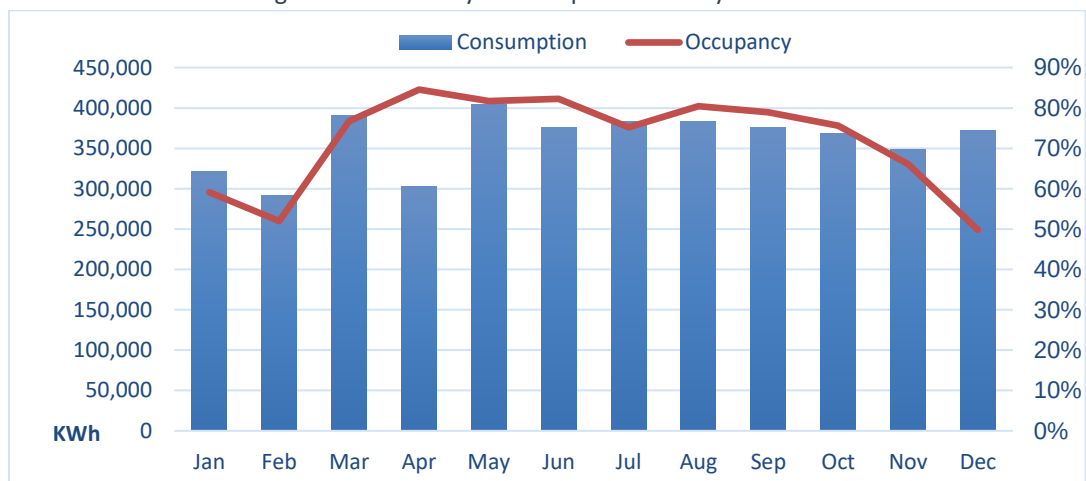
The total audited annual electricity consumption of the property is 4,319,055KWh, with the consumptions of all residences, restaurants and retail shops deducted from the total consumption of the tower. The below table provides the related monthly cost and energy.

Table 4: FSB Monthly Electricity Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	KWh	\$	\$/KWh	MJ	-
Jan	321,756	\$28,636	\$0.089	1,158,322	-12%
Feb	291,819	\$25,972	\$0.089	1,050,548	-23%
Mar	390,629	\$34,766	\$0.089	1,406,264	8%
Apr	302,987	\$26,966	\$0.089	1,090,752	-19%
May	404,833	\$36,030	\$0.089	1,457,400	11%
Jun	375,971	\$33,461	\$0.089	1,353,494	4%
Jul	383,263	\$34,110	\$0.089	1,379,748	6%
Aug	383,262	\$34,110	\$0.089	1,379,744	6%
Sep	375,772	\$33,444	\$0.089	1,352,779	4%
Oct	367,878	\$32,741	\$0.089	1,324,361	2%
Nov	348,482	\$31,015	\$0.089	1,254,535	-3%
Dec	372,403	\$33,144	\$0.089	1,340,651	3%
Total	4,319,055	\$384,396	\$0.089	15,548,599	-
Maximum	404,833 (May)	\$36,030 (May)	\$0.09 (Jan)	1,457,400 (May)	11% (May)
Average	359,921	\$32,033	\$0.089	1,295,717	-
Minimum	291,819 (Feb)	\$25,972 (Feb)	\$0.09 (Jan)	1,050,548 (Feb)	-23% (Feb)

The below figure provides the monthly electricity consumption with the occupancy line. It is clear that in most months there is a correlation between both values. However, February, and specially April and December show large deviations in consumption except for the month of April where the occupancy is quite high, but the electricity consumption is the second lowest during the baseline year.

Figure 19: Electricity Consumption Monthly Variations



3.1.3. Electricity Baseline – Regression Analysis

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationship among variables. It focuses on the relationship between a dependent variable and one or more independent variables.

In the case of FSB, for a given month, a relationship exists between the energy consumption (the main parameter) and the following variables: the occupancy expressed in Room Nights and the weather conditions (mathematically represented by the cooling degree days and heating degree days). Using the regression analysis, an empirical equation is modeled to generate projected energy consumption as a function of the above-mentioned variables. The model is linear, and the equation will be of the form:

$$KWh = AxOCC + BxCDD + CxHDD + D$$

OCC is the occupancy rate expressed in Room Nights and CDD is the Cooling Degree Day

A, B and C are the coefficients to be calculated in the process of the regression analysis

In the present regression analysis, the historical figures used for the modeling are found in the below table.

Month	Occupancy	CDD
Unit	Room Nights	Days
Jan	4,693	0
Feb	3,723	0
Mar	6,088	0
Apr	6,494	0
May	6,484	28
Jun	6,318	189
Jul	5,971	279
Aug	6,384	239
Sep	6,065	120
Oct	6,001	0
Nov	5,075	0
Dec	3,951	0
Total	67,247	855

Using these variables, and the energy consumption for the baseline period of each month, the coefficients of the regression model are obtained, leading to the below:

$$KWh(\text{per month}) = (14.5) \times OCC + (88.0) \times CDD + (272,529.8)$$

The projected consumption would be then calculated based on the above equation and the following table provides both the real baseline electricity consumption as well as the projected one if the regression equation is used.

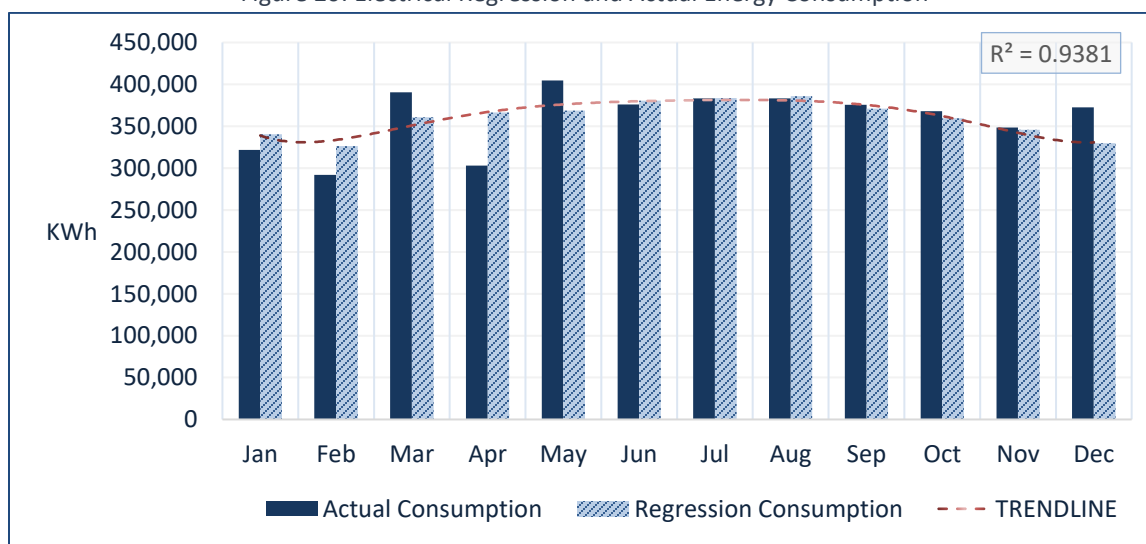
Table 5: Electricity Projected Consumption, Residuals and variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	KWh	KWh	KWh	%
Jan	321,756	340,469	-18,713	-6%
Feb	291,819	326,427	-34,608	-12%
Mar	390,629	360,664	29,965	8%
Apr	302,987	366,542	-63,555	-21%
May	404,833	368,851	35,982	9%
Jun	375,971	380,620	-4,650	-1%
Jul	383,263	383,514	-251	0%
Aug	383,262	385,948	-2,686	-1%
Sep	375,772	370,888	4,884	1%
Oct	367,878	359,405	8,473	2%
Nov	348,482	345,999	2,483	1%
Dec	372,403	329,727	42,676	11%
Total	4,319,055	4,319,055	0	0%

The residuals are the difference between the actual and the projected energy consumption for the same values of independent variables. The variation is the percentage of difference between the actual and the projected consumption. As can be seen, the annual regression provides an exact match (difference of 0); the following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

R-squared is a statistical measure linked to the regression model. It is also known as the coefficient of determination, which is the percentage of the response variable variation that is explained by a linear model (explained variation / total variation). R-squared is always between 0 and 100%: the higher the value, the better the model fits the data. In the case of FSB, R^2 was seen to be 0.9381, which means that the model can be considered as an acceptable model.

Figure 20: Electrical Regression and Actual Energy Consumption



3.2. District Cooling

3.2.1. CW Total Consumption and Cost

Chilled Water is supplied by the local District Cooling plant through two Heat Exchangers. It is used for space cooling (supplying FCUs and AHUs) as well as condenser cooling for water cooled refrigeration equipment.

The contract that was signed between the FSB development company and the chilled water supplier consists of a maximum capacity of 1150 Refrigeration Tons, with the total cost being the sum of the two charges detailed below:

- Energy Charge: Monthly cooling consumption in RTh(Refrigerated Ton Hours) x \$0.079/RTh
- Capacity subscription Charge: Monthly Charge of \$24,633

It is important to also note that these rates are dynamically changed according to the overall inflation rates and energy/utilities costs.

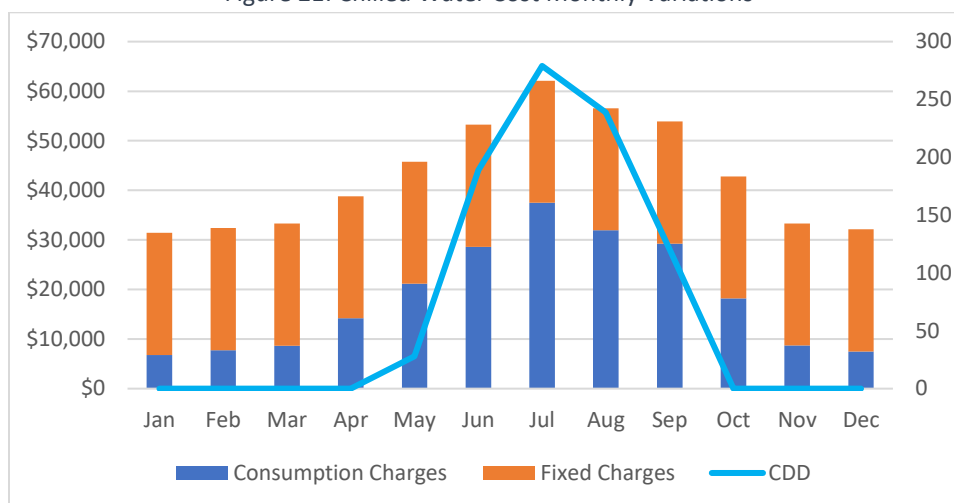
The following table depicts the FSB chilled water consumption during the baseline year with the corresponding costs.

Table 6: FSB Monthly Chilled Water Consumption and Cost

Month	Consumption	Consumption Charges	Fixed Charges	Rate	Energy Value	Deviation
Unit	RTH	\$	\$	\$/RTH	MJ	-
Jan	81,051	\$6,800	\$24,633	\$0.388	820,928	-186%
Feb	95,726	\$7,763	\$24,633	\$0.338	969,564	-142%
Mar	105,389	\$8,652	\$24,633	\$0.316	1,067,436	-120%
Apr	177,688	\$14,180	\$24,633	\$0.218	1,799,719	-30%
May	265,772	\$21,155	\$24,633	\$0.172	2,691,881	13%
Jun	362,480	\$28,600	\$24,633	\$0.147	3,671,392	36%
Jul	481,039	\$37,473	\$24,633	\$0.129	4,872,221	52%
Aug	410,018	\$31,940	\$24,633	\$0.138	4,152,882	43%
Sep	371,221	\$29,252	\$24,633	\$0.145	3,759,925	38%
Oct	225,342	\$18,185	\$24,633	\$0.190	2,282,385	-3%
Nov	109,514	\$8,695	\$24,633	\$0.304	1,109,217	-112%
Dec	96,148	\$7,509	\$24,633	\$0.334	973,838	-141%
Total	2,781,388	\$220,206	\$295,596	\$0.185	28,171,389	-
Maximum	481,039 (Jul)	\$37,473 (Jul)	\$24,633 (Jan)	\$0.39 (Jan)	4,872,221 (Jul)	52% (Jul)
Average	231,782	\$18,350	\$24,633	\$0.235	2,347,616	-
Minimum	81,051 (Jan)	\$6,800 (Jan)	\$24,633 (Jan)	\$0.13 (Jul)	820,928 (Jan)	-186% (Jan)

The below figure shows the chilled water monthly consumption along with the CDD. The profile clearly shows a parabolic shape throughout the baseline year, as the CW energy consumption logically rises in summer and drops in winter, in correlation with the blue line which represents the Monthly CDD Variations. It is also clear that during wintertime, the fixed charges account for the largest share of the total cost (up to 90% in January) due to the low cooling demand in the facility, and during summertime the two costs are more even.

Figure 21: Chilled Water Cost Monthly Variations



3.2.2. CW Regression Analysis

Building the regression equation in a similar way as explained in the electricity section would lead to the below model:

$$RT = (41.6) * CDD + (1,069.3) * OCC + (-77,732.4)$$

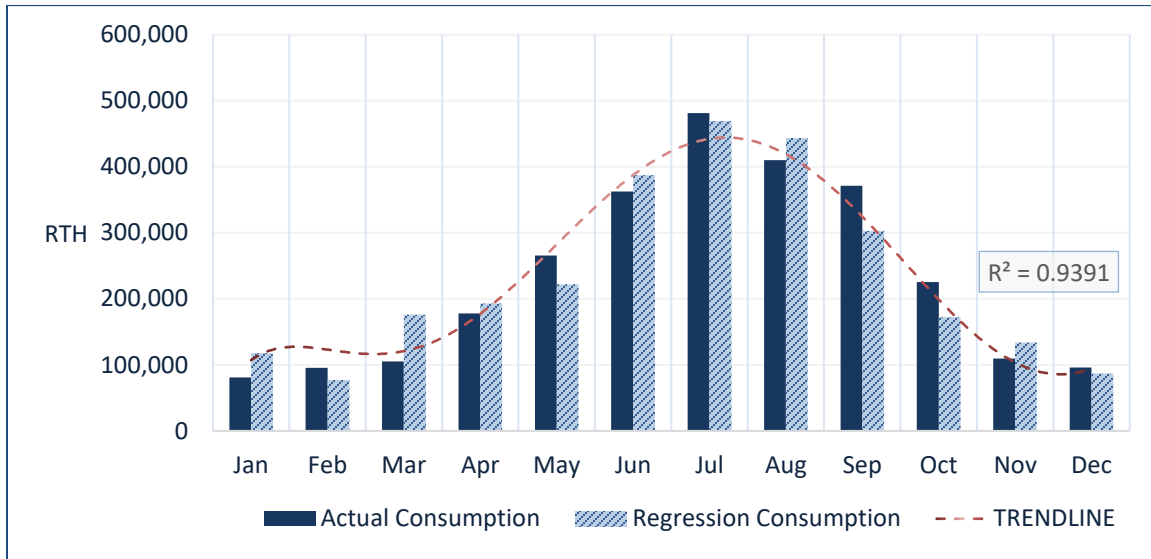
Table 7: CW Projected Consumption, Residuals and Variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	RTH	RTH	RTH	%
Jan	81,051	117,699	-36,648	-45%
Feb	95,726	77,306	18,420	19%
Mar	105,389	175,792	-70,403	-67%
Apr	177,688	192,699	-15,011	-8%
May	265,772	222,115	43,657	16%
Jun	362,480	387,460	-24,980	-7%
Jul	481,039	469,244	11,795	2%
Aug	410,018	443,351	-33,333	-8%
Sep	371,221	303,146	68,075	18%
Oct	225,342	172,169	53,173	24%
Nov	109,514	133,607	-24,093	-22%
Dec	96,148	86,800	9,348	10%
Total	2,781,388	2,781,388	0	0%

The following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

In the case of CW, $R^2 = 0.9391$, which means that the model can be strongly considered an accurate one.

Figure 22: CW Regression and Actual Energy Consumption



3.3. District Heating

3.3.1. HW Total Consumption and Cost

Hot Water is supplied from the District Heating plant to FSB through two Heat Exchangers and has three end uses: Space heating (supplying hot water to AHUs and FCUs across the tower) , heating the pools/Jacuzzis and producing domestic hot water.

The contract that was signed between the FSB development company and the district heating consists of a maximum capacity of 18.2 mmBTU/hr , with the total cost being the sum of the two charges detailed below:

- Energy Charge: Monthly usage in mmBTU x \$8.106/mmBTU
- Capacity Charge: Monthly Charge of \$19,921

It is important to also note that these rates fluctuate based on the inflation index and the energy/utilities costs.

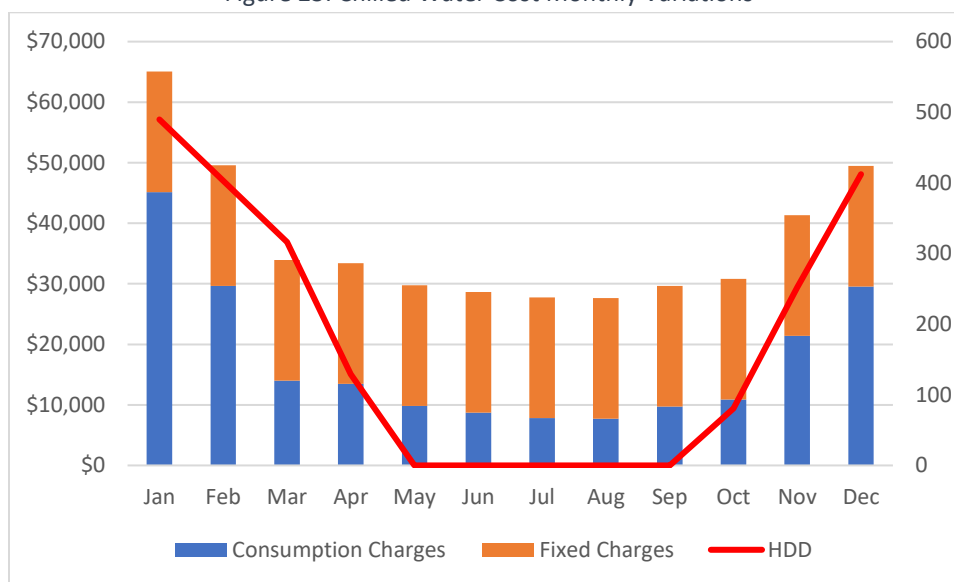
The following table depicts the FSB hot water consumption during the baseline year with the corresponding costs.

Table 8: FSB Monthly Hot Water Consumption and Cost

Month	Consumption	Consumption Charges	Fixed Charges	Rate	Energy Value	Deviation
Unit	mmBTU	\$	\$	\$/mmBTU	MJ	-
Jan	3,783	\$45,131	\$19,921	\$17.1959	3,193,021	45%
Feb	2,914	\$29,635	\$19,921	\$17.0063	2,459,546	29%
Mar	2,396	\$14,014	\$19,921	\$14.1630	2,022,331	13%
Apr	1,531	\$13,503	\$19,921	\$21.8318	1,292,232	-36%
May	1,436	\$9,837	\$19,921	\$20.7226	1,212,048	-45%
Jun	1,314	\$8,712	\$19,921	\$21.7906	1,109,075	-58%
Jul	1,164	\$7,822	\$19,921	\$23.8343	982,468	-79%
Aug	1,190	\$7,735	\$19,921	\$23.2403	1,004,413	-75%
Sep	1,469	\$9,725	\$19,921	\$20.1809	1,239,902	-42%
Oct	1,851	\$10,884	\$19,921	\$16.6423	1,562,327	-12%
Nov	2,709	\$21,401	\$19,921	\$15.2536	2,286,517	23%
Dec	3,213	\$29,527	\$19,921	\$15.3901	2,711,916	35%
Total	24,970	\$207,926	\$239,052	\$17.9006	21,075,796	-
Maximum	3,783 (Jan)	\$45,131 (Jan)	\$19,921 (Jan)	\$23.8343 (Jul)	3,193,021 (Jan)	45% (Jan)
Average	2,081	\$17,327	\$19,921	\$18.9376	1,756,316	-
Minimum	1,164 (Jul)	\$7,735 (Aug)	\$19,921 (Jan)	\$14.1630 (Mar)	982,468 (Jul)	-79% (Jul)

The below figure shows the hot water monthly consumption along with the HDD. The profile clearly shows a parabolic shape throughout the baseline year, as the HW energy consumption logically rises in winter and drops in summer, in correlation with the red line which represents the Monthly HDD Variations. It is also clear that during summertime, the fixed charges account for the biggest share of the total cost due to the low heating demand in the facility while during wintertime the two costs are more even.

Figure 23: Chilled Water Cost Monthly Variations



3.3.2. HW Regression Analysis

Building the regression equation in a similar way as explained in the electricity section (with HDD instead of CDD) would lead to the below model:

Month	Occupancy	HDD
Unit	Room Nights	°C.Days
Jan	4,693	490
Feb	3,723	403
Mar	6,088	316
Apr	6,494	129
May	6,484	0
Jun	6,318	0
Jul	5,971	0
Aug	6,384	0
Sep	6,065	0
Oct	6,001	81
Nov	5,075	252
Dec	3,951	412
Total	67,247	2,083

$$\text{mmBTU} = (-852.8) * \text{HDD} + (-44.2) * \text{OCC} + (627,770.4)$$

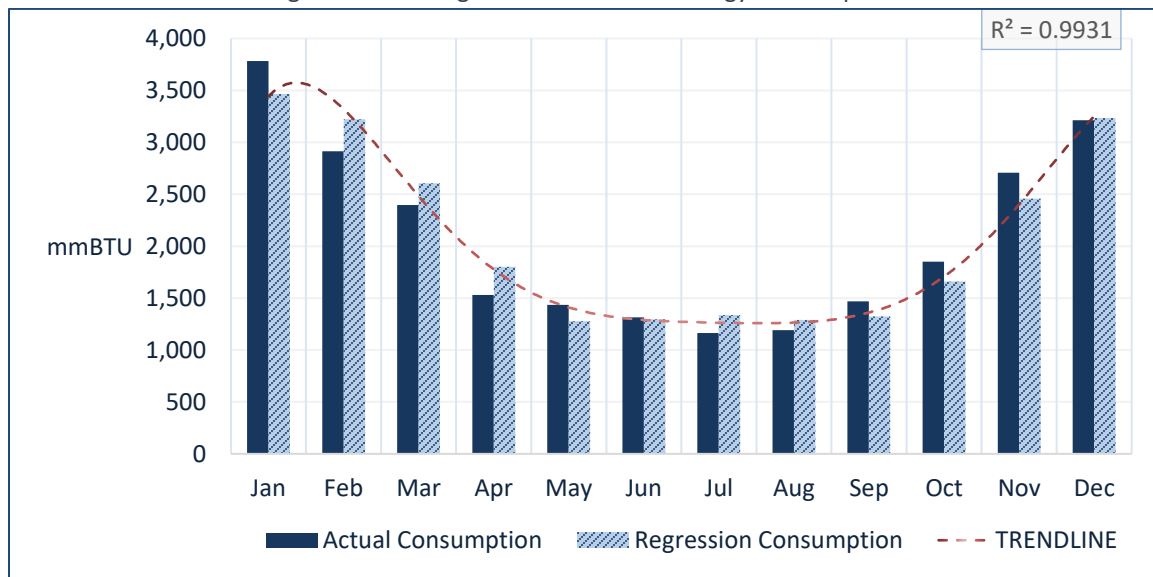
Table 9: HW Projected Consumption, Residuals and Variation

Month	Actual Consumption	Regression Consumption	Residuals	Deviation
Unit	mmBTU	mmBTU	mmBTU	%
Jan	3,783	3,466	317	8%
Feb	2,914	3,221	-307	-11%
Mar	2,396	2,607	-211	-9%
Apr	1,531	1,802	-271	-18%
May	1,436	1,278	158	11%
Jun	1,314	1,297	17	1%
Jul	1,164	1,335	-171	-15%
Aug	1,190	1,289	-99	-8%
Sep	1,469	1,324	145	10%
Oct	1,851	1,659	192	10%
Nov	2,709	2,458	251	9%
Dec	3,213	3,233	-20	-1%
Total	24,970	24,970	0	0%

The following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

In the case of HW, $R^2 = 0.9931$, which means that the model can be strongly considered an accurate one.

Figure 24: HW Regression and Actual Energy Consumption



3.4. Natural Gas

3.4.1. FSB Overall Gas Consumption and Cost

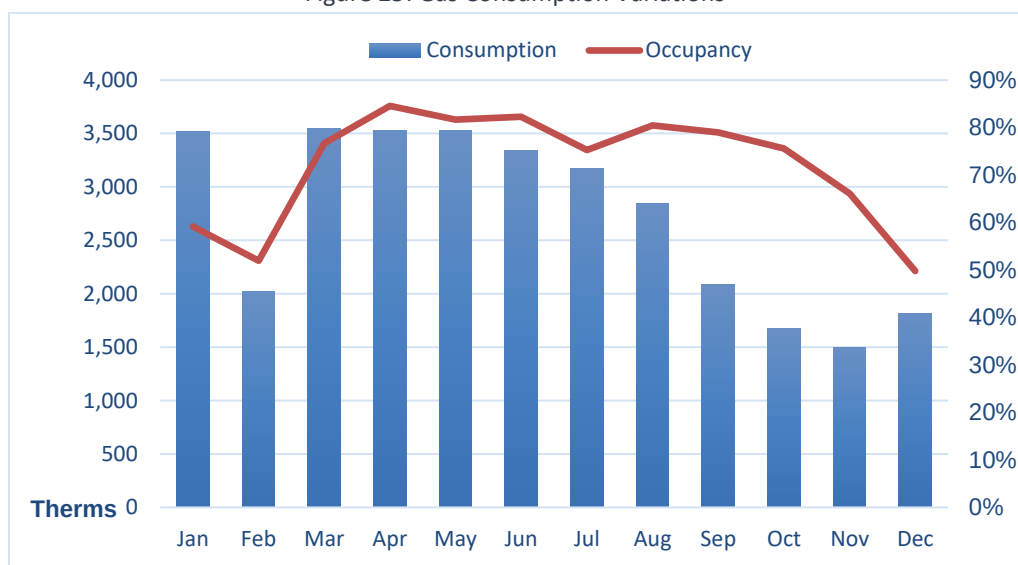
Natural Gas is mostly used to run some kitchen equipment in the hotel.

The total audited gas consumption is then **32,577 Therms** and accounts for **\$72,971** annually at an average rate of **2.24 \$/Therm**. The following table depicts the monthly Gas consumption over the baseline year.

Table 10: Baseline Gas Consumption and Cost

Month	Consumption		Rate	Energy Value	Deviation
Unit	Therms	\$	\$/Therm	MJ	-
Jan	3,515	\$7,958	2.264	92,797	23%
Feb	2,025	\$4,298	2.123	53,461	-34%
Mar	3,544	\$8,525	2.406	93,563	23%
Apr	3,526	\$6,985	1.981	93,088	23%
May	3,526	\$7,983	2.264	93,088	23%
Jun	3,345	\$7,572	2.264	88,301	19%
Jul	3,177	\$7,193	2.264	83,874	15%
Aug	2,848	\$6,448	2.264	75,188	5%
Sep	2,083	\$4,716	2.264	54,992	-30%
Oct	1,679	\$3,801	2.264	44,326	-62%
Nov	1,497	\$3,389	2.264	39,521	-81%
Dec	1,812	\$4,102	2.264	47,838	-50%
Total	32,577	\$72,971	2.240	860,037	-
Maximum	3,544 (Mar)	8,525 (Mar)	2.41 (Mar)	93,563 (Mar)	23% (Mar)
Average	2,715	\$6,081	\$2.240	71,670	-
Minimum	1,497 (Nov)	3,389 (Nov)	1.98 (Apr)	39,521 (Nov)	-81% (Nov)

Figure 25: Gas Consumption Variations



3.5. Water

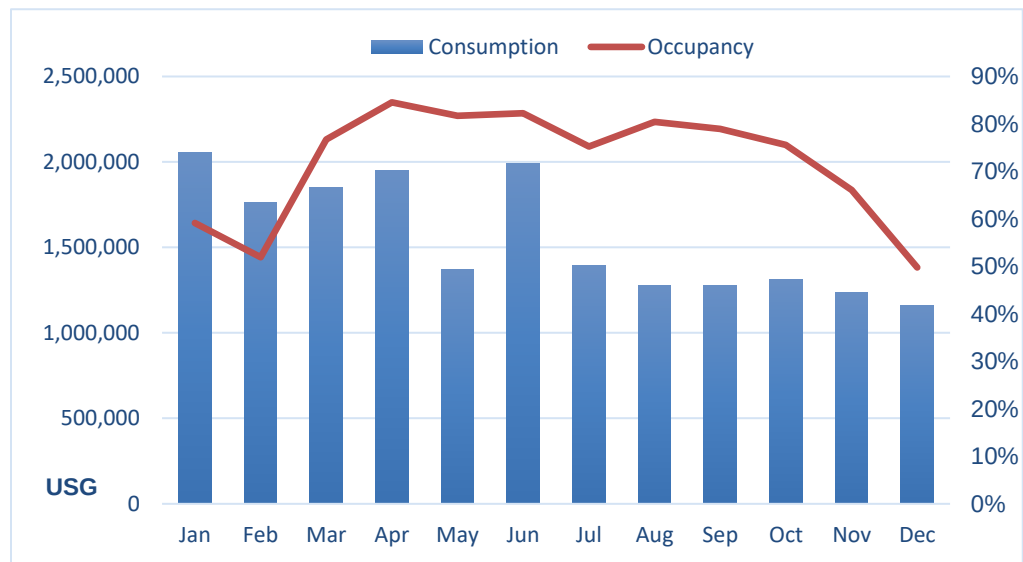
3.5.1. Water Consumption and Cost

The total water consumption for the baseline year is shown in the following table and chart.

Table 11: Monthly Water Consumption and Cost

Month	Consumption		Rate	Deviation
Unit	USG	\$	\$/USG	-
Jan	2,054,288	\$29,376	\$0.0143	24%
Feb	1,765,316	\$25,244	\$0.0143	12%
Mar	1,849,597	\$26,449	\$0.0143	16%
Apr	1,951,831	\$27,911	\$0.0143	20%
May	1,371,334	\$19,610	\$0.0143	-13%
Jun	1,988,949	\$28,442	\$0.0143	22%
Jul	1,392,843	\$19,918	\$0.0143	-12%
Aug	1,276,014	\$18,247	\$0.0143	-22%
Sep	1,277,648	\$18,270	\$0.0143	-22%
Oct	1,315,348	\$18,809	\$0.0143	-18%
Nov	1,237,040	\$17,690	\$0.0143	-26%
Dec	1,159,869	\$16,586	\$0.0143	-34%
Total	18,640,077	\$266,553	\$0.014	-
Maximum	2,054,288 (Jan)	\$29,376 (Jan)	\$0.01 (Jan)	24% (Jan)
Average	1,553,340	\$22,213	\$0.014	-
Minimum	1,159,869 (Dec)	\$16,586 (Dec)	\$0.01 (Jan)	-34% (Dec)

Figure 26: Water Monthly Variations



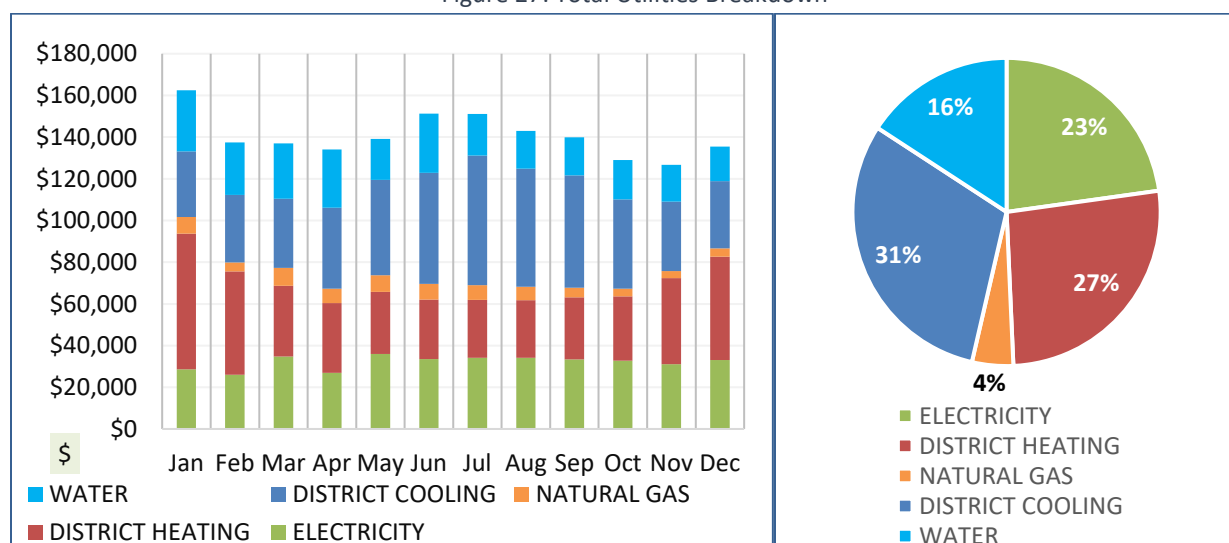
3.6. Utilities Accounting Summary

Total utilities cost is the aggregate of the cost of each of the electricity, chilled water, gas and water over the baseline period and they are shown in the following table and chart. As can be seen, district cooling accounts for 31% of the utilities expenditure while district heating accounts for 27%, electricity for 23%, water for 16% and Gas for 4%.

Table 12: Total Utilities Accounting Breakdown

MONTH	ELECTRICITY	DISTRICT HEATING	NATURAL GAS	DISTRICT COOLING	WATER	TOTAL
Unit	\$	\$	\$	\$	\$	\$
Jan	\$28,636	\$65,052	\$7,958	\$31,433	\$29,376	\$162,456
Feb	\$25,972	\$49,556	\$4,298	\$32,396	\$25,244	\$137,467
Mar	\$34,766	\$33,935	\$8,525	\$33,285	\$26,449	\$136,960
Apr	\$26,966	\$33,424	\$6,985	\$38,813	\$27,911	\$134,099
May	\$36,030	\$29,758	\$7,983	\$45,788	\$19,610	\$139,169
Jun	\$33,461	\$28,633	\$7,572	\$53,233	\$28,442	\$151,341
Jul	\$34,110	\$27,743	\$7,193	\$62,106	\$19,918	\$151,070
Aug	\$34,110	\$27,656	\$6,448	\$56,573	\$18,247	\$143,035
Sep	\$33,444	\$29,646	\$4,716	\$53,885	\$18,270	\$139,961
Oct	\$32,741	\$30,805	\$3,801	\$42,818	\$18,809	\$128,975
Nov	\$31,015	\$41,322	\$3,389	\$33,328	\$17,690	\$126,744
Dec	\$33,144	\$49,448	\$4,102	\$32,142	\$16,586	\$135,423
Total	\$384,396	\$446,978	\$72,971	\$515,802	\$266,553	\$1,686,700
Maximum	\$36,030 (May)	\$65,052 (Jan)	\$8,525 (Mar)	\$62,106 (Jul)	\$29,376 (Jan)	\$162,456 (Jan)
Average	\$32,033	\$37,248	\$6,081	\$42,983	\$22,213	\$140,558
Minimum	\$26,966 (Apr)	\$27,656 (Aug)	\$3,389 (Nov)	\$33,285 (Mar)	\$17,690 (Nov)	\$126,744 (Nov)
Percentage	23%	27%	4%	31%	16%	100%

Figure 27: Total Utilities Breakdown



3.7. Energy and Carbon Footprint

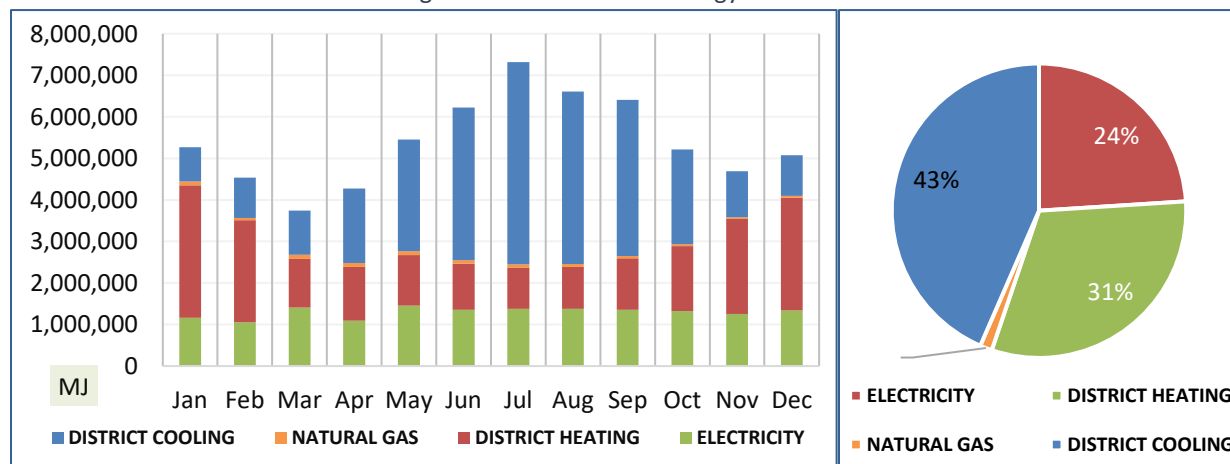
3.7.1. Energy Footprint

To be able to aggregate all the utilities energy consumption, the energy consumption of all energy sources is converted to a single energy unit MJ (mega joules). The breakdown for the energy sources consumption is shown in the following table. As can be seen, district cooling accounts for 42% of the energy sources energy consumption, followed by district heating at 30%, electricity at 26% and Gas at only 1%.

Table 13: Total Utilities Energy Breakdown

MONTH	ELECTRICITY	DISTRICT HEATING	NATURAL GAS	DISTRICT COOLING	TOTAL
Unit	MJ	MJ	MJ	MJ	MJ
Jan	1,158,322	3,193,021	92,797	820,928	5,265,068
Feb	1,050,548	2,459,546	53,461	969,564	4,533,120
Mar	1,406,264	1,175,754	93,563	1,067,436	3,743,018
Apr	1,090,752	1,292,232	93,088	1,799,719	4,275,792
May	1,457,400	1,212,048	93,088	2,691,881	5,454,417
Jun	1,353,494	1,109,075	88,301	3,671,392	6,222,262
Jul	1,379,748	982,468	83,874	4,872,221	7,318,311
Aug	1,379,744	1,004,413	75,188	4,152,882	6,612,228
Sep	1,352,779	1,239,902	54,992	3,759,925	6,407,598
Oct	1,324,361	1,562,327	44,326	2,282,385	5,213,398
Nov	1,254,535	2,286,517	39,521	1,109,217	4,689,790
Dec	1,340,651	2,711,916	47,838	973,838	5,074,242
Total	15,548,599	20,229,219	860,037	28,171,389	64,809,244
Maximum	1,457,400 (May)	3,193,021 (Jan)	93,563 (Mar)	4,872,221 (Jul)	7,318,311 (Jul)
Average	1,295,717	1,685,768	71,670	2,347,616	5,400,770
Minimum	1,050,548 (Feb)	982,468 (Jul)	39,521 (Nov)	820,928 (Jan)	3,743,018 (Mar)
Percentage	24%	31%	1%	43%	100%

Figure 28: Total Utilities Energy Breakdown



As per the above table, the **annual energy footprint** of FSB stands at: **117.83 MJ/ft²**

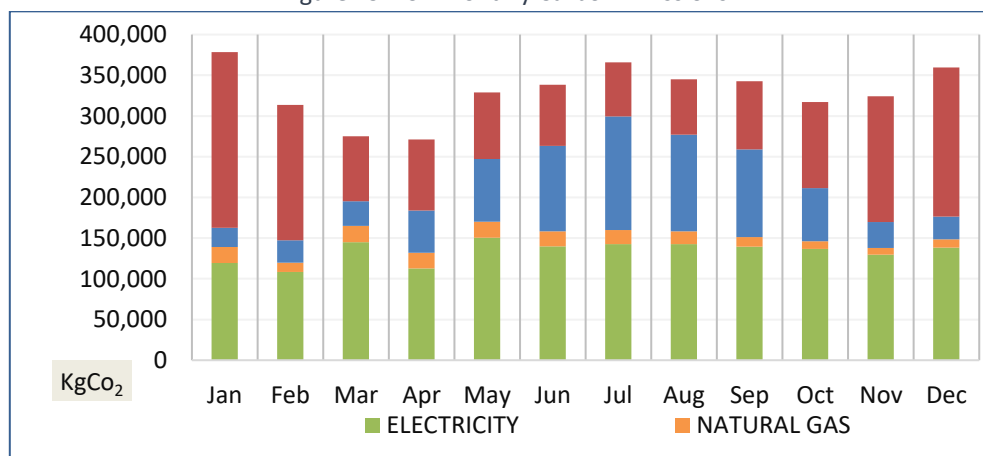
3.7.2. FSB Carbon Emissions

The carbon emissions calculation is done by multiplying the consumption of each utility (electricity, district cooling, district heating and gas) by the emissions factor of each. The electricity emissions factor varies between countries, with the USA average of 0.37 KgCO₂/KWh being taken in this report. Chilled water, hot water and Gas emission factors are 0.29 KgCO₂/RTH, 57.07 KgCO₂/mmBTU and 5.55 KgCO₂/Therm respectively. The **total annual carbon emissions** are at a total of **3,960,156 KgCO₂**.

Table 14: FSB Carbon Emissions Monthly Breakdown

MONTH	ELECTRICITY	NATURAL GAS	DISTRICT COOLING	DISTRICT HEATING	TOTAL
Unit	KgCO ₂	KgCO ₂	KgCO ₂	KgCO ₂	KgCO ₂
Jan	119,564	19,508	23,505	215,896	378,473
Feb	108,440	11,239	27,761	166,302	313,741
Mar	145,157	19,669	30,563	79,499	274,888
Apr	112,590	19,569	51,530	87,374	271,062
May	150,436	19,569	77,074	81,953	329,031
Jun	139,710	18,563	105,119	74,990	338,382
Jul	142,420	17,632	139,501	66,429	365,983
Aug	142,420	15,806	118,905	67,913	345,045
Sep	139,636	11,561	107,654	83,836	342,687
Oct	136,703	9,318	65,349	105,637	317,007
Nov	129,496	8,308	31,759	154,603	324,166
Dec	138,385	10,057	27,883	183,366	359,690
Total	1,604,956	180,801	806,603	1,367,797	3,960,156
Maximum	150,436 (May)	19,669 (Mar)	139,501 (Jul)	215,896 (Jan)	378,473 (Jan)
Average	133,746	15,067	67,217	113,983	330,013
Minimum	108,440 (Feb)	8,308 (Nov)	23,505 (Jan)	66,429 (Jul)	271,062 (Apr)
Percentage	41%	5%	20%	35%	100%

Figure 29: FSB Monthly Carbon Emissions



As per the above table, the **annual carbon intensity** of FSB stand at: **7.20 KgCO₂/ft²**

4. Electrical Energy Load Analysis & Energy Balance

4.1. Electrical Load Categories

A crucial milestone in energy audits is the buildup of the energy balance. This process starts with acquiring the load inventory – electrical end uses, and then running a complete simulation based on the operational schedule, load factors and behavioral statistics of each item in the inventory. The micro and macro energy consumption analysis of energy systems including lighting, heating, air conditioning, ventilation and building envelope, serves the purpose of identifying where and how a facility uses energy. The breakdown of the energy consumption by end-use would be the energy balance, which will provide an important baseline for the roadmap in developing the Energy Efficiency Measures (EEMs) that would optimize the overall energy costs.

Information and Database Process

FSB's Load inventory data is populated into an Energy Analysis System including assets name/category, actual electrical specification (KW) and operational schedule over 12 months (hours per day, week, month, or season). The second step is performing a total simulation and energy modeling of the facility with full bills reconciliation as per the baseline values.

In the present report, electrical loads were divided into the following categories:

1- Lighting

This end use category comprises all lighting fixtures across the property.

2- HVAC Fans

This end use category includes all ventilation loads needed for the an Indoor Air Quality operation such as AHUs, FCUs, extraction fans,...

3- Motors

This end use category includes all pumps such as the chilled water pumps, hot water pumps, water pumps (hot and domestic), circulation, pool pumps, as well as lifts.

4- Kitchen and Refrigeration

This category comprises electricity driven Kitchen and Refrigeration equipment.

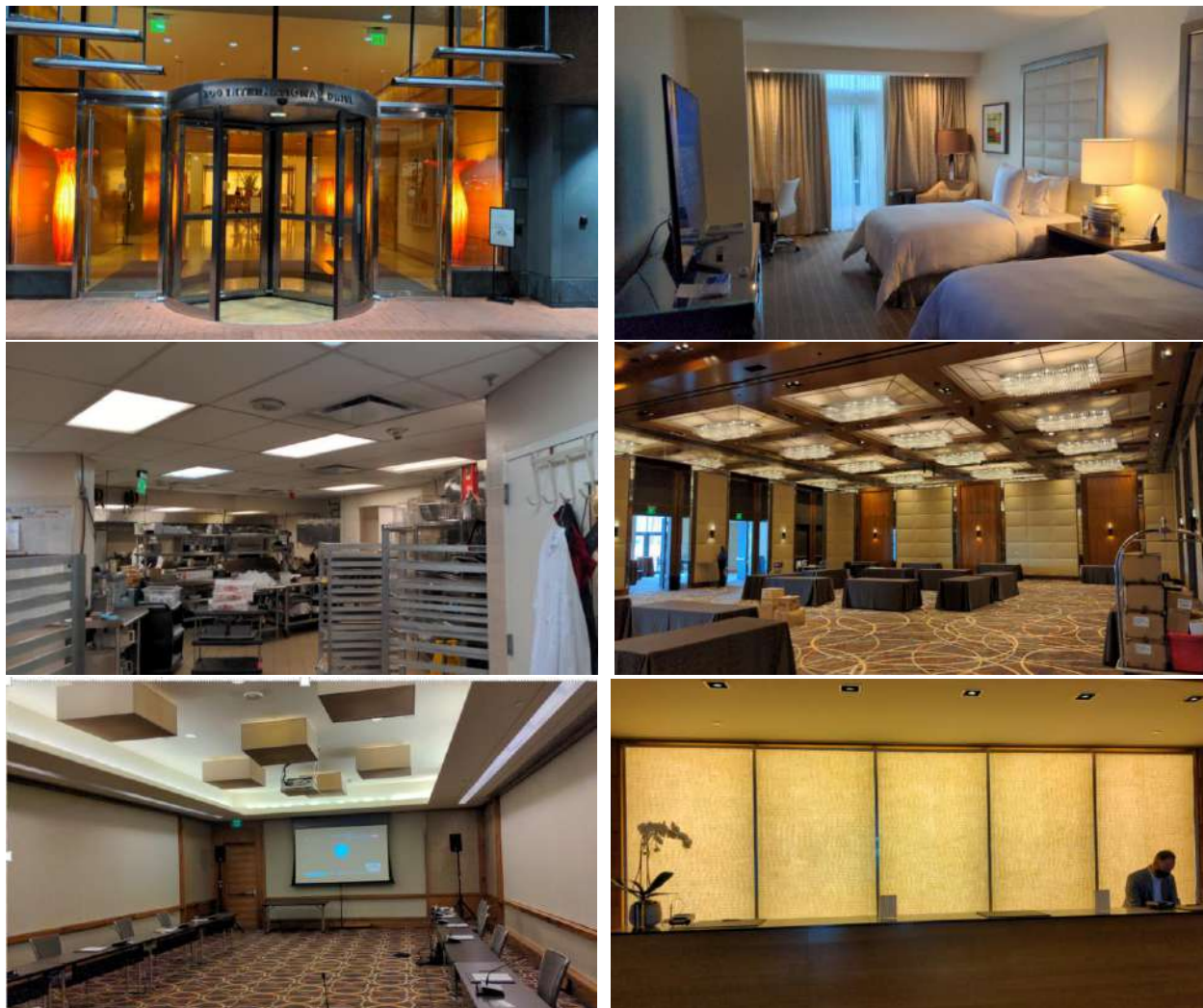
5- Miscellaneous

This category includes all loads not found in the previous sections such as low current equipment (IT/Telecom), laundry equipment, electronic/electrical appliances and other miscellaneous equipment.

4.1.1. Lighting

Lighting represents an important share of the FSB electricity consumption. Almost all lighting fixtures in the hotel are LED based including the façade lighting and all the BOH (Back of House); in fact, comprehensive LED retrofit projects took place in 2013 where around 1,000 fixtures have been replaced in the BOH and 9,400 fixtures have been replaced in the rest of the hotel. Currently, the areas that are still not of the LED type are some round PL fluorescent fixtures in the function areas (including pre-function) as well as the light walls behind the front desk in the lobby which is still equipped with T8 fluorescent fixtures.

Figure 30: Lighting Fixtures in FSB



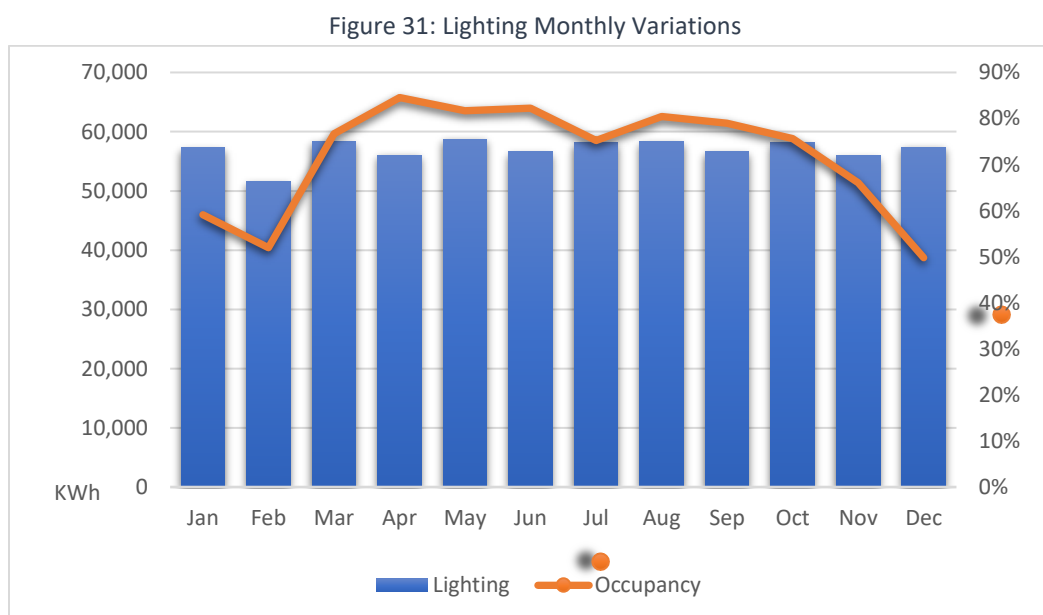
- Top (LED): Lobby Entrance (Left); Guest Room (Right)
- Middle (LED): BOH (Left); Grand Ballroom (Right)
- Bottom (Fluorescent): Second Floor meeting room (Left); Lobby Light Wall (Right)

In general, the lighting in the lobby as well as the corridors and the BOH is always operating. On the other hand, offices, façade, and pool lights are operated about 12 hours per day.

Guest rooms' lighting is controlled by the GRMS according to occupancy rate and the average presence in the room.

Compiling all the lighting inventory of FSB and their operation hours, and as per the energy simulations and bills reconciliation performed, the monthly energy consumption was found.

The total **Lighting** category energy consumption for the baseline period is **683,198 KWh** with the monthly variations as per the below chart.



4.1.2. HVAC Fans

The HVAC Fans category includes all the AHUs, FCUs, extract fans and kitchen exhaust fans installed in the hotel.

a. Air Handling Units

FSB has a total of 25 Air Handling Units (AHUs), out of which 5 are 100% FAHUs (fresh air handling units), 3 are of the VAV type and the rest are of the CAV type, grouped in three different locations, which are technical rooms located in the parking (P1), the third floor and the 16th floor. The units vary in size, type (recirculating, fresh air...) and capacity and are responsible for the proper conditioned air circulation through the whole facility.

Figure 32: AHU 17 (Left) and AHUs 18-19 (Right)



All units are connected to the JCI BMS with tracking and trending available for supply and return temperatures, along with VFD feedback when available. They are controlled via scheduling and setpoints with the note that a preheat coil is found in some AHUs.

Figure 33: AHU 4 - Lobby BMS Snapshot

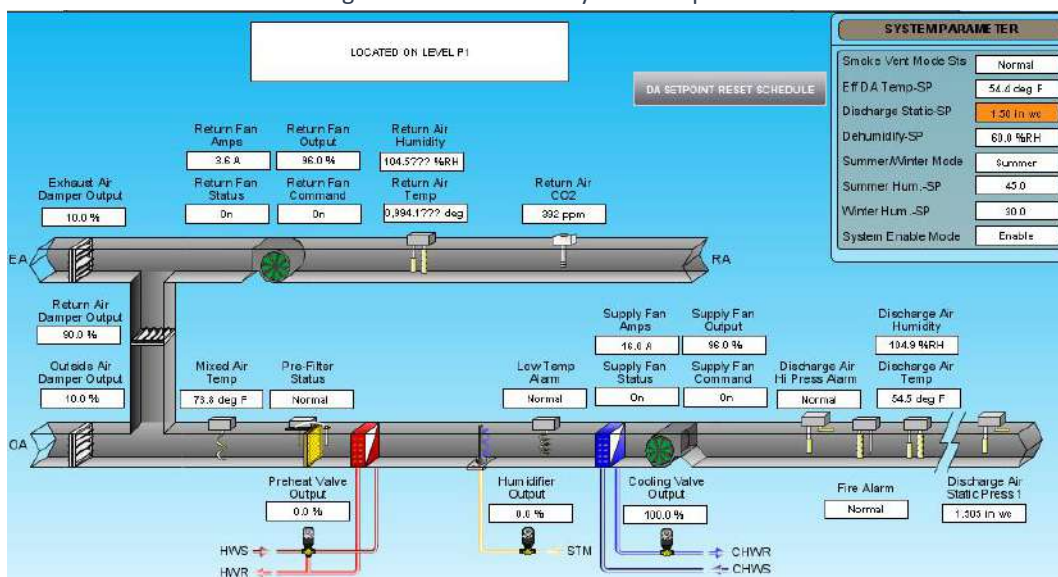
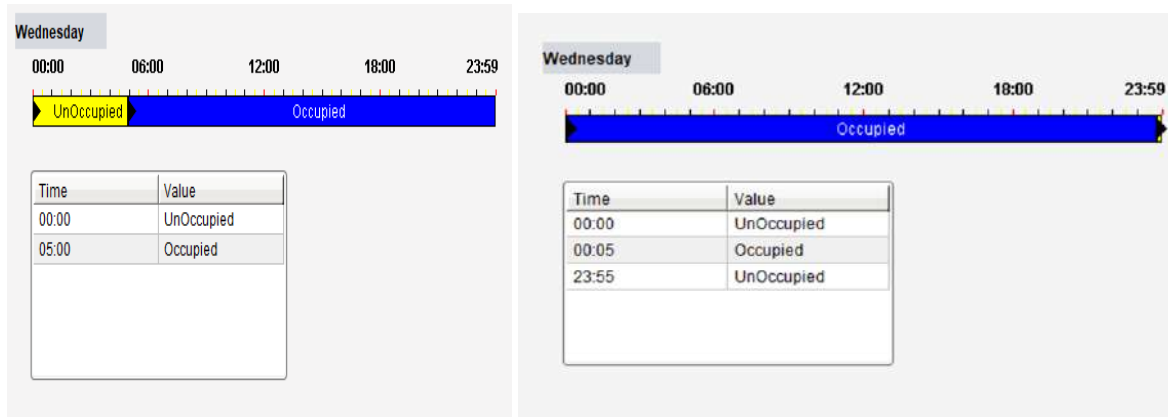


Figure 34: AHU scheduling



The below table provides the list of AHUs in the Hotel (Excluding the residences)

Table 15: List of AHUs in the FSB Hotel side

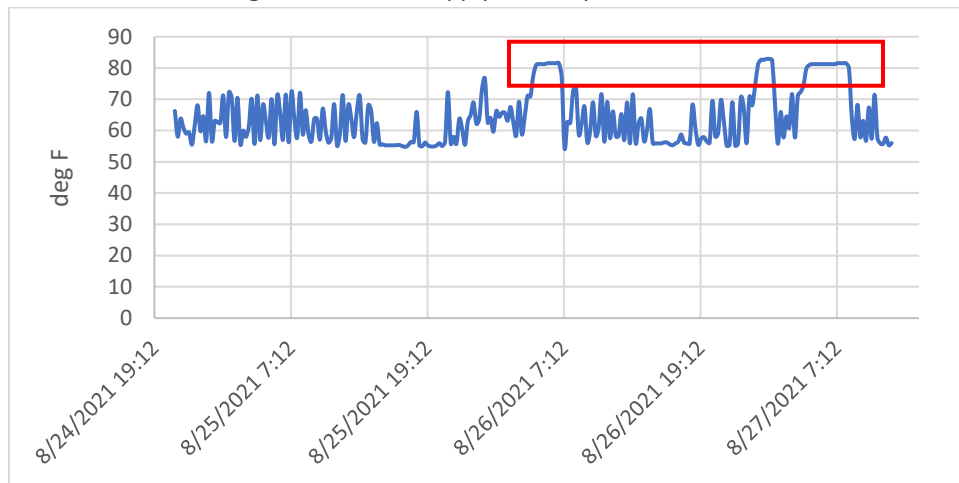
AHU#	Area Served	KW _{elec}	CFM	VFD
ACS 1	Restaurants & PDR	7.5	6800	No
ACS 2	Bar	5.5	5100	No
ACS 3	Main Kitchen MUA	7.5		No
ACS 4	Lobby and Tea Lounge	15	14300	Yes
ACS 5	Tenant Retail MUA	7.5	5000	No
ACS 6	Prefunction	18.75	21000	Yes
ACS 7	Banquet Facility	11.5	10200	Yes
ACS 8	Laundry	3.75	3000	Yes
ACS 9	Cardio Facility	5.5	8100	No
ACS 10	Ballroom 1	7.5	8300	Yes
ACS 11	Ballroom 2	7.5	8300	Yes
ACS 12	Banquet Kitchen	11.5	12500	No
ACS 13	Floor 1,2,3 back of house	7.5	7620	No
ACS 15	Meeting rooms & board room	18.75	18500	Yes
ACS 16	Spa Facilities	18.75	20000	Yes
ACS 17	18th flr Administration	1.2	950	No
ACS 18	Presidential Suite #1	3.75	2500	Yes
ACS 19	Presidential Suite #2	3.75	2500	Yes
ACS 20	Hotel Core	7.5	6240	No
ACS 21	Hotel Suite #1	11.5	10100	No
ACS 22	Hotel Corridor	7.5	7920	No
ACS 23	Hotel Suite #2	11.5	11500	No
ACS 24	Shelled area ventilation	3.75	3600	No
ACS 29	Pool Bar Kitchen MUA	2.2	3000	No

During the field audit, some trending/data recording were added to various parameters of AHUs, the below list is a sample of the gathered logging.

- **AHU 2**

The graph below highlights the supply air temperature of the main kitchen AHU for a course of three days. The AHU's operation schedule can be clearly seen in the graph (red box), since the supply temperature rises when the AHU is turned off. In fact, the AHU-2 is off from 12:00AM to 5:00AM every day, which is the exact period of time showing the maximum temperatures in the graph.

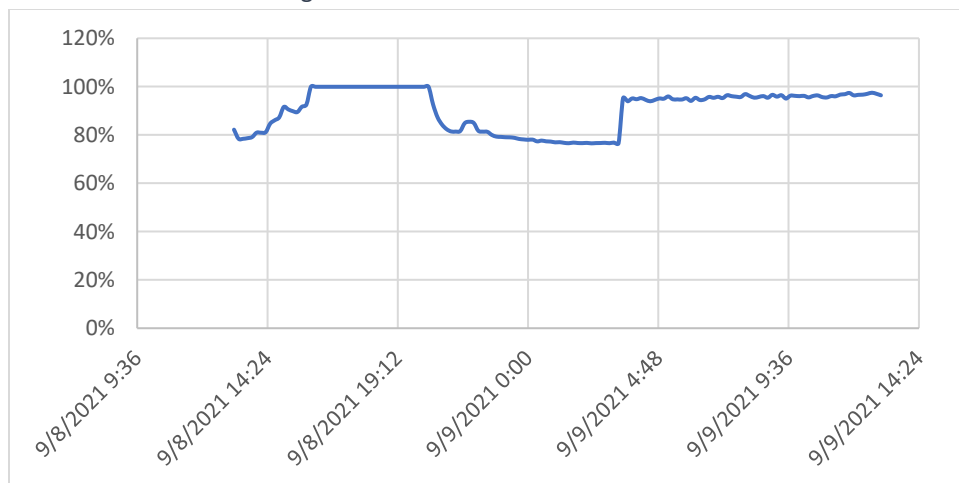
Figure 35: AHU-2 Supply Air Temperature Trend



- **AHU 4**

The graph below shows that the VFD feedback of AHU 4, which serves the lobby, always varies between 80% and 100%, with the minimum of 80% being mainly reached at night during the hours where there is no major traffic in the lobby.

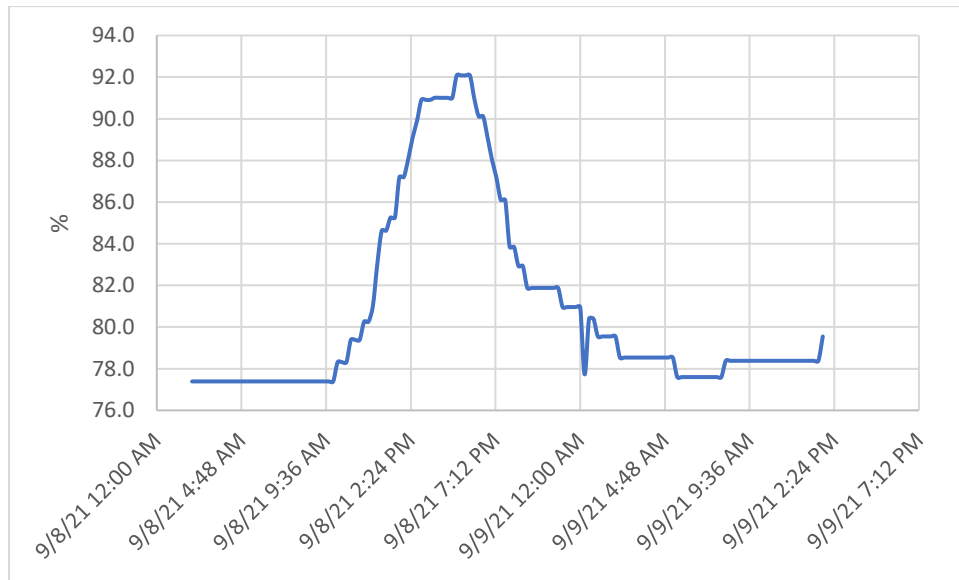
Figure 36: AHU-4 VFD Feedback Trend



- AHU 16

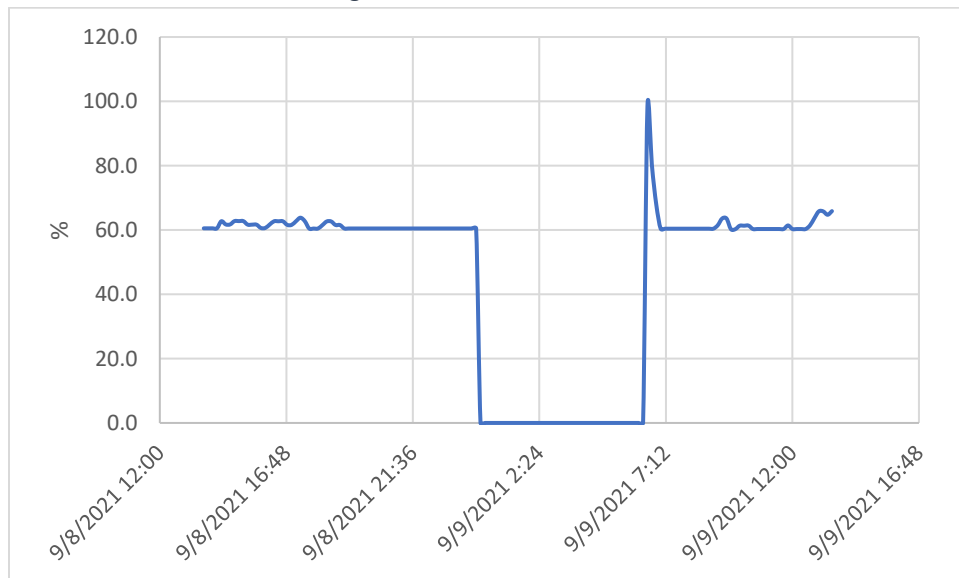
The graph below shows that the VFD feedback of AHU 16, which serves the SPA, has a VFD feedback between 77% and 92%, with the feedback being higher during the opening hours of the SPA compared to the rest of the day. In addition, it shows that the unit is running 24/7

Figure 37: AHU-16 VFD Trend

**- AHU 10**

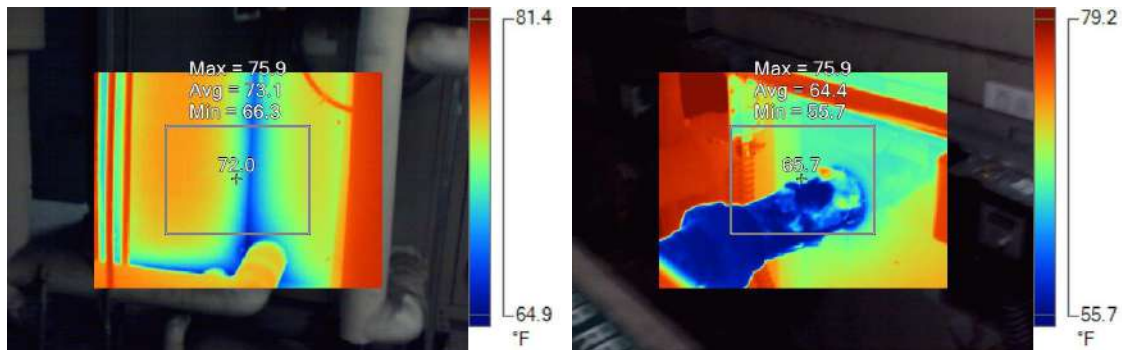
The graph below shows AHU 10, which serves the ballroom, operates on a clear schedule, where it is off from midnight until 6:00AM, and then on with a VFD feedback around 60% afterwards.

Figure 38: AHU-10 VFD Trend



Generally, the insulation of the AHUs on the enclosures and pipes was seen to be satisfactory. Nevertheless, some thermal losses were noticed in rare cases as shown in the thermal images below.

Figure 39: AHUs Thermal Images

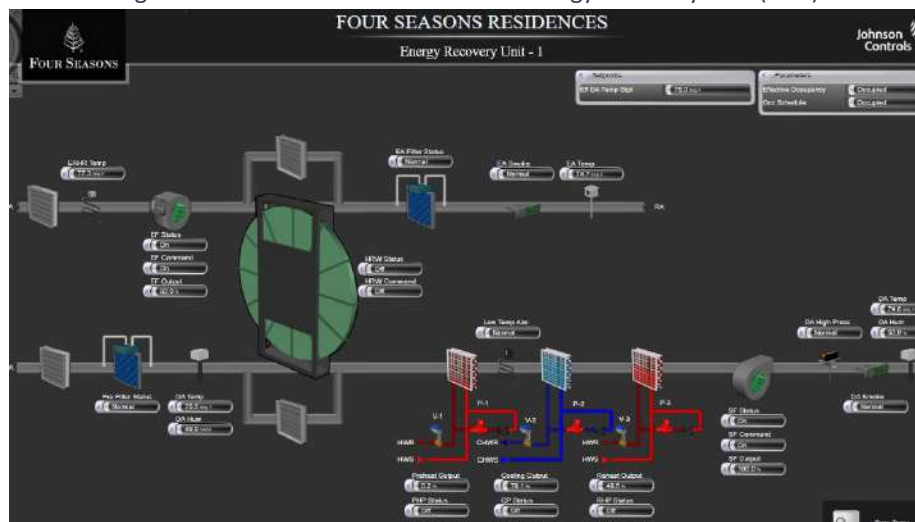


Residences AHUs:

In addition to the Hotel's HVAC units, the common areas and amenities of the residences floors are part of the present energy balance. There are in total Five AHUs in these floors, one 100% Fresh Air Handling Unit with a Heat Recovery Wheel (ERU) and four additional AHUs (RAHU 3 and 4 are seen in the left picture). The JCI BMs has been extended to accommodate them and their control is similar to those of the Hotel operation.



Figure 40: Residence Fresh AHU with Energy Recovery Unit (ERU)



b. Extract Fans

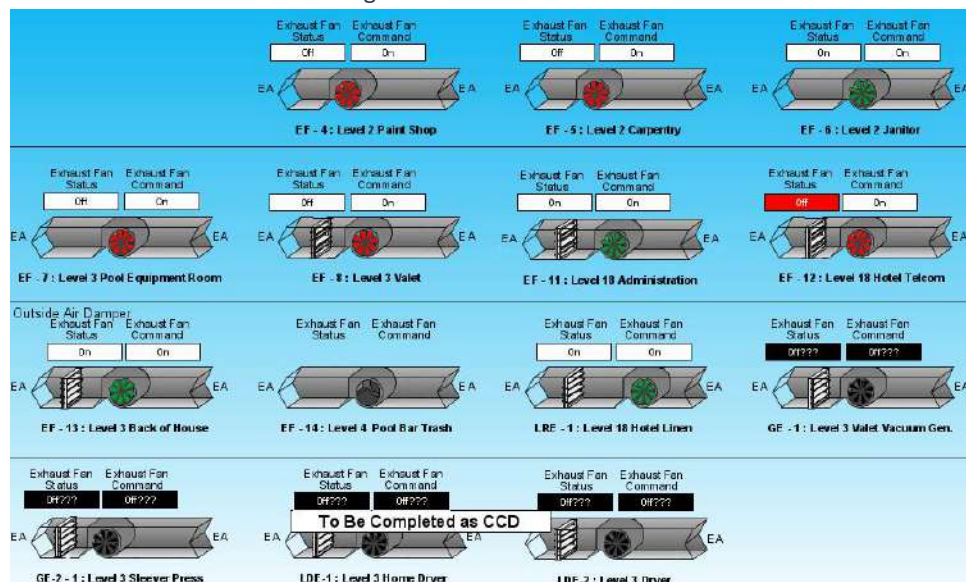
FSB has extract fans and ventilation fans spread all over the premises. These include toilet and plant rooms extract fans, as well as the kitchen extractor fans and the Ecology units which are the ones with the largest rated power and operating hours. Each kitchen has an ecology unit which are controlled by the VFD.

Figure 41: Banquet Ecology Unit (Left) and Guest Rooms Toilets Extract Fan (Right)



Most of the extract fans are connected and controlled by the BMS, as it can be seen in the snapshot below.

Figure 42: Exhaust Fans List



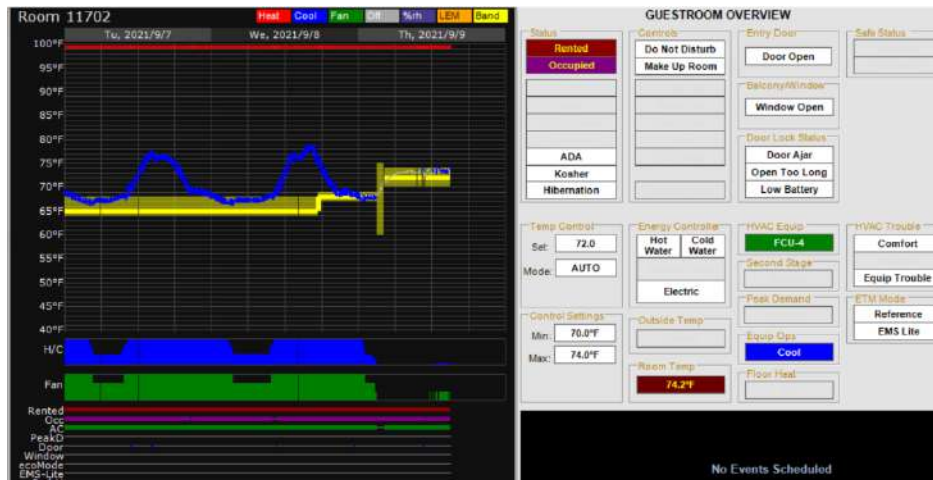
c. Fan Coil Units

There are around 80 fan coil units in FSB's common areas and BOH, all controlled by a local thermostat as seen in the right picture and remotely or remotely via the JCI BMS.

The other FCUs are located in the guest rooms and controlled by the Inncom GRMS which was seen to be properly operating.



Figure 43: GRMS controlling guest room thermostat



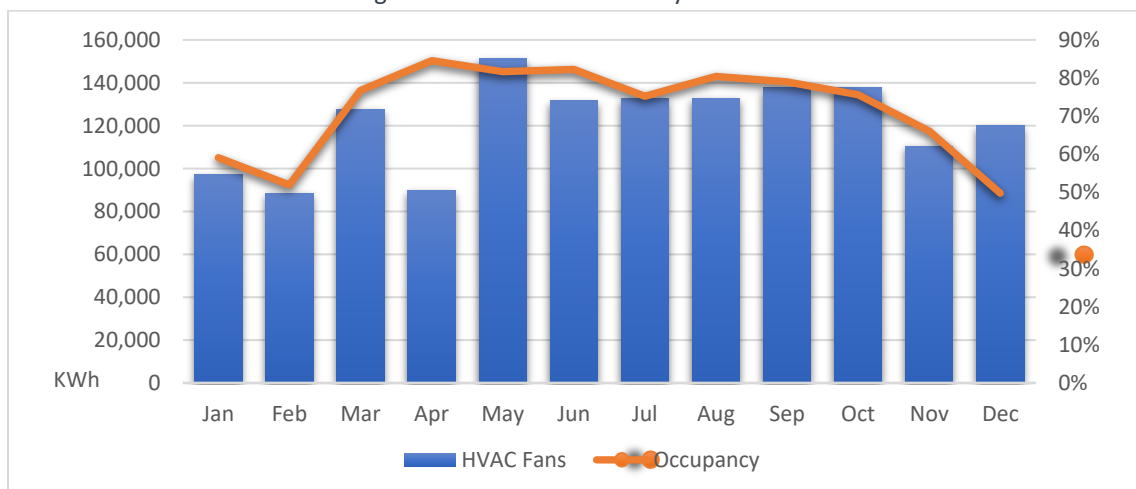
Following the energy simulation and bills reconciliation, the following table summarizes the load of each sub-category and the corresponding energy consumption for the baseline year.

Table 16: HVAC Fans Breakdown

SUB-CATEGORY	LOAD		ENERGY	
Unit	KW	%	KWh	%
Air Handling Units	207	75%	983,367	68%
Extract Fans	59	21%	386,384	27%
Fan Coil Units	11	4%	66,923	5%
Total	277	100%	1,436,674	100%

The total **HVAC Fans** category energy consumption for the baseline period is **1,436,674 KWh** with the monthly variations as per the below chart.

Figure 44: HVAC Fans Monthly Variations



4.1.3. Motors

Motors loads mainly consist of all pumps (Pool pumps, chilled water pumps, hot water pumps, circulation pumps, booster pumps ...) along with the lifts. Most Pumps are equipped with a VFD and connected to the BMS. This is the case of all Chilled and Hot Water pumps as well as all booster pumps:

The main pumps in FSB are the following:

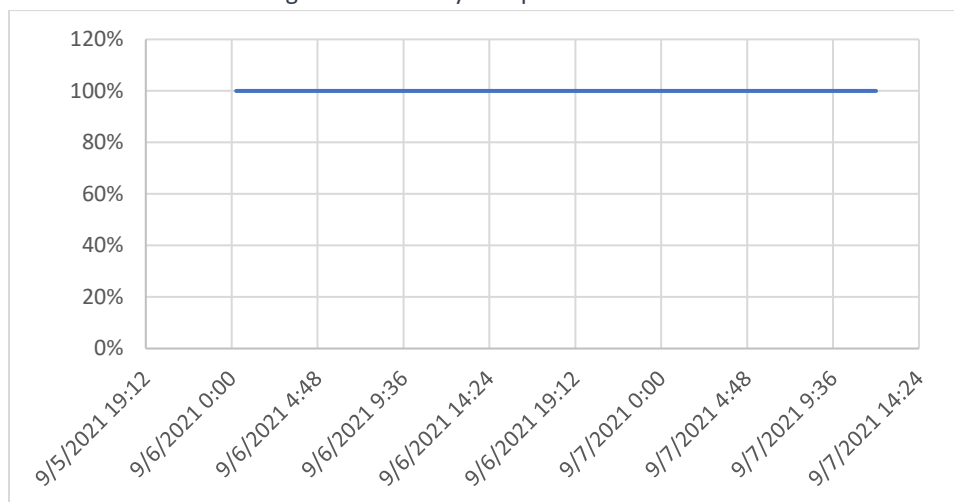
- District Cooling: 2x55KW Primary Chilled Water Pumps responsible for circulating the chilled water on the secondary side of the district cooling's heat exchangers to all the building's end uses infrastructure comprising AHUs, FCUs and cooling water pipes and networks. The pumps are connected to the facility's BMS, and both were operational during the field audit.

Figure 45: Primary Chilled Water Pumps (Left) with VFD Reading (Right)



The graph below shows the VFD feedback of one of the chilled water primary pumps during a whole day, which is always 100% meaning that the pump operates at maximum power all the time.

Figure 46: Primary Pump Chilled Water Trend



- District heating: 2x40HP Primary Hot Water Pumps responsible for circulating the hot water for the secondary side of the heat exchangers to the whole facility, as previously described for the chilled water pumps. The pumps are also connected to the BMS, and both were operational at 100% during the field audit, which is certainly wrong as hot water demand was very low . This discrepancy in the operation has been further covered in the EEM section.

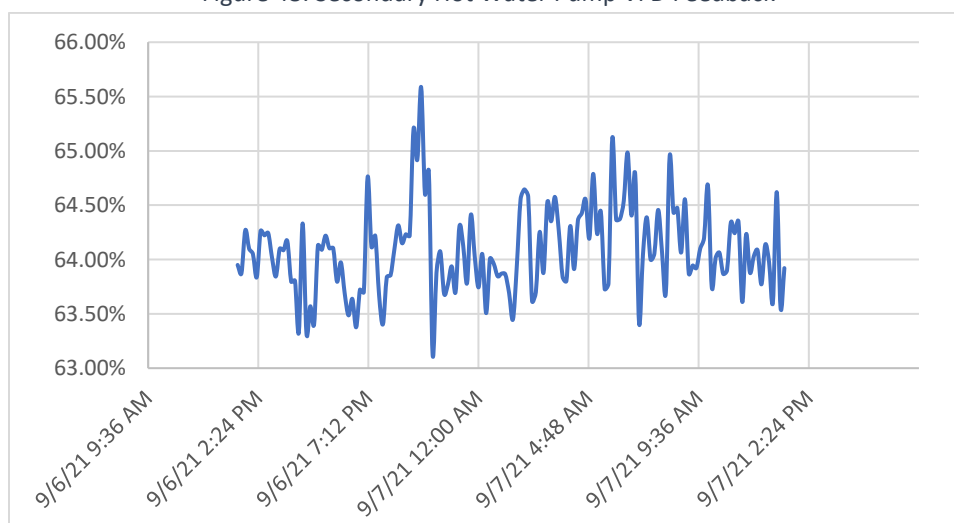
Figure 47: Primary Hot Water Pumps Nameplate (Left) with VFD Reading (Right)



The Primary Hot Water Pumps also have the same performance as the chilled water pumps where they were always returning a VFD feedback of 100%.

On the other hand, the secondary hot water pumps were seen to be operating constantly at around 64% (the operational ones) which is clear in the graph below representing the VFD feedback on one of the secondary hot water pumps.

Figure 48: Secondary Hot Water Pump VFD Feedback



- 2x5HP Return Hot Water Pumps for the hot water circulation back the district heating's heat exchangers running all the time together.

Figure 49: Return Hot Water Pumps



- 2x5KW Hot Water Pumps for the Hot Water Circulation in the AHUs, operating with VFD.

Figure 50: AHUs Hot Water Circulation Pumps



- 4x30KW Domestic Water Booster Pumps on VFD, with one continuously working.

Figure 51: Domestic Water Booster Pumps



- 2x20HP Hot Water Pumps on VFD located on the 18th floor, out of which one was only operational.

Figure 52: Hot Water Pumps



- 2x50HP Chilled Water Pumps with Heat Exchangers operating on VFD.

Figure 53: Chilled Water Pumps



- Two sets of 1x1.5KW circulation pump and 2x1.5KW blowers for the Vitality pool located in the 3rd floor plant room.

Figure 54: Vitality Pool Blowers



- 1x10HP Main Pool Filtration Pump.

Figure 55: Main Pool Filtration Pump



- 2x4KW Chilled Water and 2x4KW Hot Water Secondary Pumps serving the hotel podium, which are floors 1 to 4 of the hotel. As for the rest of the floors (5 to 21), they are directly served from the Primary Loops. All units are operational on VFD, with both the chilled water pumps running at 100%, and the hot water pumps running at 53% and 74% respectively.

Figure 56: Podium Secondary Pumps



- 8 Gen 2 Otis Elevators and 2 cable type Elevators.

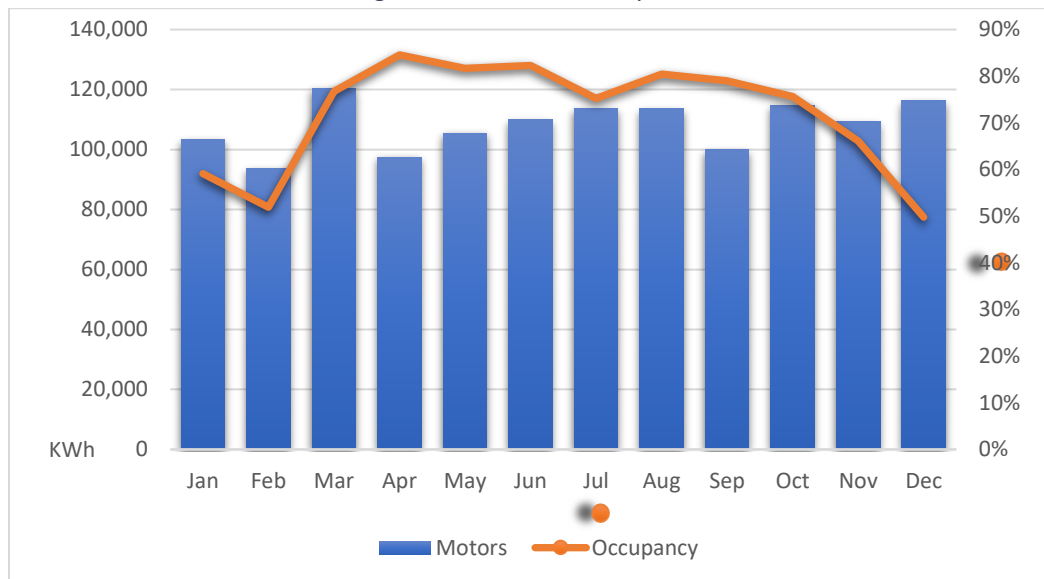
The below table provides the key subcategories of pumps with their total installed load (in KW) and the annual electricity consumption (in KWh).

Table 17: Motors Breakdown

SUB-CATEGORY	LOAD		ENERGY	
	Unit	KW	%	KWh
Chilled Water Pumps		205	37%	472,573
Hot Water Pumps		146	26%	528,217
Booster Pumps		120	22%	177,616
Pool/SPA Pumps		29	5%	74,973
Elevators		54	10%	43,782
Total		553	100%	1,297,162

The total **Motors** category energy consumption for the baseline period is **1,297,162 KWh** with the monthly variations as per the following chart.

Figure 57: Motors Monthly Variations



4.1.4. Kitchen and Refrigeration

The kitchen end use category includes all the refrigeration and electrically driven kitchen equipment across all kitchens of FSB (fridges, freezers, cold rooms, ovens, heaters, microwaves...). All the cold rooms in FSB are served by two water cooled compressors' racks, similarly, ice storage units are also of the water cooled type. As mentioned earlier, all the utilities bills for the outsourced restaurants (Bygone, Azumi, Loch Bar and Maximon) in FSB are being charged back and have been removed from the present energy balance. The in room dining and banqueting kitchens located in the 2nd floor are those considered in this end use category as well as the bakery and the pool bar .

The key kitchens in FSB are the main kitchen as well as the bakery, the pool bar, and the banquet.

The below set of pictures show one of the cold rooms' compressors racks and the schematic with the condenser cooling tapping from the return line of the Chilled water network. On the other hand, thermal images of the cold rooms' doors confirmed that the related weatherstripping/sealings are in good conditions.

Note that all remaining refrigerators and ice machines tapping on the chilled water lines are doing it from the supply side and not return like the compressors' racks.

Figure 58: Cold Rooms Compressor Racks (Up) and Kitchen Equipment (Down)

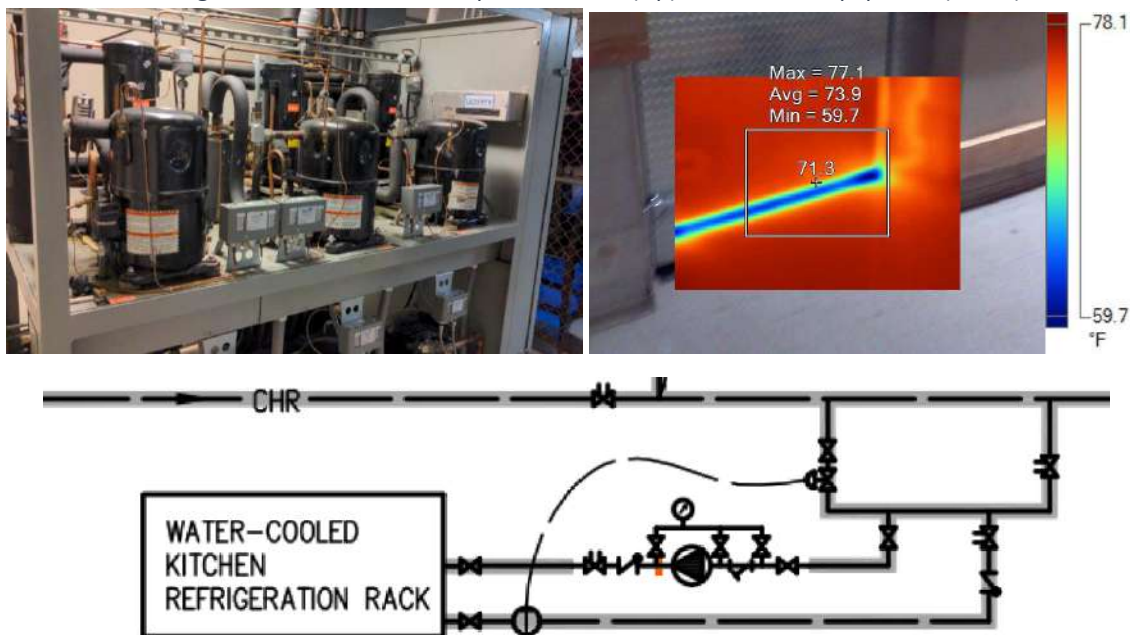
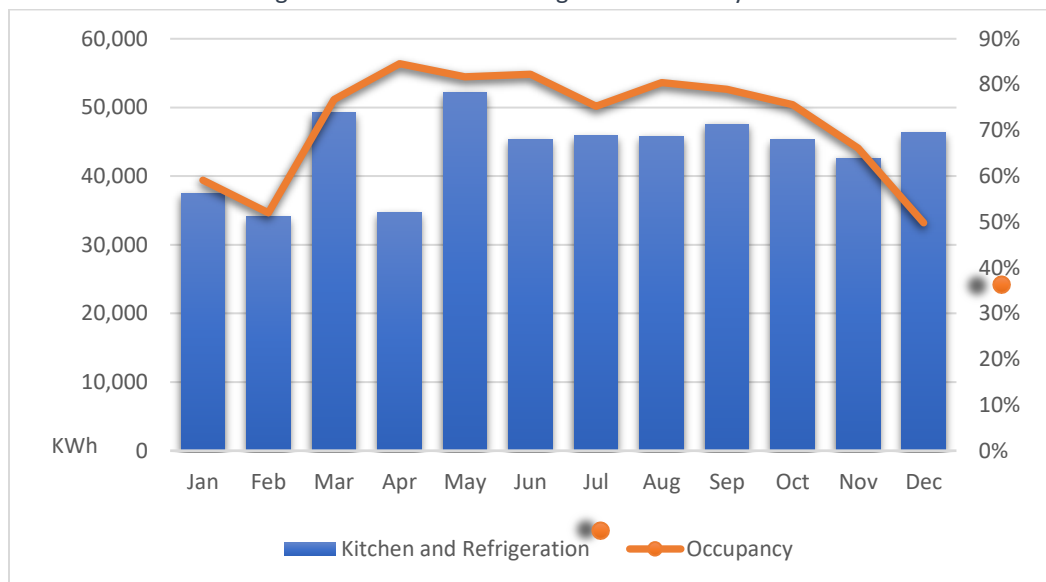


Figure 59: General kitchens view



As per the energy simulation and bills reconciliation, the total **Kitchen and Refrigeration** category energy consumption for the baseline period is **526,367 KWh** with the monthly variations as per the below chart.

Figure 60: Kitchen and Refrigeration Monthly Variations



4.1.5. Miscellaneous

Miscellaneous end use includes all remaining loads. This includes mostly all the low current loads (IT/Telecom/Security) and electronic appliances (such as TV and minibars) loads, gym equipment as well as the Sauna and Steam Room Heaters, along with the banquet booster heaters which are installed in order to boost the domestic hot water supplied at 140°F to above 165°F to comply with health department regulations.

Figure 61: Miscellaneous Equipment



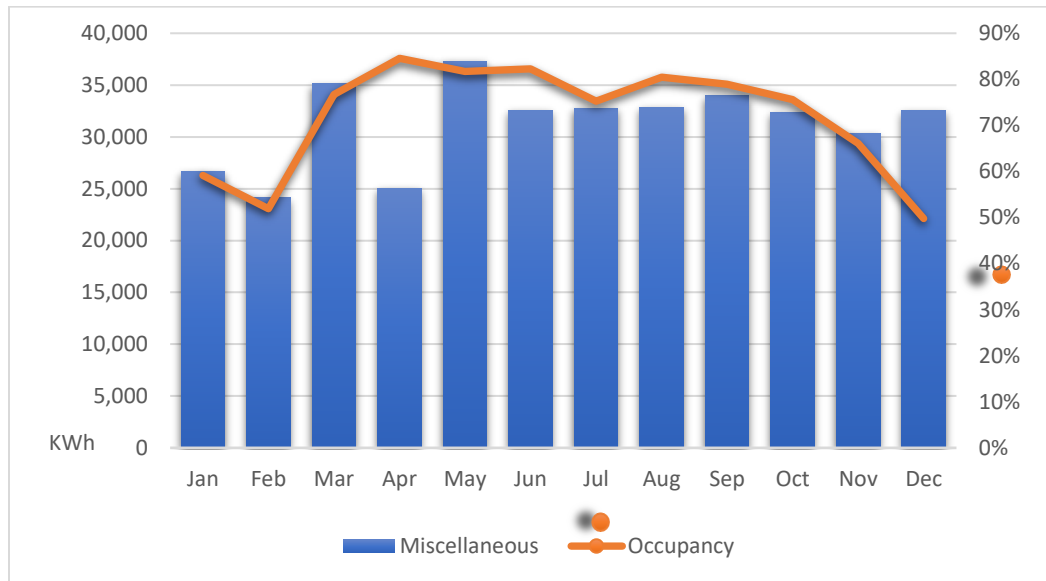
Also, FSB has a small laundry in its BOH. It includes all washing machines, dryers, and ironers. The laundry is used for guests and uniforms only, while everything else is outsourced. Before the pandemic, the laundry operated for 18 hours, which is now reduced to 10 hours without using the big washing machines.

Figure 62: Laundry Equipment



The total energy consumption of the **Miscellaneous** category is **375,654 KWh** with the monthly variations as per the below chart.

Figure 63: Miscellaneous Monthly Variations



4.2. Electrical Energy Balance

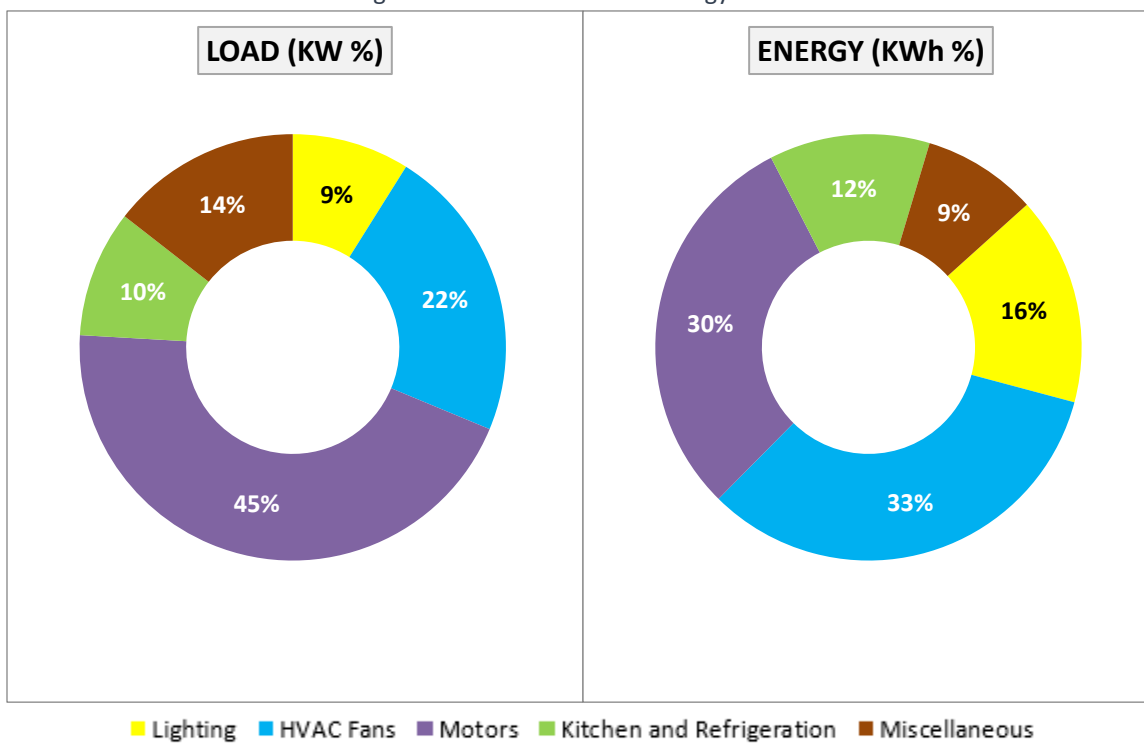
The Electrical Energy Balance provides the breakdown per chosen end-use category for both the installed electrical loads (KW) and the annual energy consumed (KWh). This is achieved following the energy simulation and bills reconciliation completion as per the load inventory, the related operations and the inspection done during the field energy audit.

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and annual energy consumption in KWh.

Table 18: Electrical Load & Energy Consumption Breakdown by Category

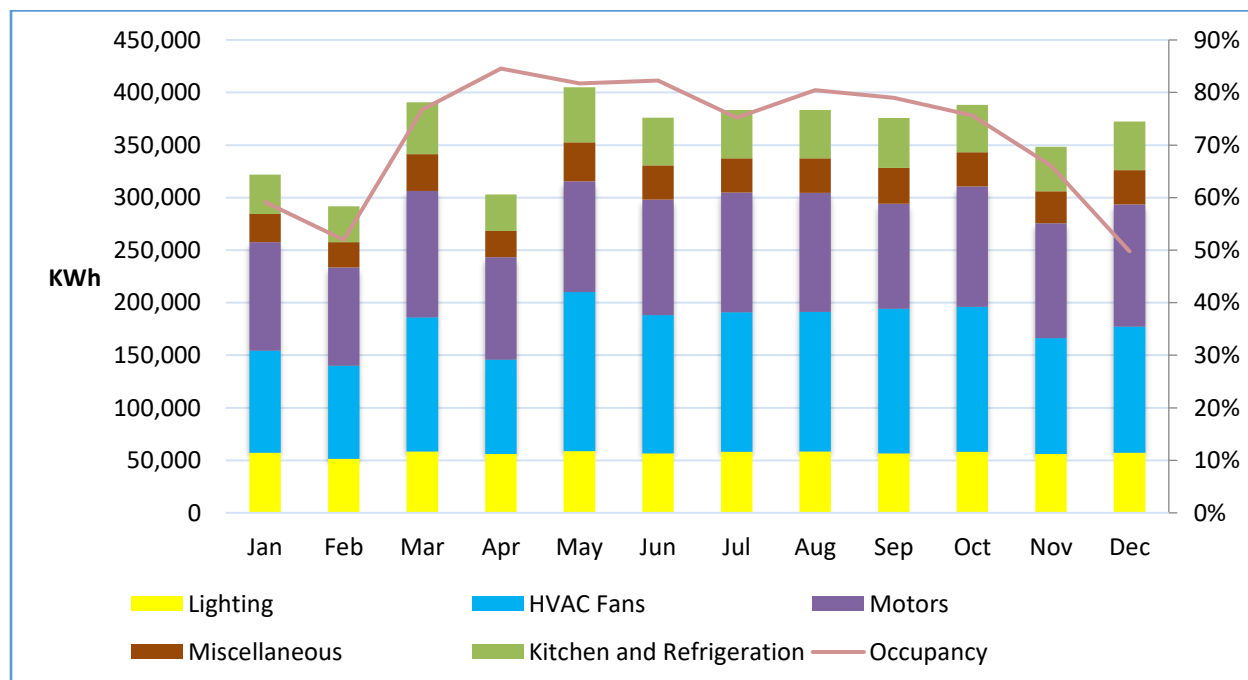
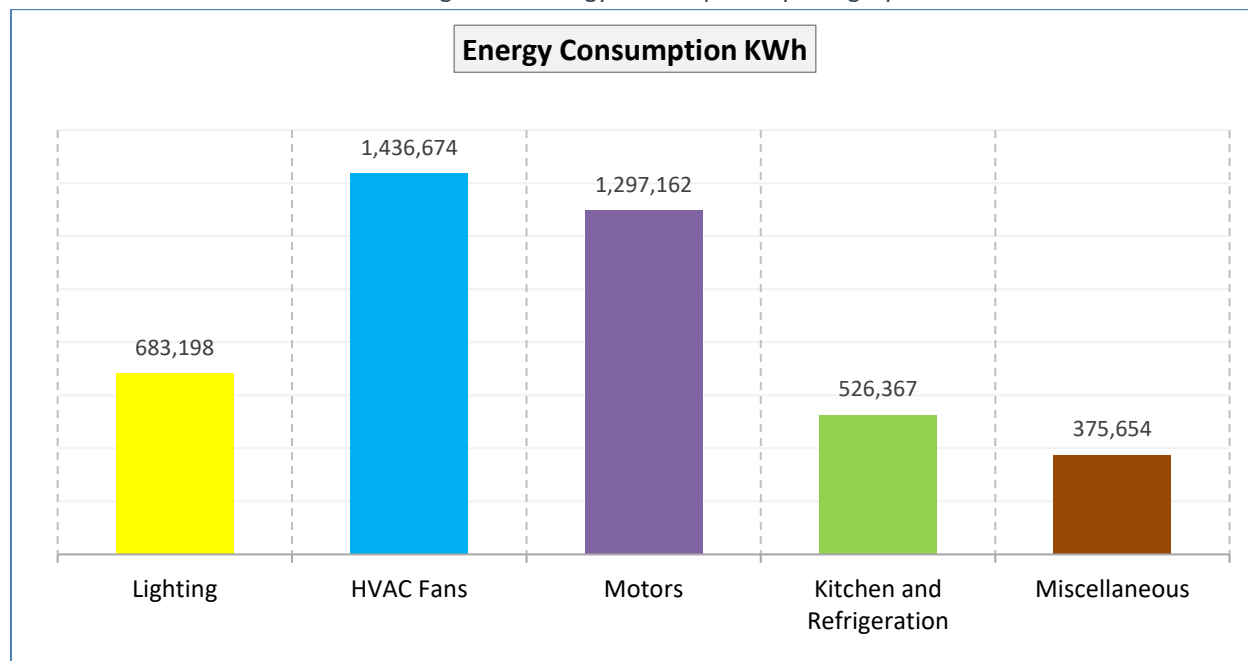
Category	Load		Energy			
	KW	%	KWh	\$	Kg. CO ₂	%
Lighting	111	9%	683,198	\$60,805	253,876	16%
HVAC Fans	277	22%	1,436,674	\$127,864	533,867	33%
Motors	553	45%	1,297,162	\$115,447	482,024	30%
Kitchen and Refrigeration	119	10%	526,367	\$46,847	195,597	12%
Miscellaneous	179	14%	375,654	\$33,433	139,593	9%
Total	1,239	100%	4,319,055	\$384,396	1,604,956	100%
Maximum	Motors - 553, 45%		HVAC Fans - 1,436,674 KWh, 127,864, 33%			
Minimum	Lighting - 111, 9%		Miscellaneous - 375,654 KWh, 33,433, 9%			

Figure 64: Electrical Load & Energy Breakdown



The Motors category shows the highest installed load at 45% of the total, followed by the HVAC Fans at 22%. Whereas the HVAC Fans category has the highest share of energy consumption at 33% of the total, followed by the Motors category at 30%.

Figure 65: Energy Consumption by Category



4.3. Thermal Energy

4.3.1. District Cooling

The District Cooling chilled water intake is used to cover all the space cooling and equipment condenser cooling demand across the tower's floors. The contractual agreement is based on a maximum subscription demand of 1150 Tons which carries a monthly fixed subscription charges. The remaining charges are purely related to the energy consumed (in RTh) while the other key contractual technical parameters are as follows

- Supply Temperature: 39.5°F +/- 2°F from May to September to heat exchanger
- 42.00 °F ± 2°F from October to April to heat exchanger
- Minimum temperature differential between supply and return 14 °F
- Supply pressure: Maximum: 150 psig | Minimum: 40 ± 5 psig
- Maximum pressure differential between supply and return 10 psid

Chilled water is supplied through two Heat Exchangers and two primary chilled water pump that is then further pushed through a set of secondary chilled water pumps covering different chilled water loops . The below schematic showcase the main chilled water line supply with the note that the key refrigeration racks' condenser cooling is connected to the return line of Chilled Water while other refrigeration/ice maker units are connected to the main supply line.

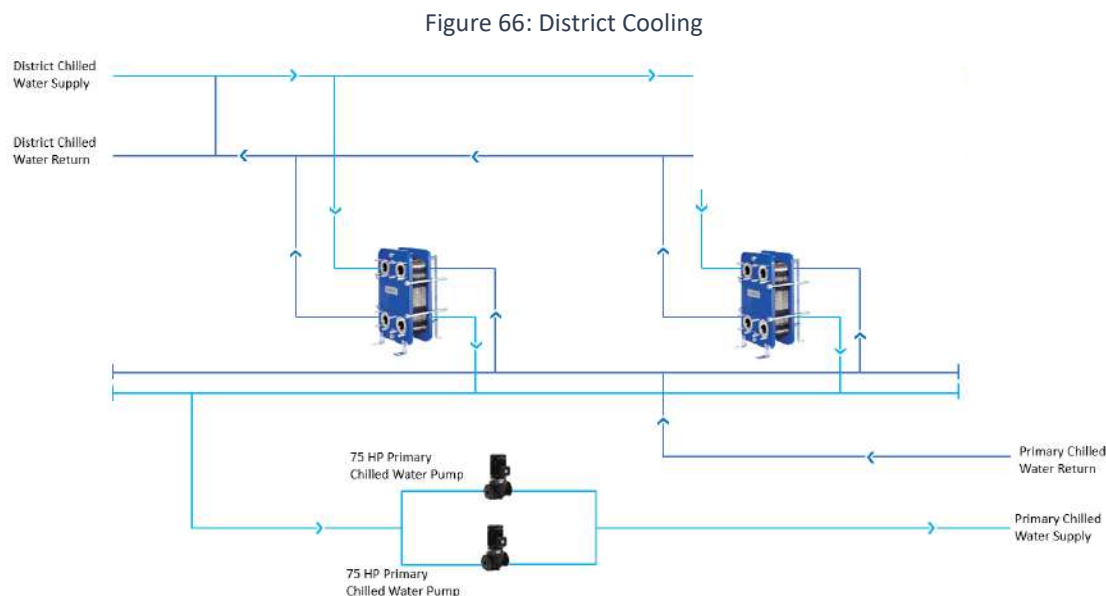
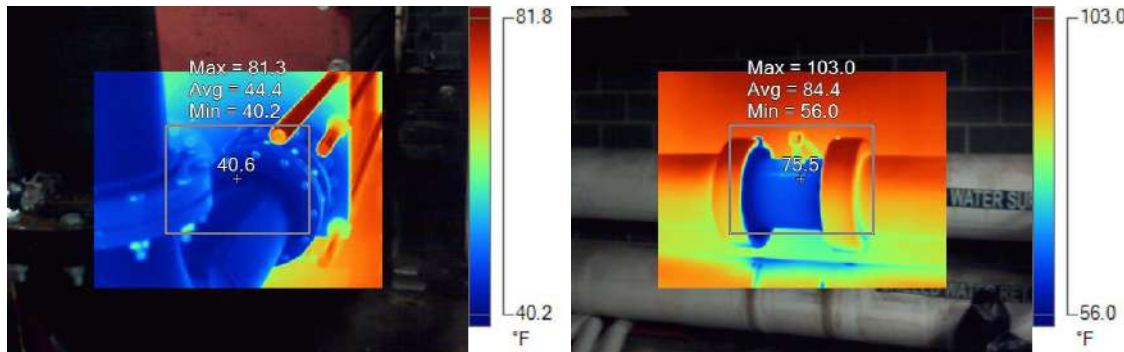


Figure 67: District Cooling Heat Exchanger (Left) and Nameplate (Right)



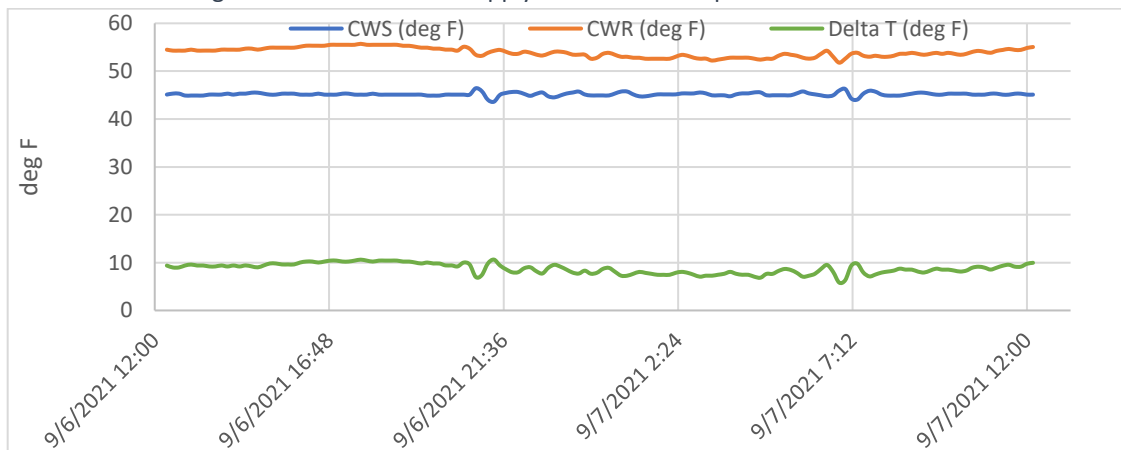
The whole system is thoroughly connected to the BMS and controlled according to the various Delta P and temperature parameters. On the other hand, thermal images showed generally a satisfactory insulation across the chilled water network except some rare cases as can be seen below.

Figure 68: Thermal Images of the Chilled Water Network



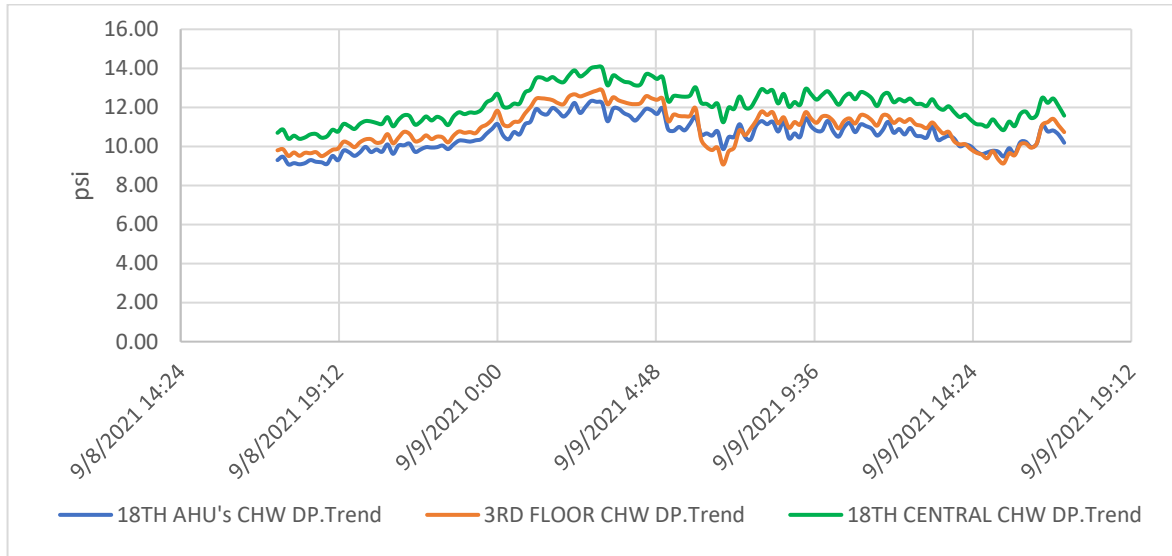
During the field audit, data recording was enacted on key parameters related to the chilled water supply and hereafter are the key charts depicting their variations. The below chart provides the chilled water supply and return temperatures as well as the total difference between the two (Delta T) which varies between 7°F in night time up to 10 °F in peak day time.

Figure 69: Chilled Water Supply and Return Temperature with Delta T



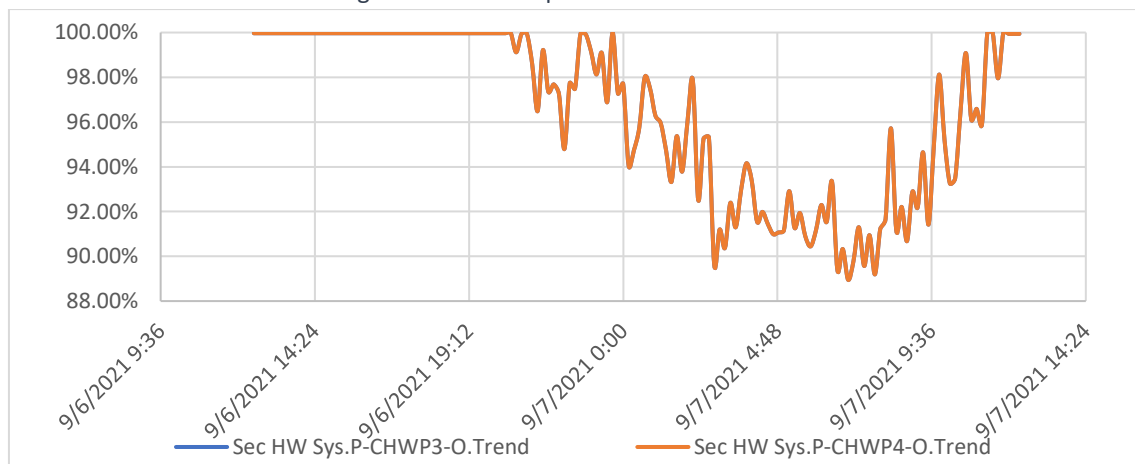
The differential pressure of the chilled water network is also tracked in three different locations in FSB, which are the third floor, the 18th floor, and the 18th floor AHUs. The graph below shows that the DP of the 18th floor is the highest between the three, with all values ranging between 9 psi and 14 psi.

Figure 70: Differential Pressure Tracking



As for the VFD feedback of the secondary chilled water pumps, the graph below shows that the third and fourth chilled water pump have exactly the same VFD feedback during the trend period, which is always between 90% and 100%.

Figure 71: CW Pumps 3 and 4 VFD Feedback



4.3.2. District Heating

Similar to the chilled water supply, hot water is provided to FSB through two heat exchangers with the primary connected to a district heating plant, The contractual agreement includes a maximum capacity of 18.2 mmBTU/hr while the other contractual parameters are as follows:

- Supply Temperature: $280^{\circ}\text{F} \pm 5^{\circ}\text{F}$ to heat exchanger
- Design temperature differential between supply and return 100°F
- Supply pressure: Maximum: 125 psi | Minimum: 65 psi
- Maximum pressure differential between supply and return 10 psid

The produced hot water has three end uses: space heating, domestic hot water production and pool heating.

Figure 72: District Heating Heat Exchangers (Left) and Nameplate (Right)



The calorifiers are distributed in the hotel as follows:

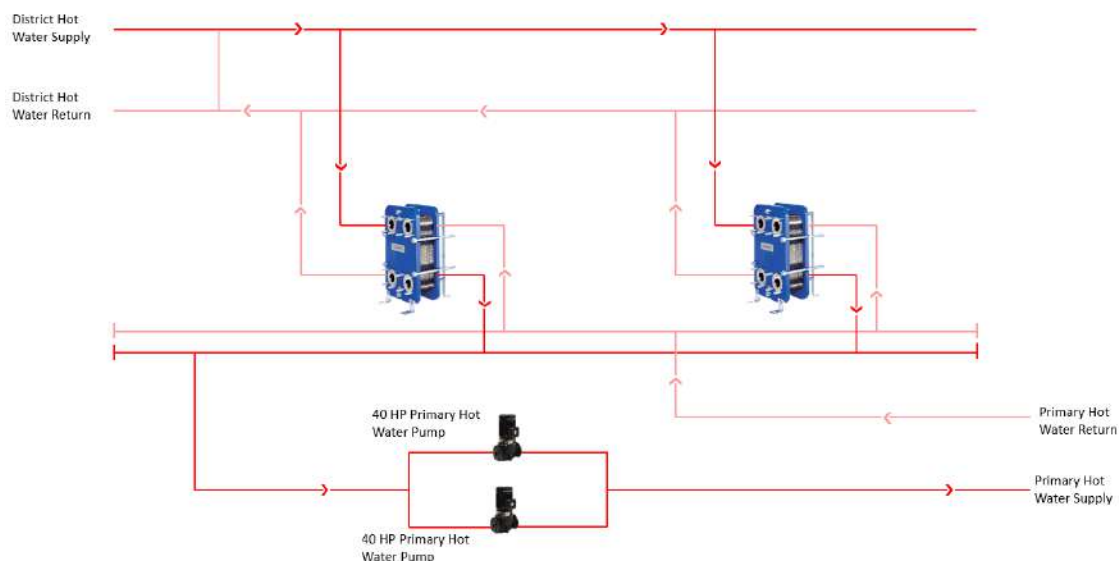
- In B1 district heating and main calorifiers: four calorifiers, two for the guest rooms and two for the kitchens and laundry.
- On the 30th floor: two calorifiers for the residences.

Figure 73: B1 (Left) and 30th Floor (Right) Calorifiers



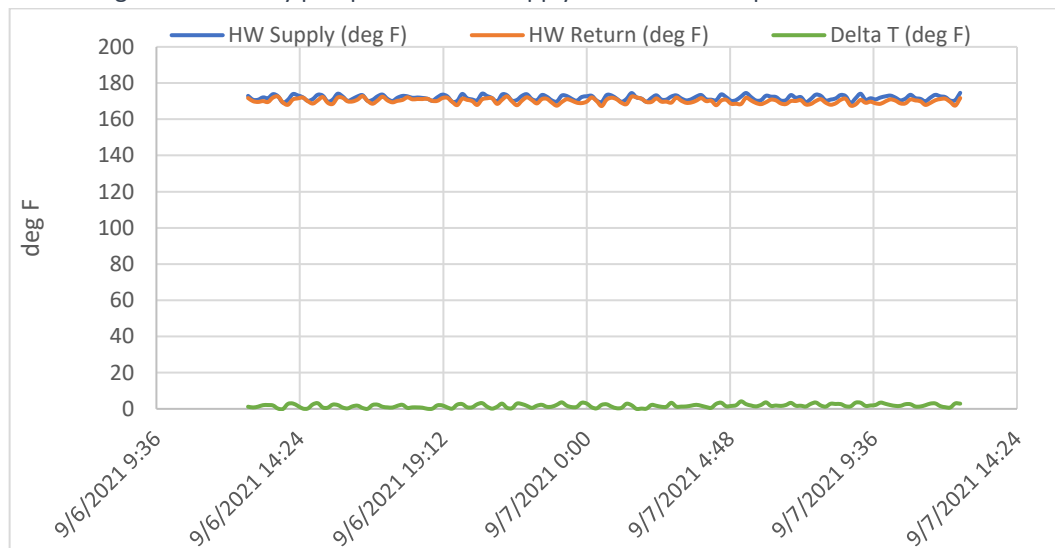
The Hot Water Network is also connected and controlled by the BMS with the below schematic showcasing the Heat Exchangers connections.

Figure 74: Hot Water Network



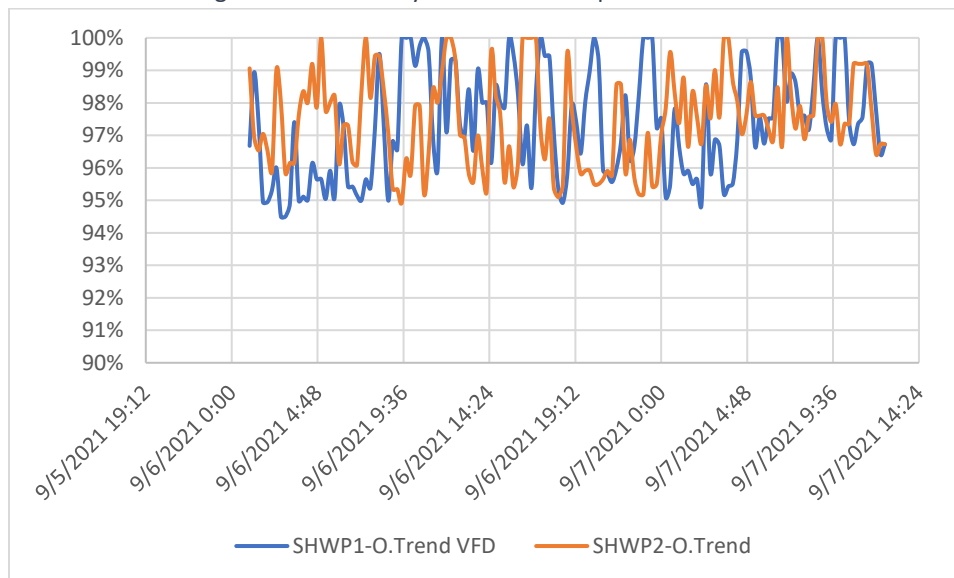
As for the District Cooling, data recording was enacted during the field audit for key hot water system parameters. As it can be seen in the graph below, which represents the supply and return hot water temperatures, Delta T is nearly 0°F with very little variations, which is a clear indication of over pumping during the time of the field audit (September 2021)

Figure 75: Primary pumps Hot Water Supply and Return Temperature with Delta T



The secondary hot water pumps VFD feedback was also be tracked, and as it can be seen in the graph below, the feedback varies between a minimum of 95% and a maximum of 100% for two pumps which is still too high.

Figure 76: Secondary Hot Water Pumps VFD Feedback



In addition, FSB has installed backup steam boilers on the 18th floor in case district heating supply fails, but the system was never used.

Figure 77: Backup Steam Boilers



As mentioned earlier, hot water has three end uses in FSB: Pool Heating, Domestic Hot Water Production, and Space Heating. The table in the following page breaks down the usage of the total hot water demand into these three categories during the baseline year.

As can be seen, domestic hot water production accounts for 60%, followed by space heating at 30% and finally pool heating at 10%. It is also important to note that all these numbers are estimates based on the facility's operation and BMS.

Table 19: Hot Water Consumption Breakdown

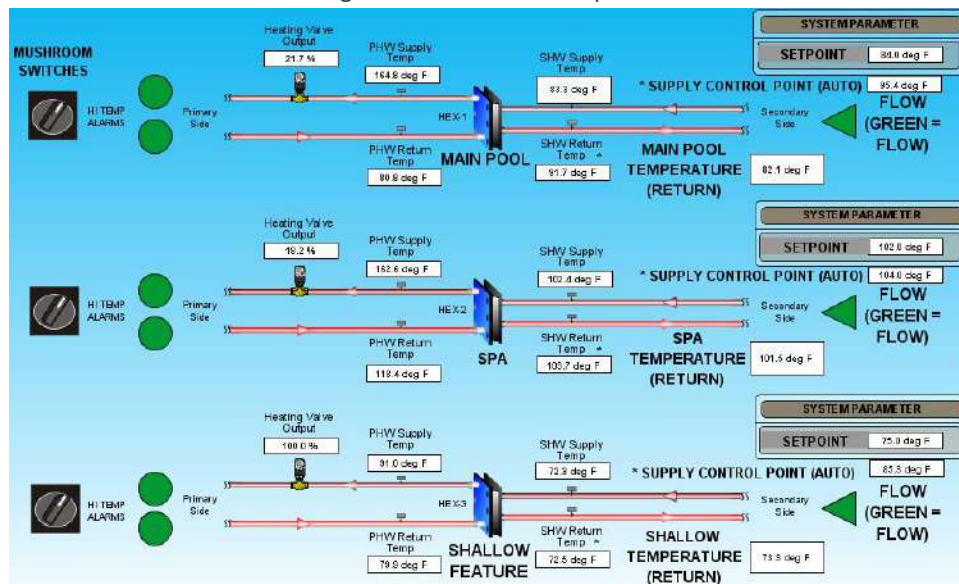
Month	Pool Heating	Domestic Hot Water	Space Heating	Total
Unit	mmBTU	mmBTU	mmBTU	mmBTU
Jan	381	1,397	2,005	3,783
Feb	287	1,209	1,418	2,914
Mar	254	1,339	803	2,396
Apr	184	1,239	108	1,531
May	159	1,222	55	1,436
Jun	123	1,126	0	1,314
Jul	127	1,164	0	1,164
Aug	127	1,164	0	1,190
Sep	154	1,126	0	1,469
Oct	191	1,280	380	1,851
Nov	246	1,295	1,168	2,709
Dec	318	1,397	1,499	3,213
Total	2,550	14,959	7,436	24,970
Maximum	381 (Jan)	1,397 (Jan)	2,005 (Jan)	3,783 (Jan)
Average	212	1,247	620	2,081
Minimum	123 (Jun)	1,126 (Jun)	00 (Jun)	1,164 (Jul)
Percentage	10%	60%	30%	100%

Pool Heating

Pool Heating in the hotel is done through three heat exchangers, one for the main pool, one for the shallow pool and the last one for the hot tub next to it, with all these systems connected to the BMS. The specifications of the three heat exchangers are as follows:

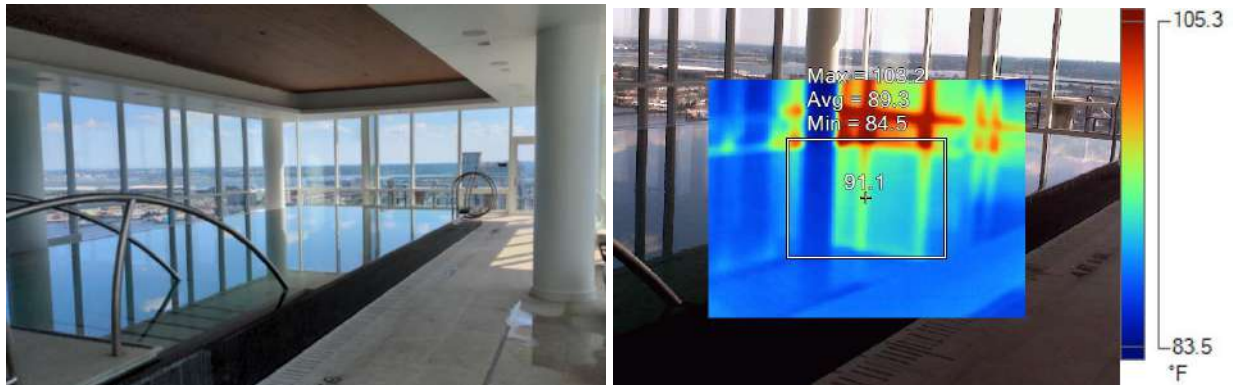
- For the main pool:
 - HOT SIDE: 168.3 GPM of 190 deg F EWT to 160 deg F LWT
 - COLD SIDE: 195.3 GPM of 59 deg F EWT to 84 deg F LWT
- For the hot tub:
 - HOT SIDE: 30.2 GPM of 190 deg F EWT to 160 deg F LWT
 - COLD SIDE: 35 GPM of 59 deg F EWT to 84 deg F LWT
- For the shallow pool:
 - HOT SIDE: 15.4 GPM of 190 deg F EWT to 160 deg F LWT
 - COLD SIDE: 17.9 GPM of 79 deg F EWT to 104 deg F LWT

Figure 78: Pools BMS Snapshot



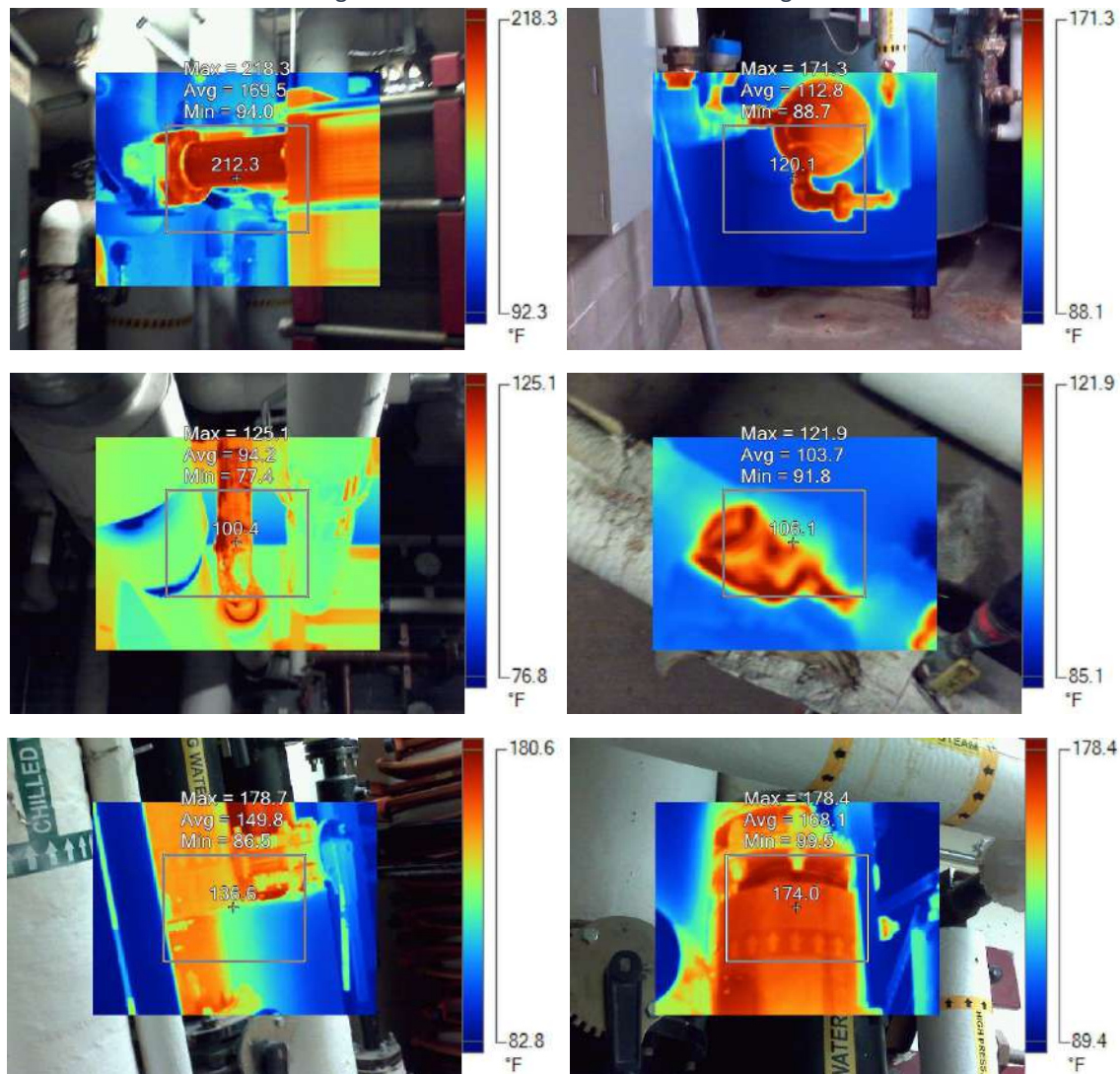
In addition to the above three pools' heating, the system is also used to heat the residence indoor pool all year long at 83°F.

Figure 79: Residences Indoor Pool (Left) and Thermal Image (Right)



The hot water network's insulation has been inspected thoroughly during the field audit and has shown a number of areas with weaknesses including damaged insulation layers or uninsulated parts. These comprise components like Heat Exchangers and valves as well as general piping.

Figure 80: Hot Water Network Thermal Images



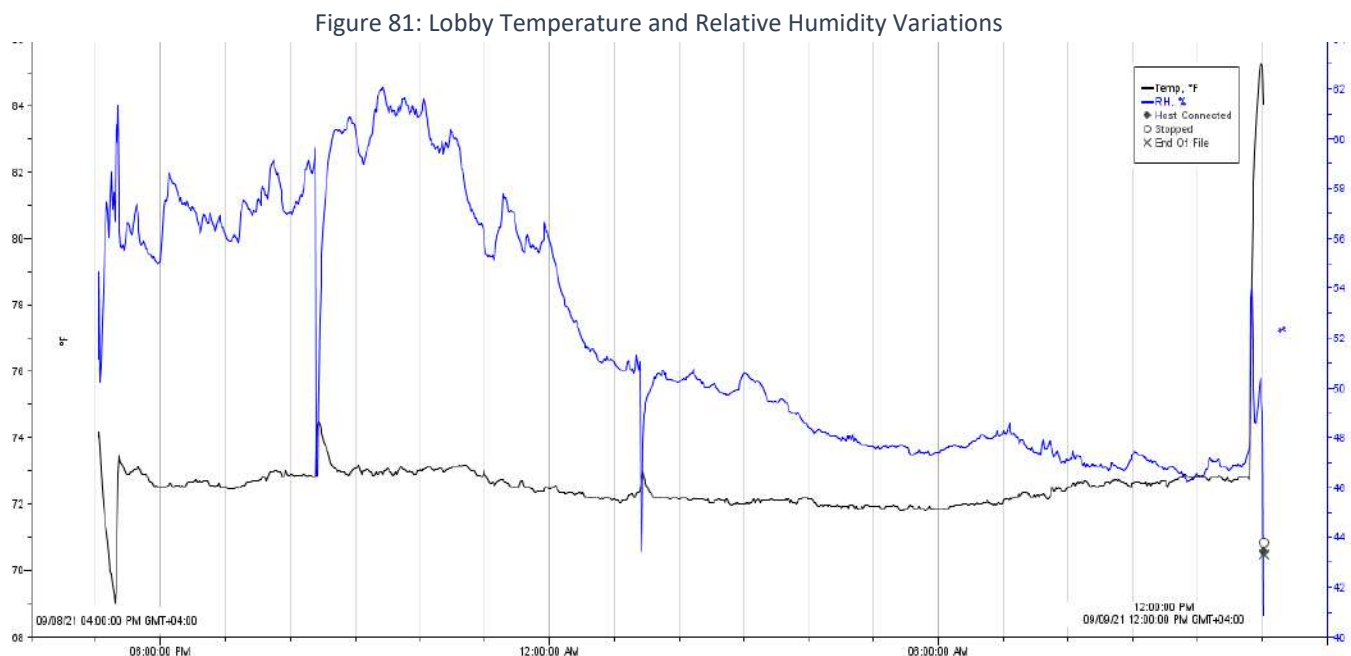
5. Temperature and Relative Humidity Logging

Indoor air quality (IAQ) focuses on airborne contaminants including thermal comfort quality and air quality, covering therefore temperature, humidity, CO₂, CO and VOC (Volatile Organic Compounds) levels.

When it comes to standards, the main guidelines refer to standards promulgated by the American Society of Heating, Refrigerating and Air-conditioning Engineers Inc. (ASHRAE). These are found in the ASHRAE documents Ventilation for Acceptable Indoor Air Quality (ASHRAE 62-2001) and Thermal Environmental Conditions for Human Occupancy (ASHRAE 55-2004).

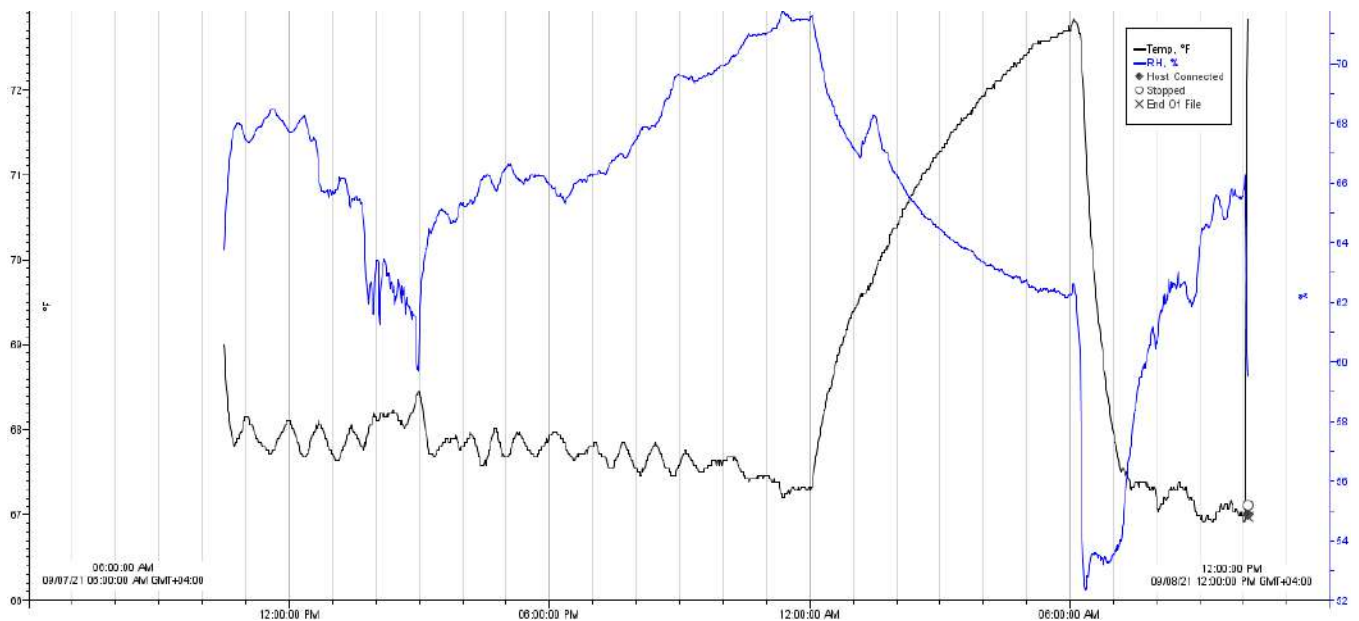
Humidity and Temperature

There is no “ideal” humidity level and temperature suitable for all building occupants. Many factors, such as personal activity and clothing may affect personal comfort. Acceptable relative humidity levels should range from 20 percent – 65 percent year-round. Levels less than 20 percent in the winter and greater than 65 percent in the summer should be considered off the standards. Elevated relative humidity can promote the growth of mold, bacteria, and dust mites, which can aggravate allergies and asthma. To achieve maximum occupant comfort, relative humidity should be maintained between 30% and 65%. The following are graphs, showing temperature and humidity plots in some spaces of FSB. These measurements were taken during the field audit of FSB in September 2021.



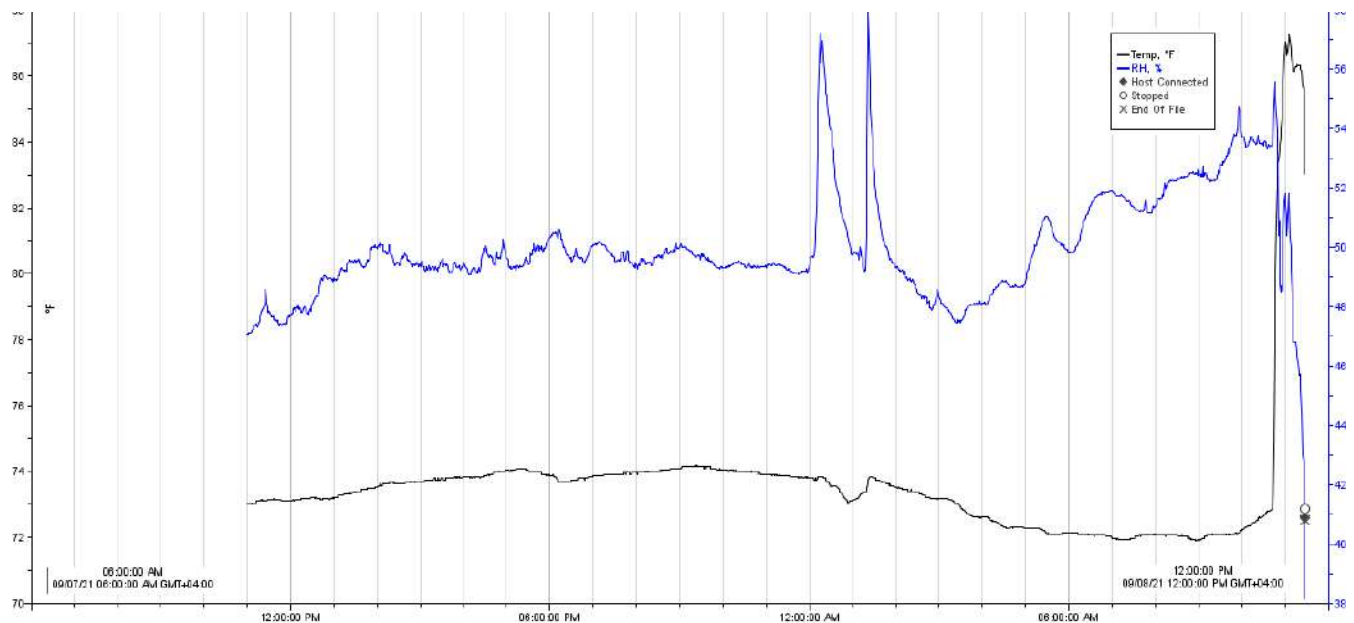
- ➔ The above chart shows that the lobby temperature is quite constant around 73°F which is within the comfort range. Relative Humidity (RH) is seen to drop significantly at night .

Figure 82: Grand Ballroom Temperature and Relative Humidity Variations



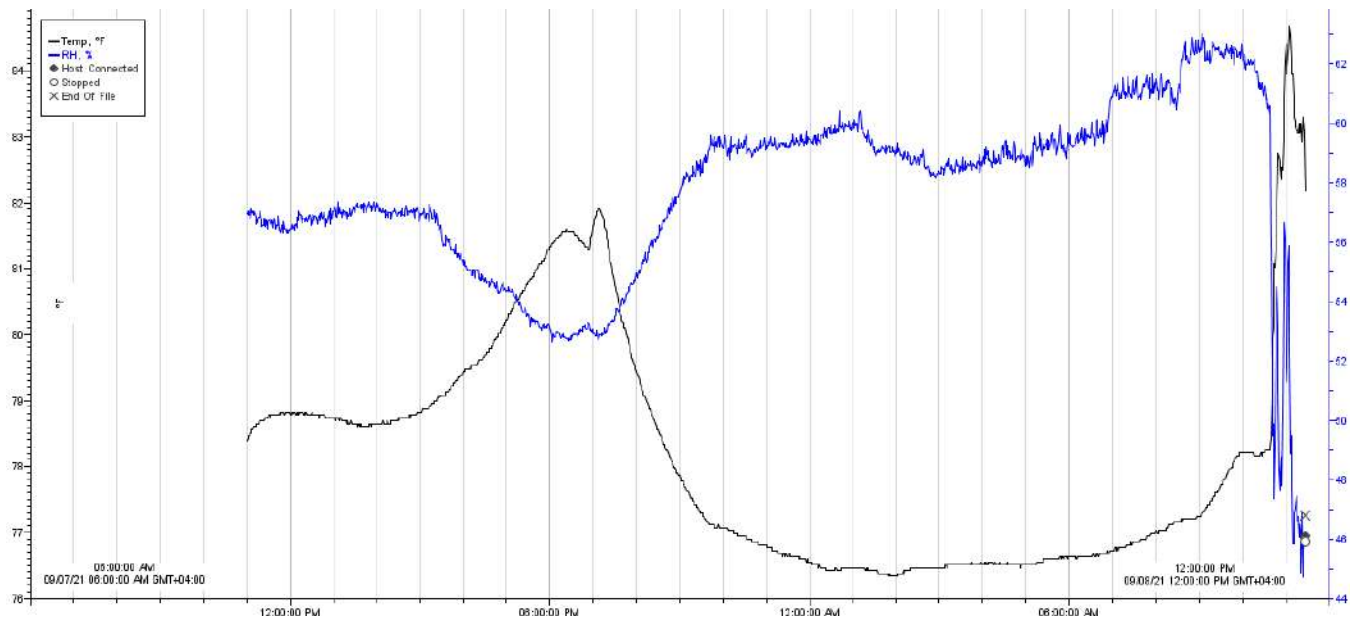
- ➔ The grand ballroom temperature and relative humidity are both within comfort range throughout the whole day. Temperature is seen increasing at 12hAM and dropping after 6hAM in line with the schedule of operation of the related AHU.

Figure 83: SPA Entrance Temperature and Relative Humidity Variations



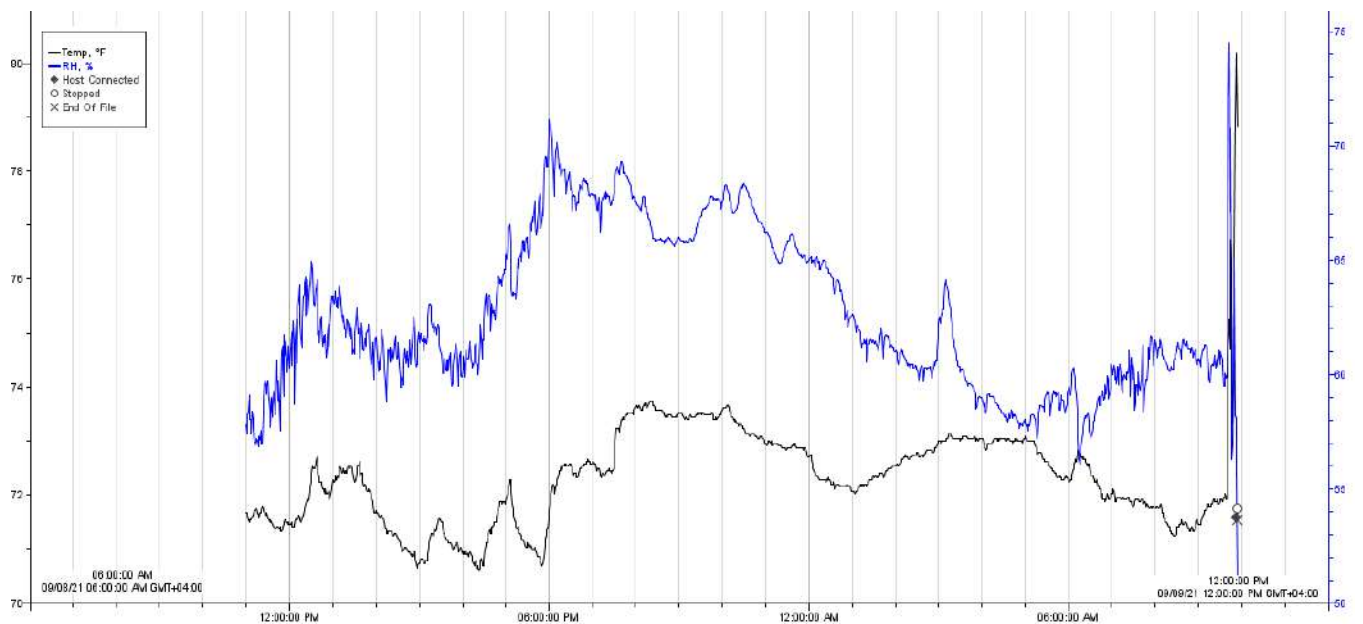
- ➔ The SPA entrance temperature and relative humidity are both within the comfort range throughout the whole day. The chart confirms a 24/7 operation.

Figure 84: 28th Floor indoor Pool Temperature and Relative Humidity Variations



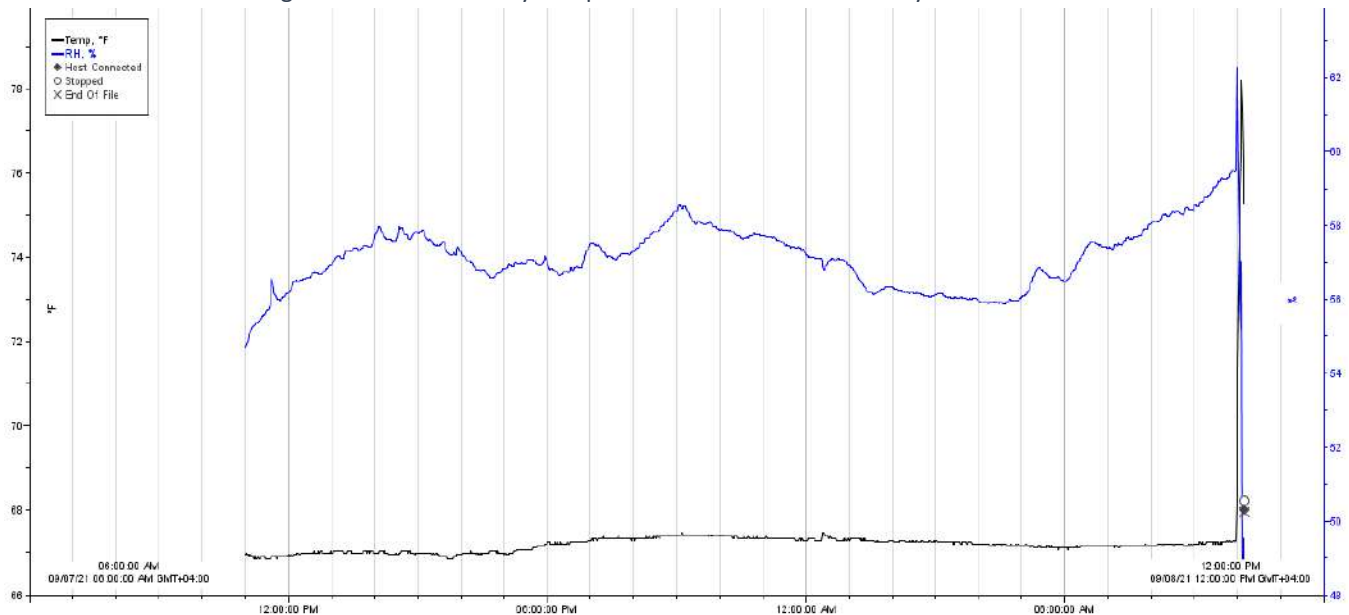
- ➔ The indoor pool temperature shows a quick but short increase during the afternoon as it reaches a maximum of 82°F around 7PM due to the direct sunlight, but goes down to 76.5 °F afterwards. RH is maintained at about 60% levels which showcases a good operation of the AHU serving the area.

Figure 85: Kitchen Corridor Temperature and Relative Humidity Variations



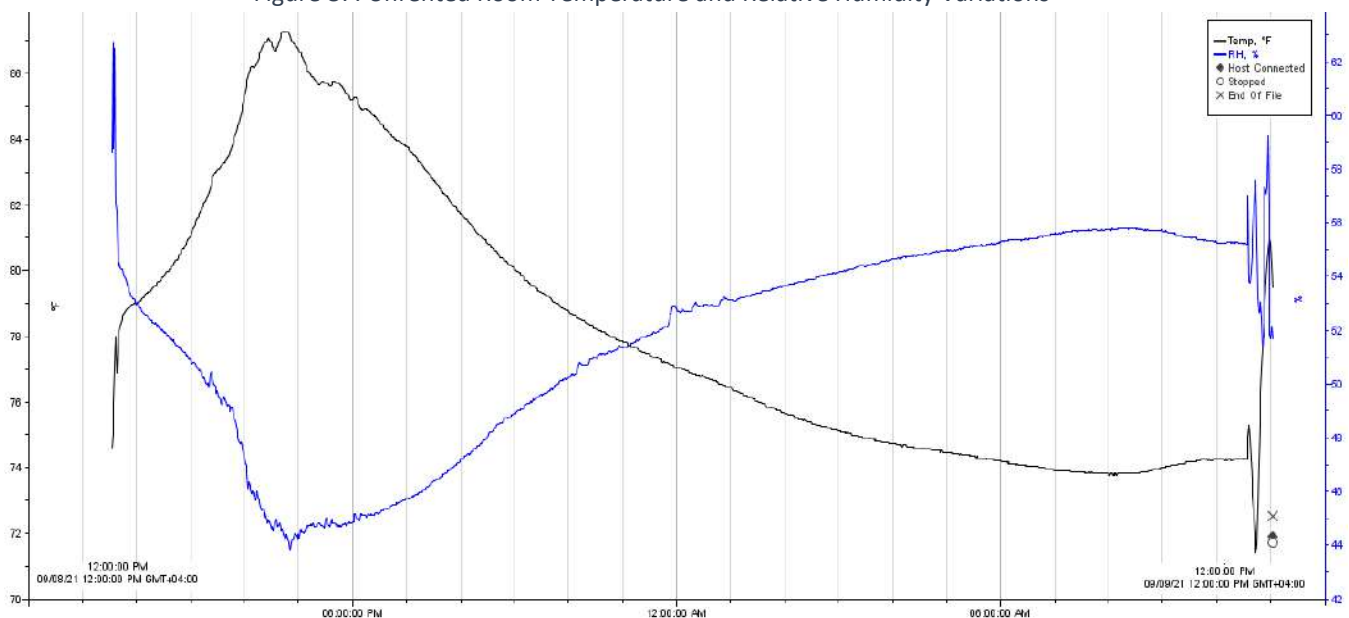
- ➔ The Kitchen Corridor temperature and relative humidity show little variations throughout the day, as the temperature is constant around 72°F and the relative humidity around 65%.

Figure 86: Elevator lobby Temperature and Relative Humidity Variations



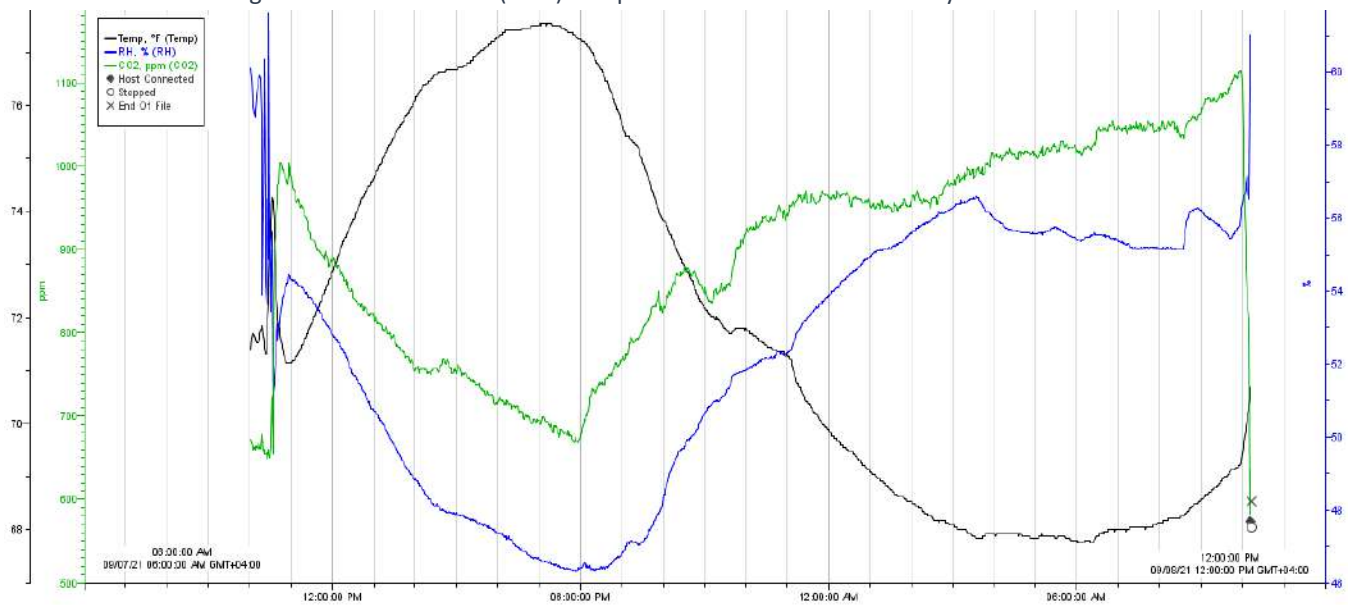
➔ The above chart is for the recordings in an elevator lobby in a guest room floor . Temperature and relative humidity are always with the temperature maintained at about 67 °F that is on the colder side of the settings.

Figure 87: Unrented Room Temperature and Relative Humidity Variations



➔ The unrented room shows a decrease in the temperature in the afternoon, which is caused by the conditioned air delivery to the unrented space since the temperature drops from a high of 88°F (31°C) to around 74°F (23°C) which was seen in the previous graphs of conditioned spaces. RH varied between 48% and 58%.

Figure 88: Rented Room (68°F) Temperature and Relative Humidity Variations



- ➔ The graph above represents the temperature, relative humidity and CO₂ level of a rented room with has a temperature setting of 68°F. As it can be seen, the room temperature takes substantial time to reach the desired temperature due to the high solar heat gains , and then remains constant at it. As for the CO₂ level, it is between 700ppm and 1000ppm, which is within Ashrae Std 62 (the 1000ppm being the acceptable limit)

6. Energy Efficiency Measures

The main objective of this section is to provide a roadmap that would include a number of Energy Efficiency Measure (EEM) which are all the actions that would reduce the utilities consumption. In this aim, EEG investigated a considerable number of energy efficiency solutions, which could yield to substantial energy cost and carbon footprint reductions in the event of their implementation. These opportunities include those who require very little or no CAPEX at all – Low Cost/No Cost – [LC/NC] as well as those needing capital expenditure [CAPEX]. The assessment of an EEM is its financial feasibility and technical viability. The present EEMs are complementing all the efforts done and pushing efficiencies higher. Each EEM is attributed with a color code which is a priority or urgency indicator for the EEM's implementation. The color codes adopted in this report along the proposed measures are found in the below tables:

Table 20: Energy Efficiency Measures Color Codes

Color Code	Description	Priority
	Measures with very small CAPEX and excellent payback (<1 year) – Low Cost/No Cost	A
	Measures with good payback period (<3 years)– Immediate Implementation	A
	Measures with midterm payback period (3 to 6 Years) – Strategic Implementation	B
	Measures to be done during a planned retrofit/replacement	C
	Measures with high payback period (>6 years) – Upon CAPEX Availability	D

Table 21: Energy Efficiency Measures

CODE	TYPE	TITLE
EEM - 01	LCNC	Reduce Cooling and Heating subscription Capacity
EEM - 02	LCNC	Lighting Control
EEM - 03	CAPEX	Motion Detectors Installation
EEM - 04	CAPEX	Lighting Retrofit
EEM - 05	CAPEX	Improve Thermal Insulation of the Hot Water Network
EEM - 06	LCNC	HVAC Control
EEM - 07	LCNC	Hot Water Primary Pumps Operation
EEM - 08	CAPEX	Installation of VFDs on Fans and Pumps
EEM - 09	CAPEX	Pool Cover - Indoor Pool residences
EEM - 10	LCNC	Antifreeze control management for outdoor pool
EEM - 11	LCNC	Building Envelope - Heat gain mitigation for the first 21 floors
EEM - 12	CAPEX	Operation of the Blending Mixing Valve on Chilled Water Line
EEM - 13	CAPEX	Recovery Wheels for FAHUs
EEM - 14	CAPEX	Solar Water Heaters Installation
EEM - 15	CAPEX	Photovoltaic Electricity Generation
EEM - 16	CAPEX	Re-Commissioning and Water balancing across the Chilled and Hot Water Loops

6.1. EEM – 01: [LCNC] Reduce Cooling and Heating subscription Capacity ■

6.1.1. Rationale

As previously mentioned, FSB has signed contracts with the district cooling and heating providers with a maximum capacity of 1150 Tons and 18.2 mmBTU/hour respectively. This EEM studies the possibility of renegotiating both of these contracts to reduce the subscribed capacity, thus reducing the fixed capacity charges.

6.1.2. District Cooling

The table below provides the actual monthly FSB chilled water consumption and the calculated average demand which was then used to estimate the peak demand based on a 40% increase from the average value.

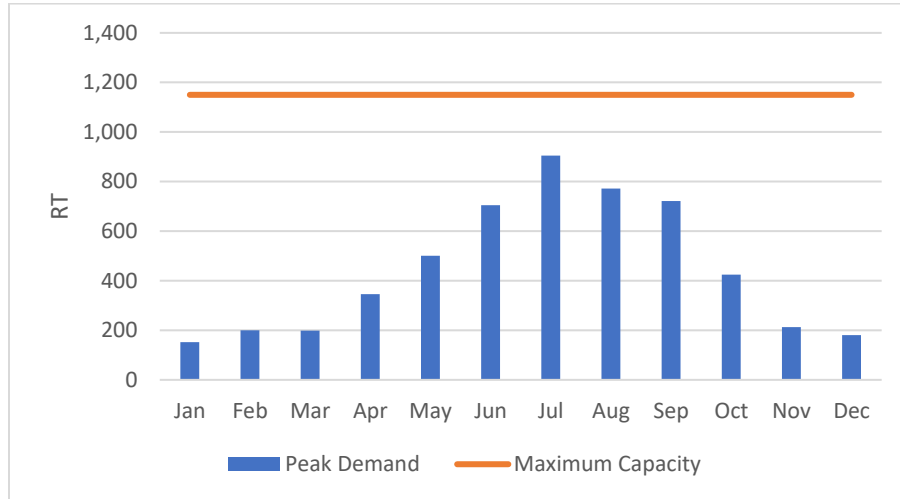
Figure 89: Chilled Water Demand Breakdown

Month	Consumption	Average Demand	Peak Demand
Unit	RTH	RT	RT
Jan	81,051	109	153
Feb	95,726	142	199
Mar	105,389	142	198
Apr	177,688	247	346
May	265,772	357	500
Jun	362,480	503	705
Jul	481,039	647	905
Aug	410,018	551	772
Sep	371,221	516	722
Oct	225,342	303	424
Nov	109,514	152	213
Dec	96,148	129	181

As can be seen the highest maximum demand reached was 905RT in July which is still 234 RT lower than the subscribed cooling capacity of 1150RT.

This means that the district cooling contract can be easily renegotiated to reduce the maximum capacity from the actual 1150RT to around 1000RT, which would reduce the monthly capacity fixed charges.

Figure 90: Peak Demand vs. Maximum Capacity



6.1.3. District Heating

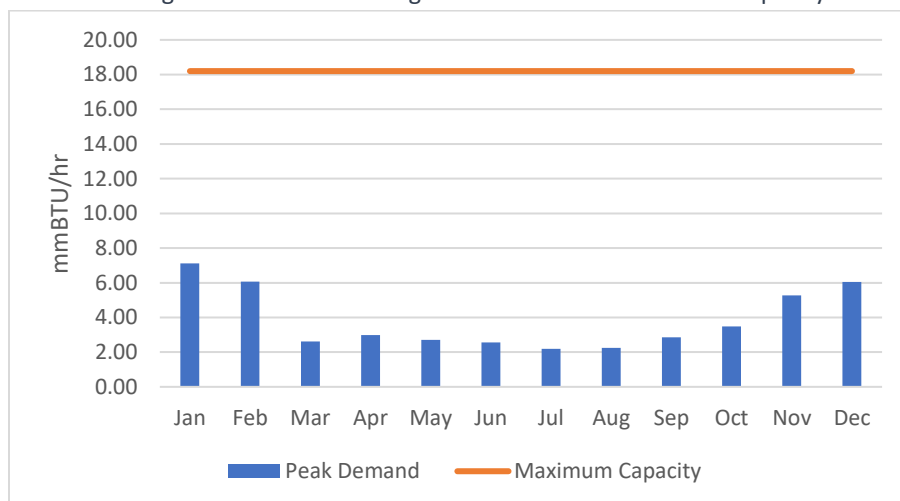
In a similar way to the district cooling, the table below breaks down the actual FSB hot water consumption into an average demand per month and the peak demand, which was estimated to be 40% higher than the average demand.

Figure 91: Hot Water Demand Breakdown

Month	Consumption	Average Demand	Peak Demand
Unit	mmBTU	mmBTU/hr	mmBTU/hr
Jan	3,783	5.08	7.12
Feb	2,914	4.34	6.07
Mar	1,393	1.87	2.62
Apr	1,531	2.13	2.98
May	1,436	1.93	2.70
Jun	1,314	1.83	2.56
Jul	1,164	1.56	2.19
Aug	1,190	1.60	2.24
Sep	1,469	2.04	2.86
Oct	1,851	2.49	3.48
Nov	2,709	3.76	5.27
Dec	3,213	4.32	6.05

Comparing the estimated peak demand to the contract's subscribed heating capacity in the graph below shows that the peak demand is more than 50% lower as the highest heating demand was found to be in January with 7.12mmBTU/hr while the subscribed demand is 18.2mmBTU/hr.

Figure 92: District Cooling Peak Demand vs. Maximum Capacity



Accordingly, it is recommended to renegotiate the district heating contract if possible with a new subscribed power 10mmBTU/hr, which would reduce the monthly capacity fixed charges.

6.1.4. Energy Savings

Clearly, this EEM depends on the flexibility to renegotiate the contract upon any new terms. Typically, as the connections are now more than 10 years old, the district energy suppliers should allow such a reduction as their investment has been surely depreciated over this time. By reducing the maximum capacity of district cooling to 1000 RT and district heating to 10 mmBTU/hr, the following savings can be achieved.

EEM - 01		
Review Cooling and Heating Capacity Charges in Contracts		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT COOLING	CO ₂
KWh	RTH	Kg
0	0	0
0.0%	0.0%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$146,260
Annual Share of Cost Savings [%]		8.7%
Implementation Cost [\$]		\$0

6.2. EEM – 02: [LCNC] Lighting Control ■

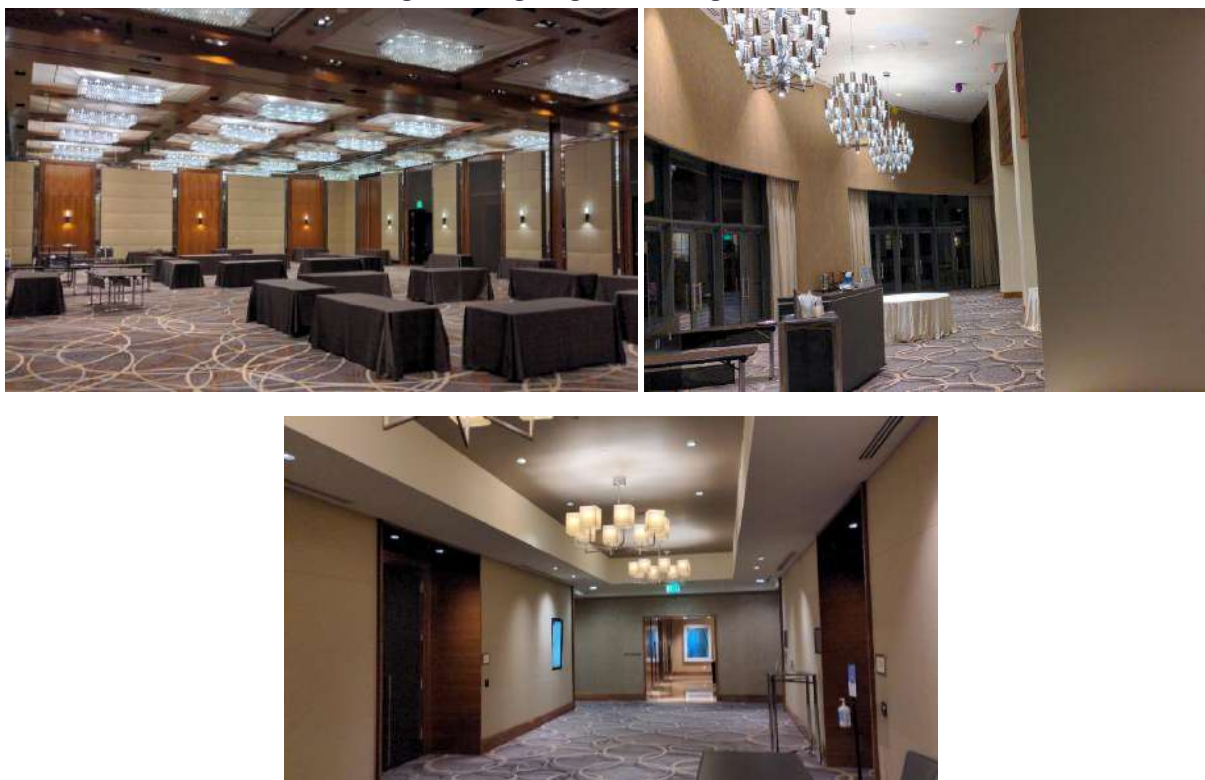
6.2.1. Rationale

FSB's lighting energy consumption represents 16% of the total energy consumption. Therefore, any substantial reduction on the lighting loads' energy consumptions would lead to a positive impact on the total facility's energy and power demands. This measure highlights the opportunities available for lighting consumption reduction through an improved control – manually or through the BMS.

6.2.2. Turn off all the lighting in functions rooms during vacancies (no events)

During nighttime, lighting fixtures in meeting rooms and ballrooms were seen to be mostly on while there were no events or preparation for events. A more stringent control of all these lighting – mostly by the banqueting/F&B department would lead to substantial electricity savings across the year. Below pictures are examples of these cases, all taken at 11h30PM.

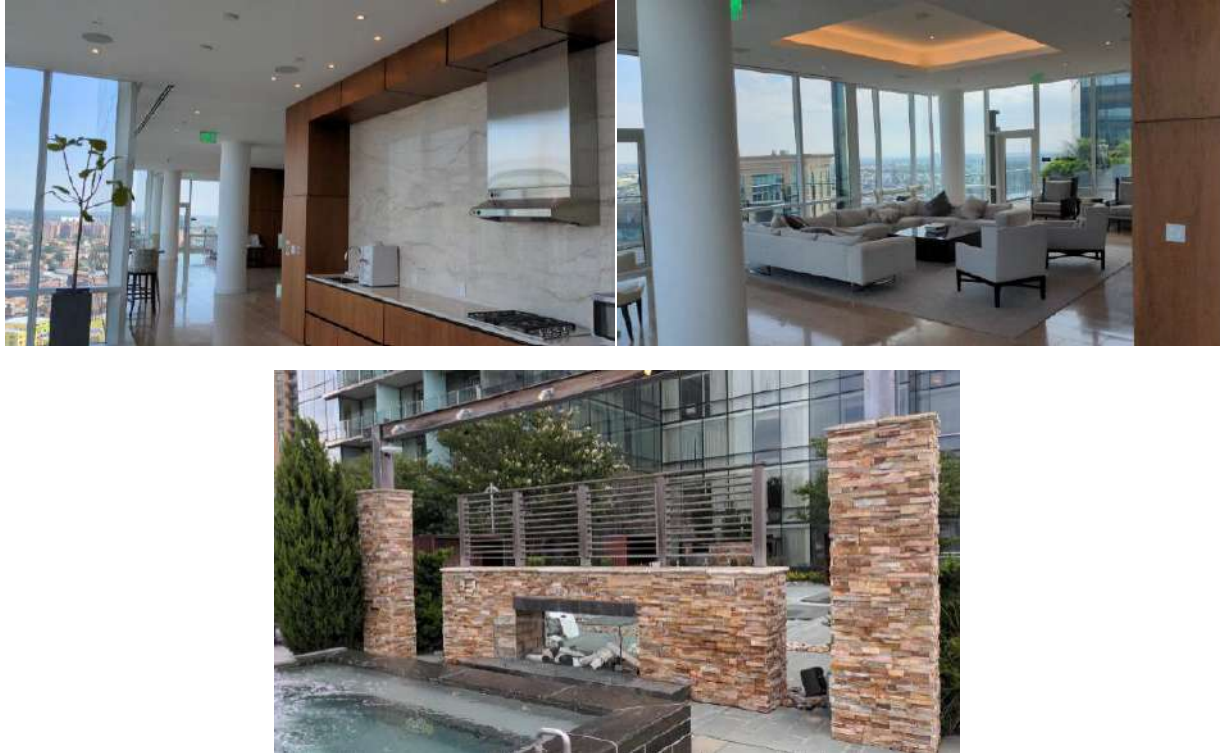
Figure 93: Lighting Control Targeted Areas



6.2.3. Turn off lighting fixtures during daylight

Many lighting fixtures were seen to be kept on during the day in areas where daylight is ample to meet and exceed all lighting standards; this was mostly seen in the residences lounges which has an all glazed façade. There were also many outdoor fixtures kept on during daytime. The pictures below show these areas and the lighting fixtures which were on during the day, with the efficiency action being to simply turn them off during the day.

Figure 94: Areas where Lighting Fixtures were kept on during the day



Combining the electricity savings of both actions would lead to the below key indicators.

EEM - 02		
Lighting Control		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT COOLING	CO ₂
KWh	RTH	Kg
10,950	0	4,069
0.3%	0.0%	0.2%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$975
Annual Share of Cost Savings [%]		0.1%
Implementation Cost [\$]		\$0

6.3. EEM – 03: [CAPEX] Motion Detectors Installation ■

6.3.1. Rationale

Lighting Control's concept is based on introducing new lighting control technologies in the existing lighting system. It provides a major opportunity for any facility to reduce its overall lighting energy consumption. Lighting control examples are: Time control, Daylight control and Occupancy control.

The suggested lighting control for FSB is the installation of motion detectors in all BOH's technical rooms (>5 lighting fixtures), storage rooms and housekeeping rooms/lobby in every guest room floors. All these areas were seen to have frequently lights on with full vacancy. Examples are found in the below images.

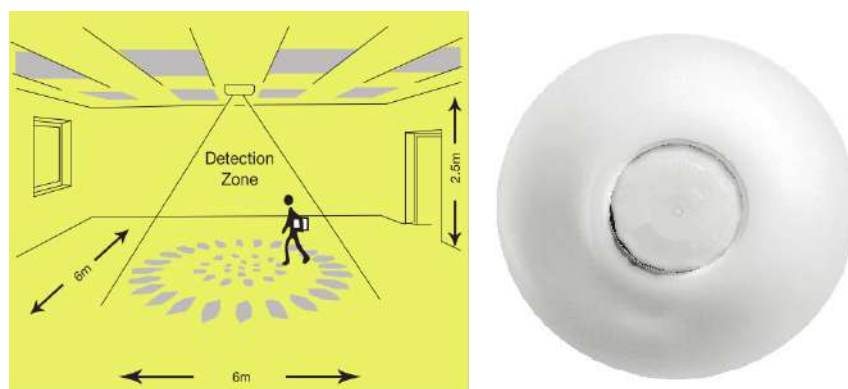
Figure 95: Motion detectors Targeted areas



6.3.2. Efficiency Action and Energy Savings

Motion sensors will shut the controlled lights OFF when the space is vacant and will put them ON when they sense a human presence. Typically motion sensors can provide more than 20% energy savings in areas that are partly unoccupied during the day. Their installation is rather easy and usually, the objective is to control 70% to 85% of the total lighting fixtures of the controlled area. You will find here below an example of coverage of the 360° sensor that is the model recommended.

Figure 96: Motion Detector 360° Coverage



A total of 60 possible areas have been considered but a total review of targeted rooms can be done and the implementation expanded. You will find in the table below the key indicators of this EEM.

EEM - 03		
Motion Detectors Installation		
ENERGY SAVINGS		
ELECTRICITY	CHILLED WATER	CO ₂
KWh	RTH	Kg
27,094	0	10,068
0.6%	0.0%	0.4%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$2,411
Annual Share of Cost Savings [%]		0.1%
Implementation Cost [\$]		\$3,000
Simple Payback [Years]		1.2
Net Present Value [\$]		\$16,080
Internal Rate of Return [%]		80%

6.4. EEM – 04: [CAPEX] Lighting Retrofit ■

6.4.1. Rationale

FSB has completed a through LED retrofit to the tower back in 2013 while the residences floors have all been built with LED lighting.

However, there are two areas where LED retrofit is still possible: The lighting wall behind the front desk in the lobby and the CFL recessed lighting as well as the CFL round fixtures found in the meeting rooms and pre function areas in the 2nd floor.

6.4.2. Efficiency Action and Energy Savings

The actions to be pursued are as follows:

- Replace the lighting wall with LED lights on a one to one basis: Currently, the wall is composed of about 50 T8 fluorescent tube of 58W each , they can be replaced by similar T8 LED tubes of 22W, 3000K , The key challenge is the removal and installation of the decorative wall . As these lighting are operating 24/7 every day. Their replacement will have a substantial savings .
- Replace the recessed CFL fixtures/bulbs in some meeting rooms by LED strips dimmable 3000K , compatible with existing dimmer.
- Replace the round CFL fixtures in the pre function areas by equivalent diameter and angle LED fixtures 2700K, dimmable.



Figure 97: CFL recessed light (Left) | CFL round fixtures (Right)



Completing these lighting retrofit actions would ensure that FSB has a fully LED lighting and the EEM key indicators are found in the below table:

EEM - 04		
Lighting Retrofit		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
30,594	0	11,369
0.7%	0.0%	0.4%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$2,723
Annual Share of Cost Savings [%]		0.2%
Implementation Cost [\$]		\$4,000
Simple Payback [Years]		1.5
Net Present Value [\$]		\$17,545
Internal Rate of Return [%]		68%

6.5. EEM – 05: [CAPEX] Improve Thermal Insulation of the Hot Water Network

6.5.1. Rationale

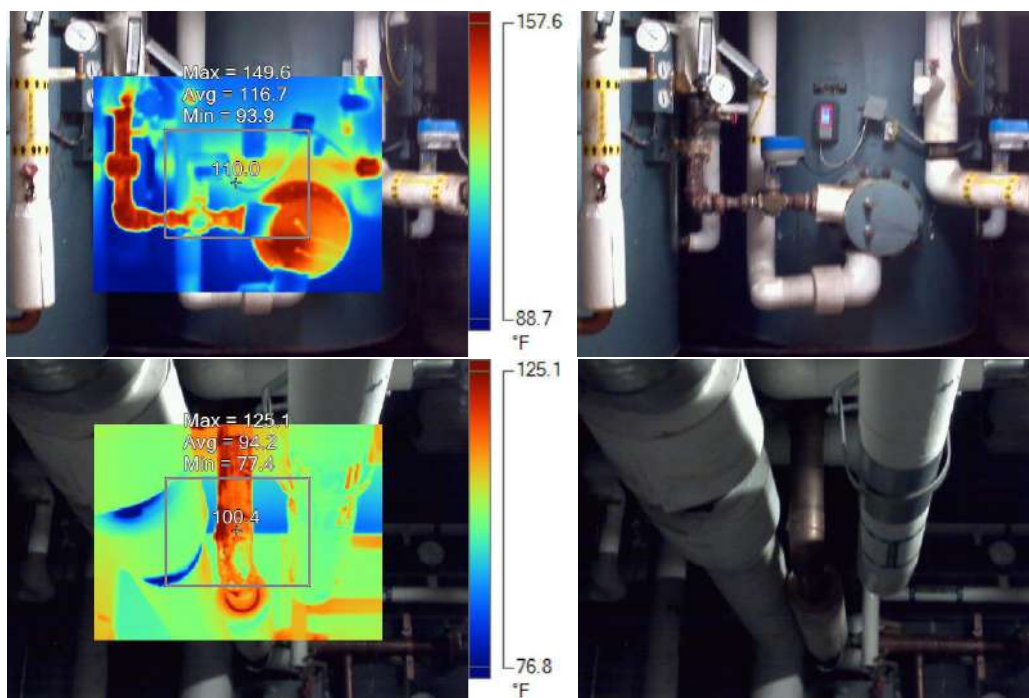
Poor insulation of the network and components of a thermal energy system is a cause of energy waste that can be easily avoided. As seen in the present report, the chilled water network was generally satisfactory in terms of pipes and components insulation – except very rare cases; however, the extensive thermal imaging done on the hot water infrastructure network showed large areas of insulation weaknesses that need to be mitigated.

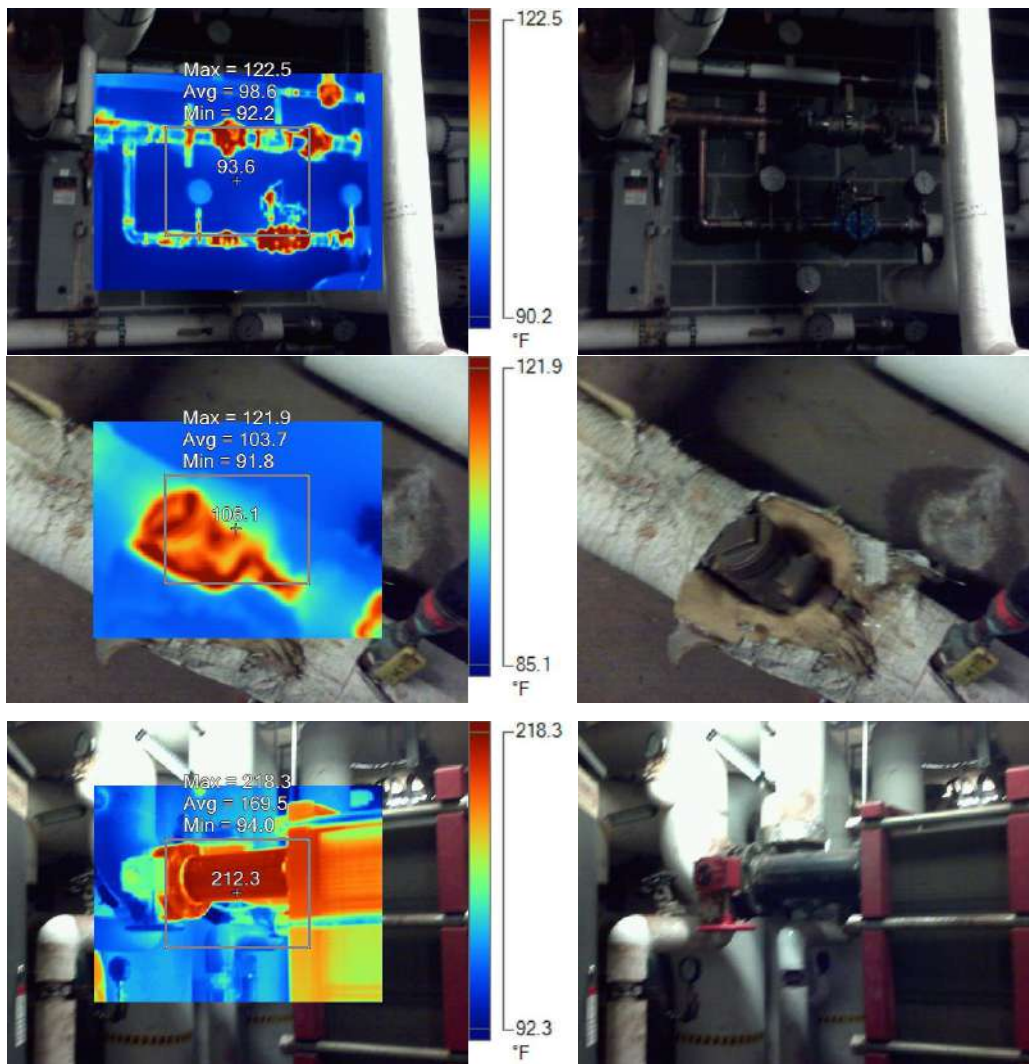
6.5.2. Efficiency Action and Energy Savings

The below thermal images and their visible ones provide some indications on the areas to be tackled. It is recommended to review all hot water pipes from the main intake from the heat exchangers to all visible distribution branches and connections to AHUs. Insulation improvement can be done only at the weaknesses points as per the below areas in red in the thermal images.

This could be a phased action done over a period of time whereby a comprehensive review through thermal imaging can be done and poorly insulated parts improved.

Figure 98: Thermal Images of Hot Water Pipes System





The picture on the right shows the key connections at the level of the two calorifiers serving the residences and which have large parts without any insulation



Pipes insulation is the easiest energy efficiency measure that could be applied to any thermal energy system's distribution. In addition, the insulation cost (rock wool, fiberglass...) is rather quite low.

It is recommended to have a review of all the thermal insulation for the hot water pipes in line with what was mentioned before and install properly sized insulation of fiberglass or thermobreak (foam insulation). NAIMA (North American Insulation Manufacturers Association) can provide very good guidance on the insulation type and sizes. (Ref: <https://insulationinstitute.org/im-a-building-or-facility-professional/piping/pipe-insulation-codes-standards/>)

Note that thermal insulation should also be considered on all heat exchangers (the main district heating ones and all pools' heating ones) and valves.

The below table provides the Key indicators of this EEM.

EEM - 05		
Improve Thermal Insulation of the Hot Water Network		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
0	375	21,376
0.0%	1.5%	0.7%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$3,119
Annual Share of Cost Savings [%]		0.2%
Implementation Cost [\$]		\$10,000
Simple Payback [Years]		3.2
Net Present Value [\$]		\$14,679
Internal Rate of Return [%]		31%

6.6. EEM – 06: [LCNC] HVAC Control

6.6.1. Rationale

HVAC Fans are the heart of the space cooling and heating in a building. In FSB they were seen to represent the largest electricity end use consumer. In fact, with 30 AHUs (Hotel + Residences) , a large number of extractor fans and all the FCUs, FSB has a myriad of ventilation loads to manage.

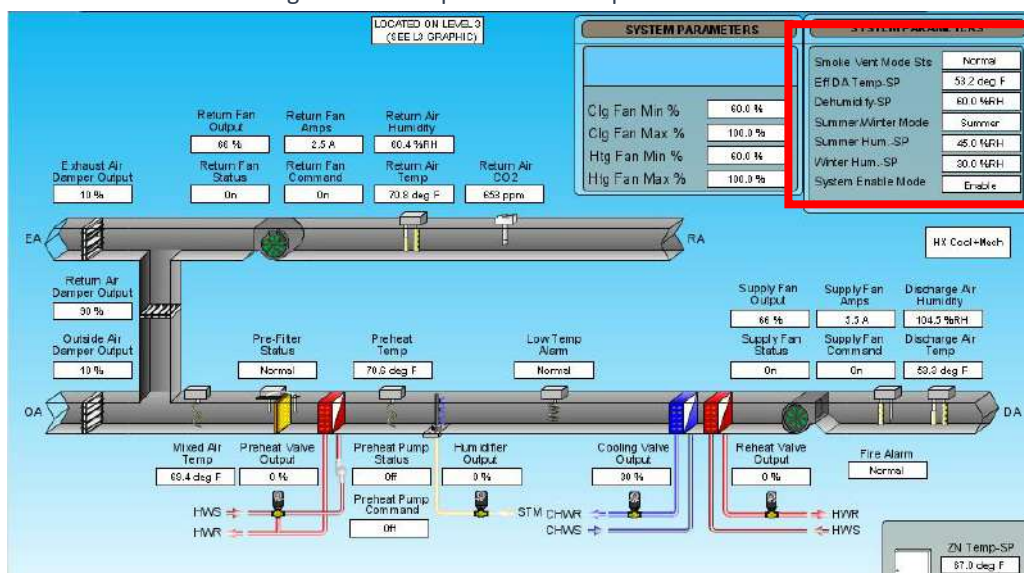
A proactive and aggressive control of all these loads will allow an optimization of not just the electricity consumption – due to the reduction of the fans loads- but also the district cooling and heating. In fact, managing HVAC fans will have a direct impact on the reduction of the thermal energy used is pursued. The following sections cover various strategies that were seen possible in FSB.

6.6.2. Reset setpoints

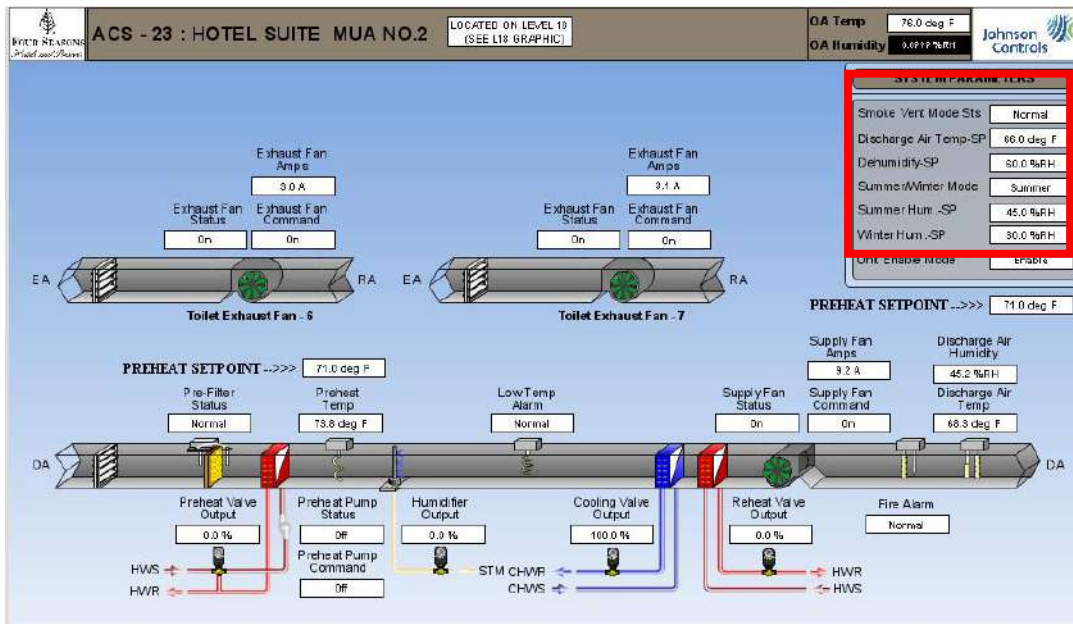
Many AHUs in the facility can have the temperature set point increased to a level temperature that keeps guests and staff comfort, as well as decreasing the AHU's energy consumption by operating it more efficiently. Obviously, setpoints can be more aggressively set in off-peak seasons. Below are two examples that can be generalized across all AHUs. The main point is to be proactive and continuously adapt to conditions without maintaining a constant value across the board. .

The snapshot below is for the AHU 10, which feeds the banquet facility, and it shows a temperature set point of 53.2°F which can be increased since the banquet facility isn't always occupied.

Figure 99: Example of AHU Temperature reset



Similar to AHU 10, AHU 23 can have its setpoint raised easily from 66°F to 70°F which will reduce the cooling demand.



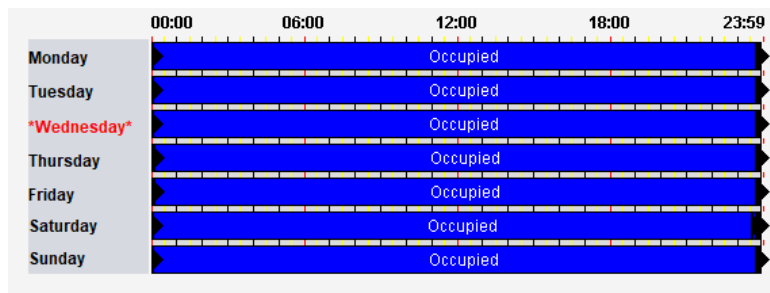
Virtually all AHUs can be controlled in a similar way whereby the temperature setpoint can be readjusted proactively on a daily basis according to the occupancy, ambient conditions and events.

6.6.3. Aggressive Scheduling control

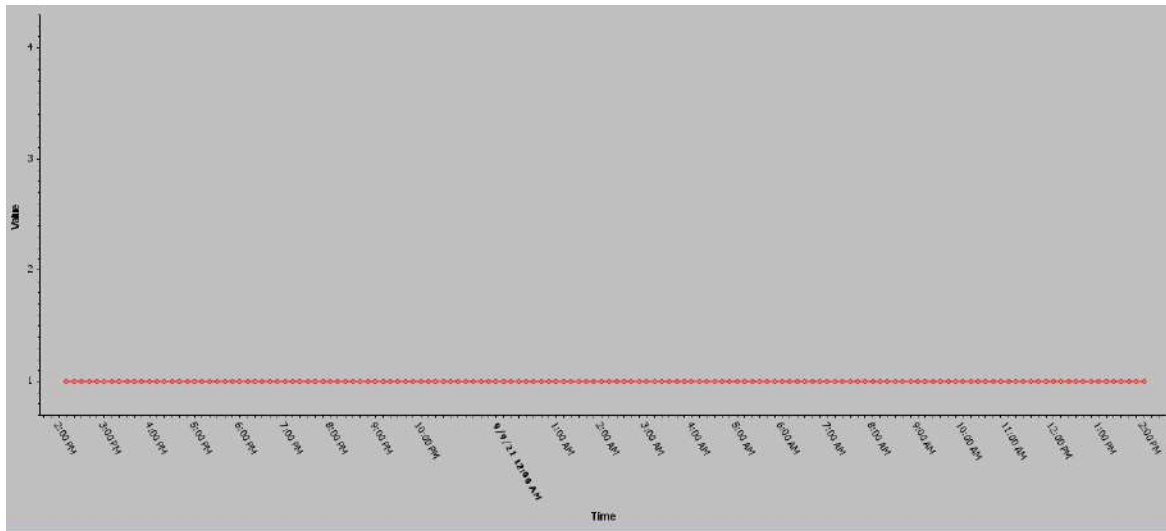
As discussed in section 4.1.2, each AHU connected to the BMS has a certain schedule, based on the need and the zone fed by the AHU. Some AHUs should be operating 24/7 such as the ones feeding the lobby and the BOH, but other units can have changes to their schedules

For example, the AHUs serving the GF restaurants have their AHUs running 24/7 while they are on the energy loads of the hotel (i.e: their electricity consumption is not charged back). They all can be shut down at least 3 to 5 hours every day. BMS screenshot below represents the schedule of AHU 1, feeding the Azumi Restaurant.

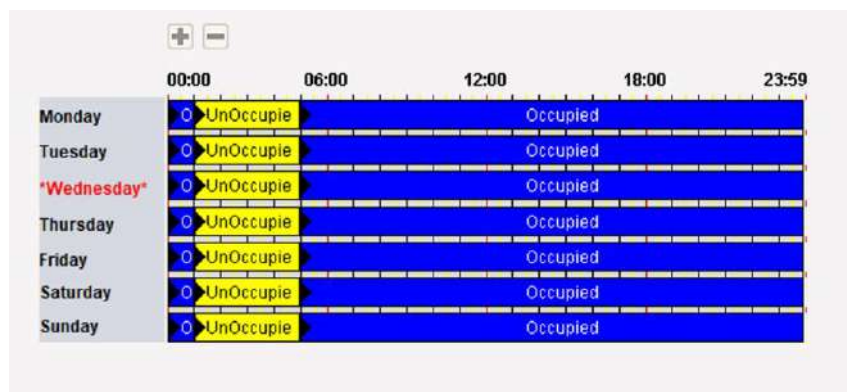
Figure 100: AHU 1 Schedule



The below chart showcases the VFD feedback of the supply fan of AHU 18 serving the presidential suite and operating 24/7 while the suite was vacant. Both AHU 18 and AHU 19 can be stopped at least 50% of the time when the suites are vacant.



Another example are all AHUs serving the banquet areas and meeting rooms (2nd and 4th floors). They have currently a scheduling off between 12hAM/1hAM until 5hAM/6hAM. However, they can easily be shut down one to two hours before until one to two hours after depending on the events and utilization. The below example is for the unit serving the meeting rooms of the 4th floor.



The key mitigation actions are the following:

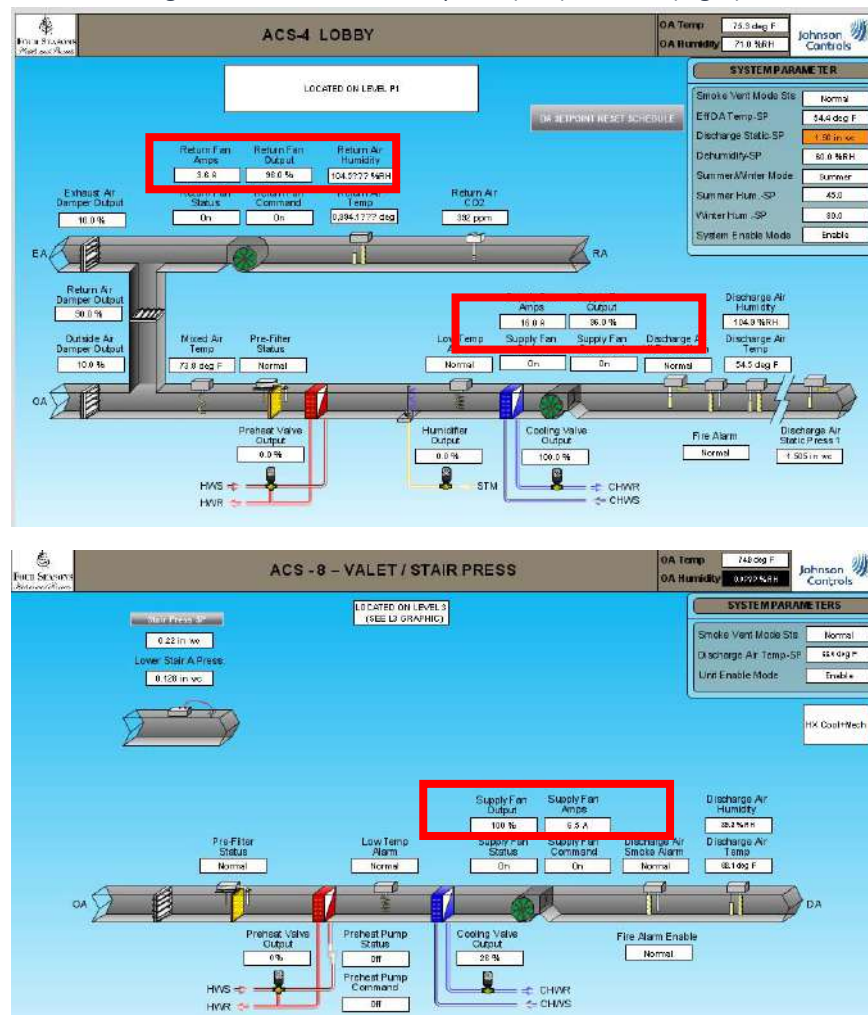
- i. Review the operation of each AHU as well as the area it services.
- ii. Implement an aggressive scheduling that should be proactively modified every day in some units, every week in others.

6.6.4. Proactive Control of VFDs

FSB has VFDs installed on a large number of its AHUs population. However, during the field audit, it was noticed that while some are varying correctly, others seemed to have a fixed speed continuously. Knowing that a 10% speed reduction would lead to savings up to 27% from the original consumption, while a reduction of 20% leads to savings up to 49%, the optimization of improvement is quite large.

The below pictures provide examples of AHUs' VFDs with a constant speed.

Figure 101: AHU 4 BMS Snapshot (Left) and VFD (Right)



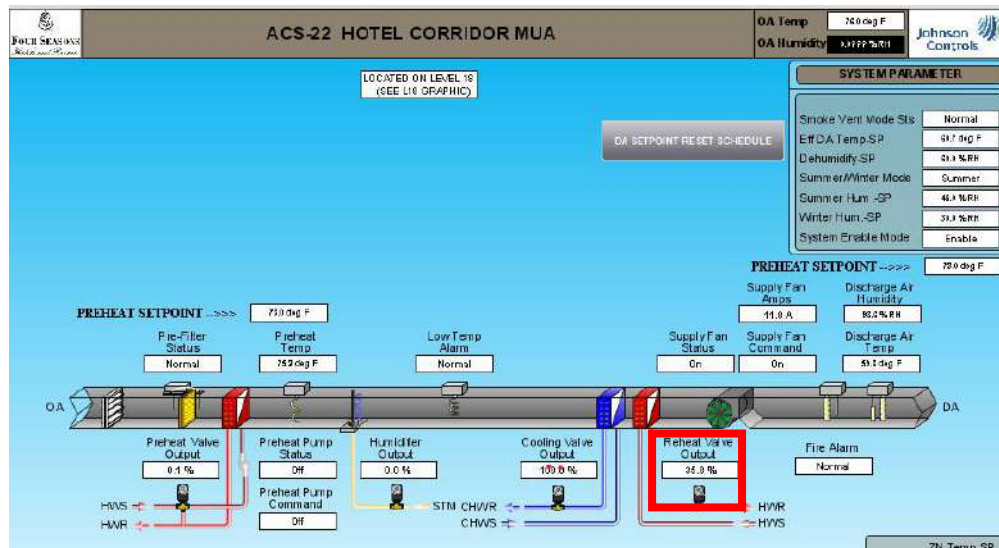
The key mitigation actions are the following:

- Review all the PID loops of the VFD of VAV AHUs ensuring an aggressive programming. Then test the impact of the pressure on the VFD feedback.
- Alternatively, implement a scheduled based VFD operation with a lower speed in off peak seasons or specially at night time like for the above cases AHU 4 and AHU 8

6.6.5. Remove unnecessary reheat

Reheating coils are used for dehumidification, but when the ambient RH is below 60%, there is no needed to have any dehumidification in the AHU process.

Figure 102: AHU 22 with reheat coil open

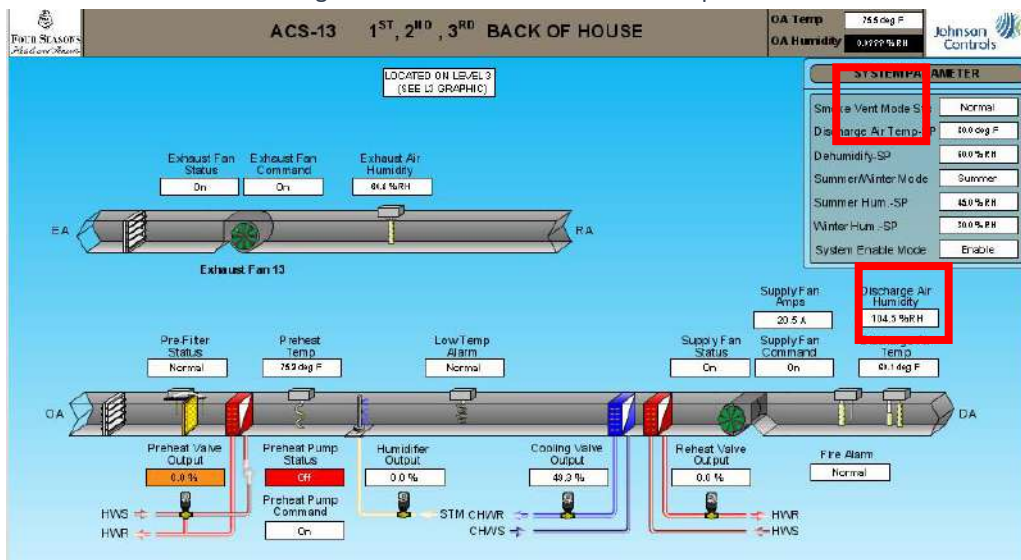


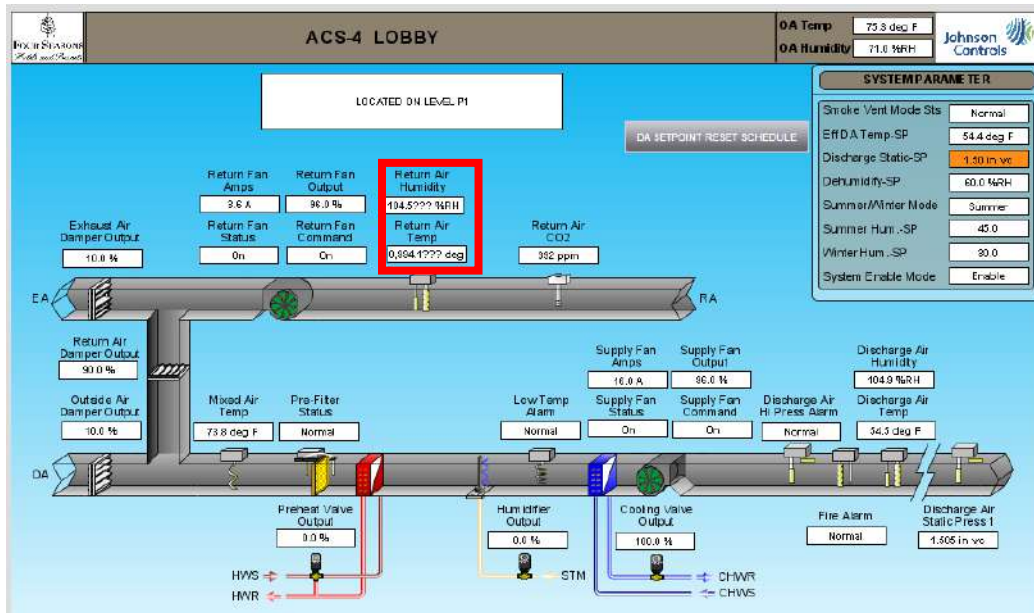
6.6.6. Re-commission/calibrate sensors

Sensors are the heart of a BMS. It is crucial to ensure all faulty sensors are repaired or replaced while a program of calibration to be planned over a timeline to ensure gradually that all sensors are well operating with accuracy.

The below are two examples with highlighted sensors errors.

Figure 103: AHU 22 with reheat coil open





6.6.7. Kitchen Hoods and Extractor Fans

The operation of kitchen hoods and dishwasher fans is also very important in the path of HVAC Fans optimization. It was mentioned that kitchen hoods in the outsourced restaurants have their hoods operating 24/7 while it was seen during the field audit that dishwasher fans were sometimes operating while there were no dishwashing.

In line with the previous sections, it is recommended to review all kitchen hoods operation and then align with each kitchen department for a reduced operation – even with those outsourced restaurants and then implement the adequate scheduling on the BMS.

For dishwashers, ideally, the solution would be to connect the operation of the dishwasher extraction to the one of the dishwasher itself. Only when the latter is effectively operating, the extractor would work.

Figure 104: Dishwasher (Left) and Kitchen Hood (Right)



6.6.8. GRMS programing during unrented room

Temperatures settings of guest rooms' thermostat during unrented scenario can be readjusted higher in the summer (79°F) and lower in winter (65°F) pushing the FCUs operation lower as well as the related thermal energy.

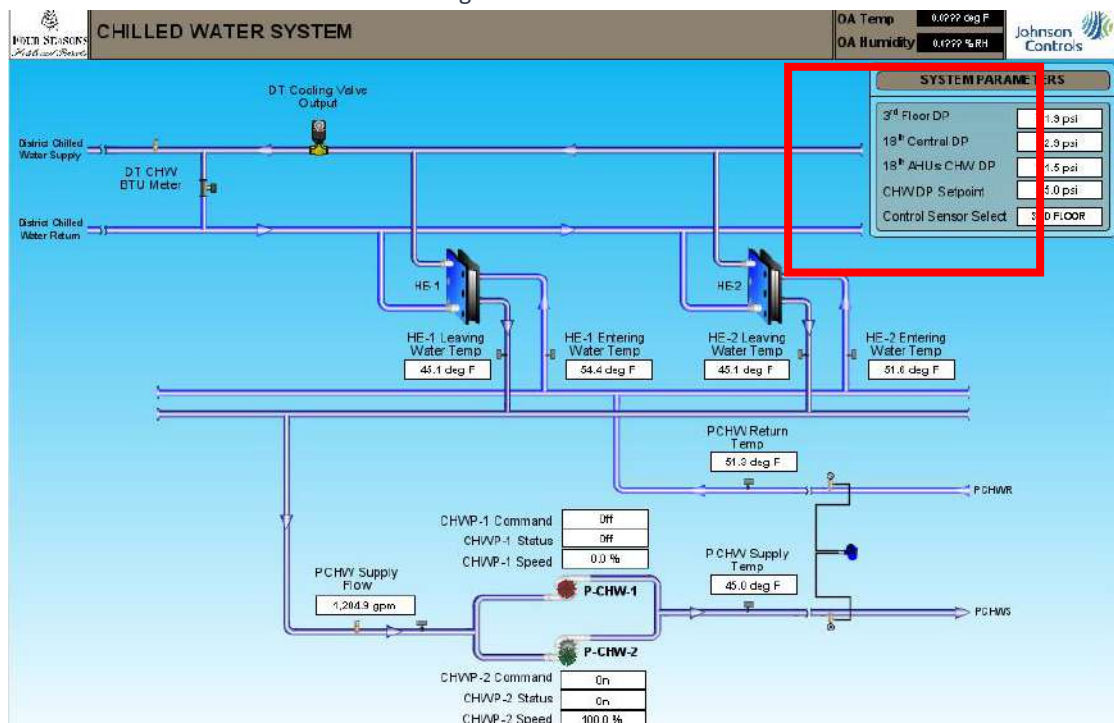
6.6.9. Proactive Delta P settings at secondary chilled water pumps

The present EEM covers the proactive setpoint change for the Delta P (Pressure Difference). During the field audit it has been confirmed from BMS that the Delta P (DP) setpoints across all different chilled water loops were kept constant.

The importance of the Delta P is at two levels:

- I. It enhances the operation of the secondary pumps further by reducing the chilled water flow demand.
- II. It enhances the Delta Temperature as an over pumping will reduce Delta T and will increase the capacity/consumption.

Figure 105: Chilled Water Delta P



The recommendation is to reduce the chilled water Delta P as we move away from the peak hot season . This can be a 10% reduction per month (moving from September to October, then to November ...) and then increasing it back as you move out of winter and back to summer.

6.6.10. Energy Savings

Applying the various efficiency actions discussed above would yield to savings on both the electricity consumption of AHUs and fans as well as the district cooling's chilled water energy and the district heating's hot water energy.

The below table provides the estimated total savings on both utilities.

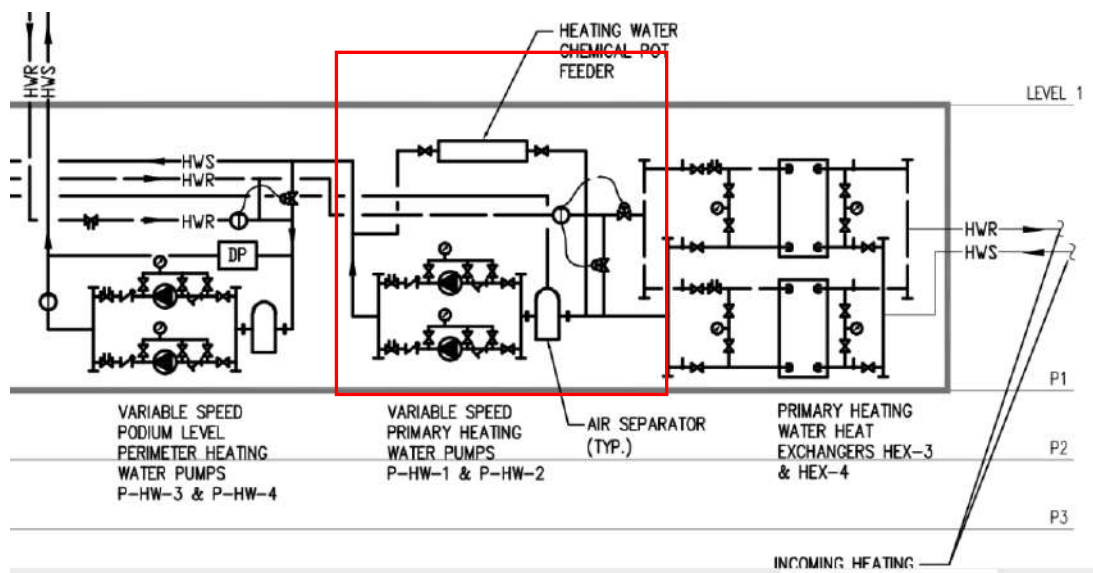
EEM - 06			
HVAC Control			
ENERGY SAVINGS			
ELECTRICITY	DISTRICT COOLING	DISTRICT HEATING	CO ₂
KWh	RTH	mmBTU	Kg
242,368	144,097	744	174,289
5.6%	5.2%	3.0%	4.5%
ECONOMIC PERFORMANCE			
Annual Cost Savings [\$]			\$39,171
Annual Share of Cost Savings [%]			2.3%
Implementation Cost [\$]			\$0
Simple Payback [Years]			-
Net Present Value [\$]			\$309,950
Internal Rate of Return [%]			-

6.7. EEM – 07: [LCNC] Primary Hot Water Pumps Operation

6.7.1. Rationale

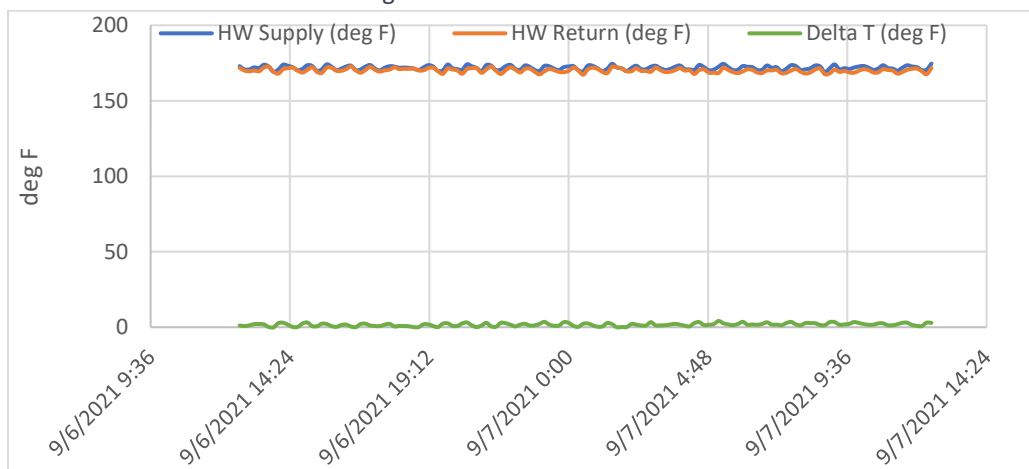
During the field audit in September 2021, when the ambient temperature was still topping the 82°F, the two district heating primary pumps were seen to be running at 100% speed on a 24/7 basis.

Figure 106: Primary Hot Water Pumps



It appears that this is a problem recurrent since the beginning of the operation of the property . The result is an excessive electricity consumption with 2x30HP Pumps running at full speed 24/7 while also the poor Delta T as seen in the below chart reduces the efficiency of the while District Heating system

Figure 107: Hot Water Delta T



6.7.2. Efficiency Action and Energy Savings

The Engineering Department engaged a company to review the situation in 2018 and a thorough analysis was done with the main outcome being that the problem of having the VFD on the primary pumps not operating – and eventually having both pumps running at full speed is that the hot water loop is serving the preheating coils in a large number of AHUs . As a matter of fact, these preheating coils are all equipped with a 3 way valve and have a total of 689gpm of pumping requirements .

These findings are validated herein and the 3 way valves don't allow a proper flow variation and accordingly a speed control. The ultimate solution would be to replace them by 2 way valves and then re-commission the VFDs and control of the primary hot water pumps. An alternative intermediate solution would be to manually close the bypass line in the 3 way valves and effectively turning them into 2 way valves. This could be achievable and will carry a very little cost.

In all cases, even if the above is not done, it is recommended to reduce the speed of the two pumps manually by up to 20% gradually as we move out of the cold season and into the intermediate and hot seasons, Accordingly, the months of June to September will have both pumps running at 80% speed maximum. This can be implemented immediately and on a 'trial and error ' basis to test the limits of the system.

Reducing the operation of both primary hot water pumps would provide substantial savings as they represent a net 60HP of running load. The implementation of a gradual reduction of speed in milder and hotter months with a 5% reduction in winter months would lead to the below savings.

EEM - 07		
Hot Water Primary Pumps Operation		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
204,651	1,249	147,300
4.7%	5.0%	4.9%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$28,610
Annual Share of Cost Savings [%]		1.7%
Implementation Cost [\$]		\$0
Simple Payback [Years]		-
Net Present Value [\$]		\$226,385
Internal Rate of Return [%]		-

6.8. EEM – 08: [CAPEX] Installation of VFDs on Fans and Pumps ■

6.8.1. Rationale

FSB has already VFDs installed on a large population of AHUs and pumps. The usage of VFDs will allow important energy savings to be achieved as a 10% reduction of speed will lead to a 27% reduction of the KW drawn by the motor. The targeted loads in this EEM are the remaining AHUs (not currently equipped by VFDs) and the pool pumps (except the one of the main pool which has a VFD)

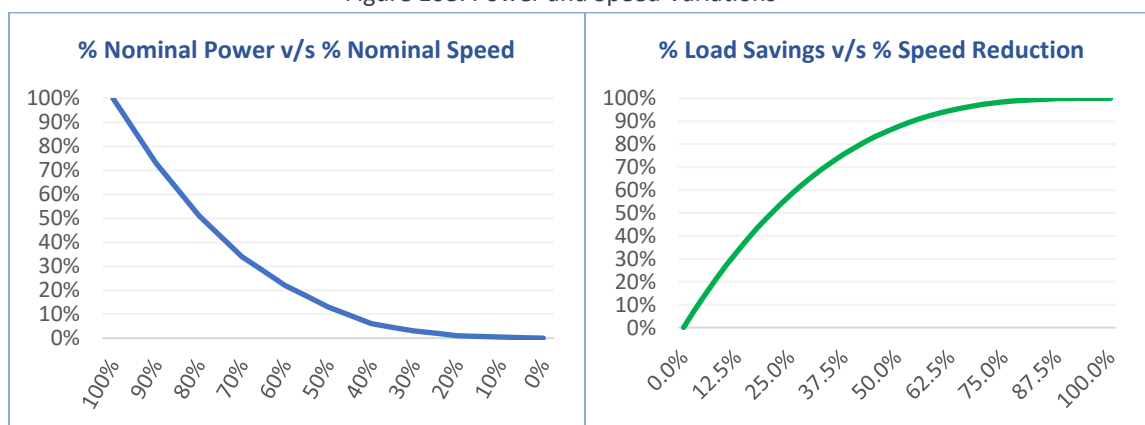
6.8.2. VFDs on AHUs

The present EEM is to expand the VFD installation which will allow implementing the proactive speed control bringing direct savings on the Pumps & Fans electricity consumption as well as the chilled water demand.

Obviously, the usage of VFD will allow substantial energy savings (if the speed reduction is well implemented in a PID/Automatic look or proactively changed manually or via a schedule time setting). A 10% speed reduction would lead to a 27% savings on the Power used.

VFDs on AHUs can be implemented on a time-based operation while they could be operated similarly in on CAV AHUs or alternatively on a PID Loop.

Figure 108: Power and Speed Variations



Based on an average reduction of 20% of the speed of controlled AHUs (i.e.: Frequency to drop from 60Hz to 48 Hz on the average per day), the estimated energy savings can be calculated as per the following table:

Table 22: VFDs on AHUs

Asset	Existing Annual Energy Consumption	Overall Average Speed Reduction	Load/Energy Reduction	Savings	Cost Savings
Unit	KWh	%	%	KWh	\$
AHU 3	39,867	20.0%	49%	19,455	\$1,732
AHU 9	29,236	20.0%	49%	14,267	\$1,270
AHU 17	6,379	20.0%	49%	3,113	\$277
AHU 20	39,867	20.0%	49%	19,455	\$1,732
AHU 21	61,130	20.0%	49%	29,831	\$2,655
AHU 23	61,130	20.0%	49%	29,831	\$2,655
AHU 24	19,934	20.0%	49%	9,728	\$866
Total	257,542	-	-	125,680	\$11,186

6.8.3. VFDs on Pumps

The same procedure can also be applied on the pool pumps, with the savings being calculated in the table below:

Asset	Existing Annual Energy Consumption	Overall Average Speed Reduction	Load/Energy Reduction	Savings	Cost Savings
Unit	KWh	%	%	KWh	\$
Circulation Pump - Residences Pool	7,005	20.0%	49%	3,418	\$304
Pool Pumps	7,659	20.0%	49%	3,738	\$333
Total	14,664	-	-	7,156	\$637

6.8.4. Energy Savings

The below table provides the key indicators of this EEM

EEM - 08		
Installation of VFDs on Fans and Pumps		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
132,836	0	49,362
3.1%	0.0%	1.6%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$11,822
Annual Share of Cost Savings [%]		0.7%
Implementation Cost [\$]		\$28,000
Simple Payback [Years]		2.4
Net Present Value [\$]		\$65,548
Internal Rate of Return [%]		42%

6.9. EEM – 09: [CAPEX] Pool Cover for residences' indoor pool ■

6.9.1. Rationale

FSB residences have an indoor heated pool on the 28th floor and that has also a rectangular form. Generally swimming pools lose energy in a variety of ways, but evaporation is by far the largest source of water and energy losses. On the other hand, being in an indoor environment, the pool is a source of high humidity which pushed the AHU serving the area to have an increased load.

6.9.2. Installation of pool cover

A pool cover would ensure minimizing water evaporation at night while maintaining the water temperature and reducing the space humidity.

There are many types of pool covers, but having a pure rectangular form, a floating cover can be easily used and would have the lowest CAPEX. The below pictures show the existing pool and an example of a floating cover.

Figure 109: Existing Pool (Left), Floating Pool Cover Example (Right)



6.9.3. Pool pumps duty cycling and Energy Saving

The implementation of the pool cover ensures a better temperature control, with minimum evaporation rate. Therefore, it creates a potential possibility for implementing an energy efficient action on pool pumps which are scheduled to be running overall 24 hours per day. A duty cycling on the pumps can be applied whereas each pump, on its duty can be cycled by 25% meaning the pool pump on duty mode will be running for 45 minutes and off for 15 minutes each hour. This will lead to about 25 % saving on pool pumps electricity consumption. This can be done either by implementing the adequate scheduling on the BMS or simply by adding adjustable timers. Duty cycling will be done during the longest time of inactivity of the pool (a combination of nighttime and peak day time during the week would be optimal).

The financial indicators of this EEM are shown in the table hereafter with the note that the benefits of the pool cover are beyond the electricity and water savings as it will also reduce chemical usage.

EEM - 09		
Pool Cover - Indoor Pool residences		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
1,401	274	16,163
0.0%	1.1%	0.0%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$2,407
Annual Share of Cost Savings [%]		0.1%
Implementation Cost [\$]		\$5,000
Simple Payback [Years]		2.1
Net Present Value [\$]		\$14,047
Internal Rate of Return [%]		48%

6.10. EEM – 10: [LCNC] Outdoor Pool freeze protection control ■

6.10.1. Rationale

The outdoor pool in FSB has a total volume of 67000 gallons of water and is heated to 83°F from mid-May to mid Sept then kept 60°F across winter to avoid freezing. With temperature dropping below 50°F as of November, the related hot water demand is high across the year and provide also options for energy efficiency as the pool is unused during water but is only kept operating for purely the anti-freezing protection.

Figure 110: FSB Outdoor Pool



6.10.2. Efficiency Action and Energy Savings

Ideally, installing a pool cover would help in reducing energy losses during winter as well as allowing a lower temperature threshold setting. However as this is deemed difficult in light of the specific outdoor pool's shape design, the optimal solution would be to install a 'Freeze Protection Control' which works with a sensor that measures the outside air temperature and turns on the pool pump when the low threshold is reached. More advanced pool controllers can also open any closed valves, and turn on other pumps, like booster pumps or waterfall pumps.

These can be also developed within the JCI BMS by adding a winter control for the pool pumps.

As the outdoor pool's circulation pump is equipped with a VFD, substantial savings can be achieved with the appropriate control strategies that should be implemented by JCI in the BMS.

The main schemes of the strategies would be as follows:

- i. Install an outdoor ambient temperature sensor just for the pool operation
- ii. Calibrate the indoor pool temperature sensor to ensure readings are 100% accurate
- iii. Program a control that would use heating only when ambient temperature is below 42°F. Heating temperature could be set at 50°F instead of 60°F
- iv. VFD on the pump should be used to lower the speed to minimum 70% as the pool is unused and the scheme is just to circulate hot water.

- v. When there are freezing ambient temperatures only, use pool pumps at 100% speed and 24/7 until temperature warms up and you revert to the previous condition.
- vi. In intermediate months (October and March to May), pool pump would therefore be used at 50% of the time (or at an even higher reduced speed of 50%) and without heating as there is no need to warm the water.

Implementing the above control strategies through the BMS would require little CAPEX as there is already a DDC controlling the pool and the investment is just on adding temperature sensor and implementing the control strategies. It will provide however substantial energy savings during 8 months of the year at the levels of both reductions of pool heating demand and electricity consumed by the pool pump.

The below table provides the key indicators of this EEM.

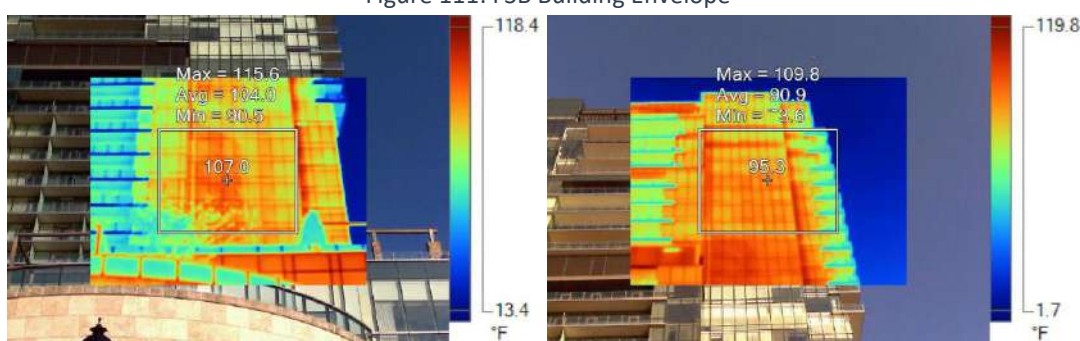
EEM - 10		
Antifreeze control management for outdoor pool		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
11,000	1,420	85,127
0.3%	5.7%	0.1%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$12,803
Annual Share of Cost Savings [%]		0.8%
Implementation Cost [\$]		\$6,000
Simple Payback [Years]		0.5
Net Present Value [\$]		\$95,310
Internal Rate of Return [%]		213%

6.11. EEM – 11: [LCNC] Building Envelope – Heat gain mitigation for the first 21 Floors ■

6.11.1. Rationale

As discussed in section 2.2, the thermal efficiency of the glazed façade of the first 21 floors is lower compared to those installed in 2017 for the upper 10 floors. This difference in the surface temperature means that there are more heat gains in the building envelope of the first 21 floors which causes an increase of the space temperatures, leading to higher cooling demand. This has also been verified in Figure 88 whereby temperature logging in a rented room showed a substantial increase in the early afternoon when direct solar radiations are hitting the façade.

Figure 111: FSB Building Envelope



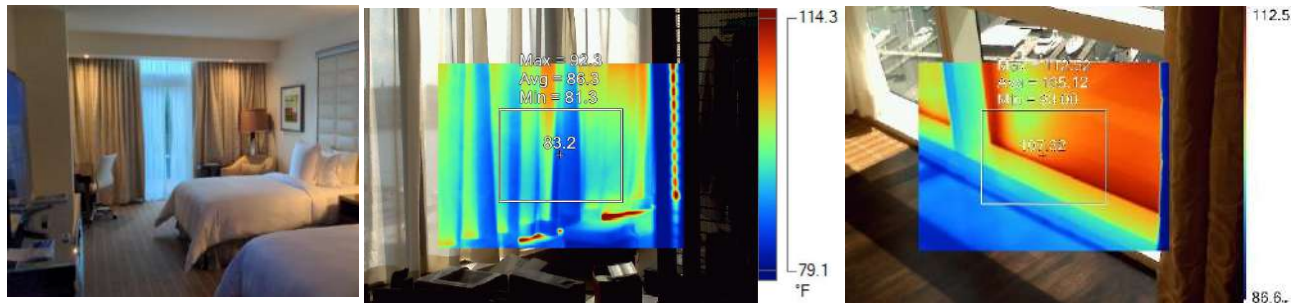
6.11.2. Efficiency Action and Energy Savings

Ideally, the property should plan a review of all the glazing of river facing guest rooms and consider upgrading the window films to appropriate ones with higher efficiency. As this could prove to be challenging in time, execution and investment, the alternative would be to enforce closing the curtains (blackout) in all unrented rooms and ensure at least the white translucent curtains are closed in all rented rooms.

Note that curtains are not connected to the GRMS and this action needs to be enforced through housekeeping.

The pictures in the following page validates the impact of curtains with the translucent white one providing already a better barrier. The usage of blackout curtain will ensure reducing the heat gains to the maximum possible.

Figure 112: Curtains in guest rooms and their heat gain mitigation impact



The below table provides the Key indicators of this EEM .

EEM - 11		
Building Envelope - Heat gain mitigation for the first 21 floors		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT COOLING	CO ₂
KWh	RTH	Kg
5,000	30,000	8,700
0.1%	1.1%	0.4%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$2,820
Annual Share of Cost Savings [%]		0.2%
Implementation Cost [\$]		
Simple Payback [Years]		-
Net Present Value [\$]		\$22,315
Internal Rate of Return [%]		-

6.12. EEM – 12: [CAPEX] Operation of the Blending Mixing Valve on Chilled Water Line

6.12.1. Rationale

FSB is using a District Cooling connection for its chilled water system. While at the present time there may not be any Delta T (Temperature) penalty for a lower Return Temperature, increasing the latter provides core advantages when it comes to controlling the cooling consumption and limiting the buildings' supplied chilled water temperature. This would be much like a chilled water temperature reset in winter in a normal chilled water plant, and if effectively used, it would reduce both the cooling energy consumption and the chilled water flow. Optimizing the return chilled water temperature depends on a good control of the secondary chilled water pump and the effective utilization of a blending valve that seems to have been installed as per the original schematic, but not seen on the BMS control.

6.12.2. What is a blending mixing valve?

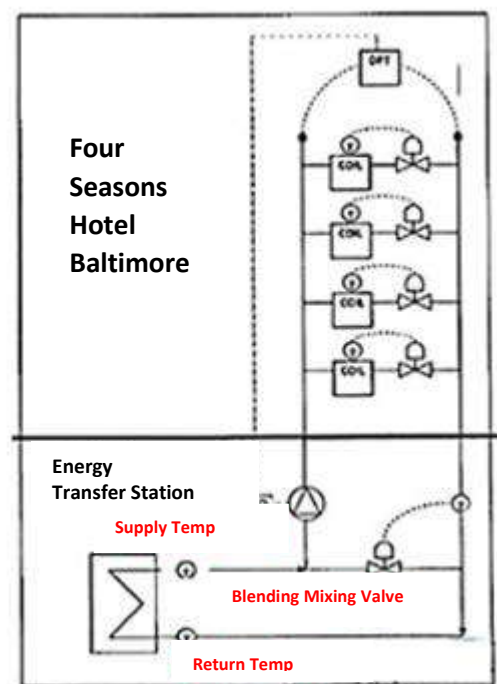
Typically, a blending mixing scheme is designed to ensure that its user can avoid low chilled water return temperature in the district cooling system. When the chilled water return temperature drops to a low limit, return water is blended into supply water to reduce primary chilled water flow and finally increase chilled water return temperature. This is ensured by the installation of a blending mixing valve.

The Blending Mixing Valve is put in between the supply and return CHW lines as seen in the schematic on the right.

If automated per the return temperature, the valve will allow a portion of the returned chilled water to be mixed with the supplied water, thus leading to a higher supply temperature, therefore a higher return temperature.

If it is automated adequately, it can act as implementing a chilled water reset temperature in normal chilled water system by raising the supply temperature and reducing the consumption of chilled water energy.

It will mostly be applied during 8 months of the year when ambient temperatures are mild or cold.

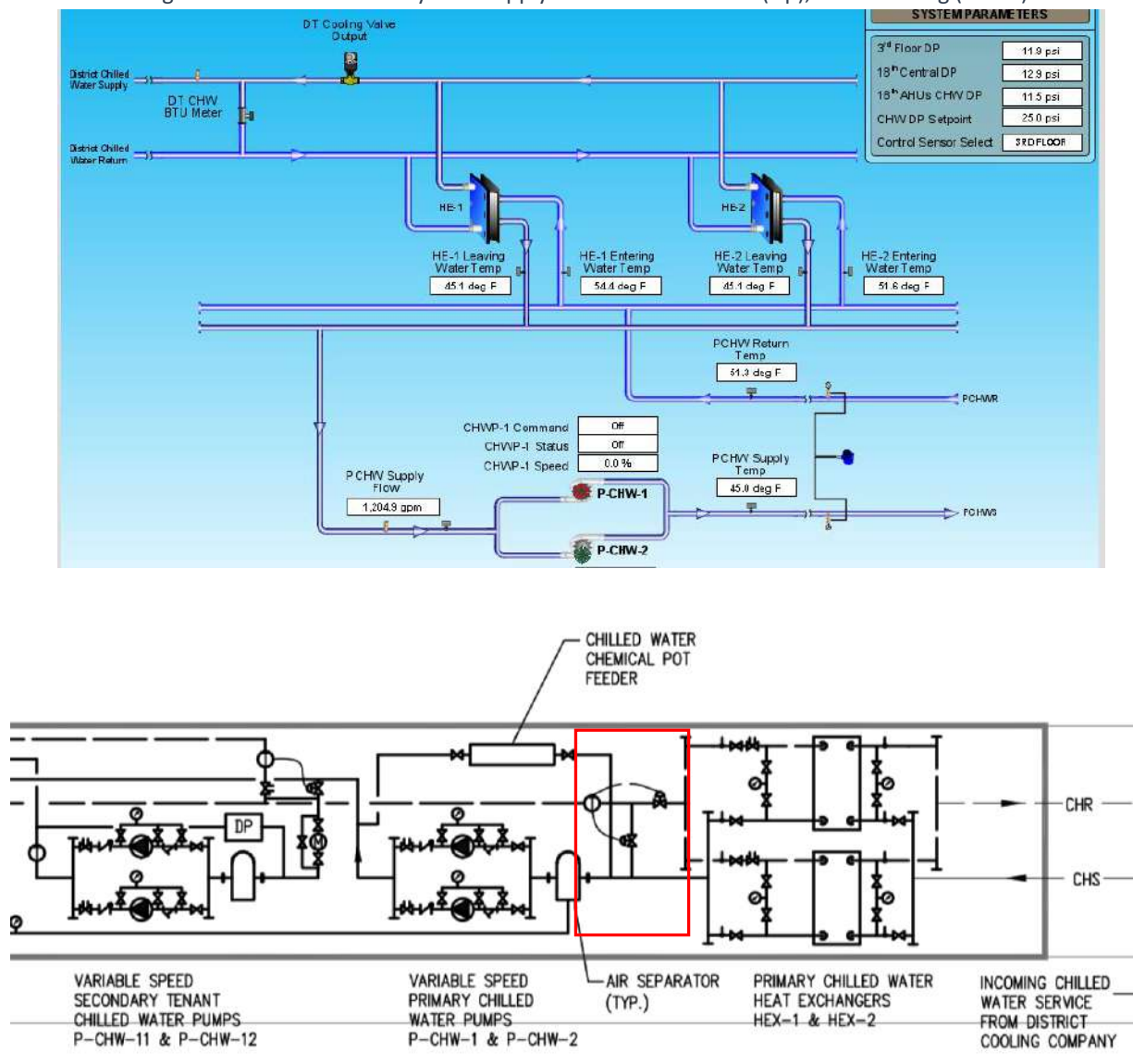


Return temperature can be raised up to 10°F optimizing the overall chilled water usage and the mixing valve could theoretically mix up to 2/3 of the returned chilled water with the supplied one.

6.12.3. Recommended Action

The below two figures showcase the District Cooling system at FSB. The first one is from the BMS which doesn't indicate the presence of any blending valve nor have any control of the return chilled water recycling, while the second is from the original drawings which indicate the presence of valves that could be used for blending purposes if controlled (shown in red).

Figure 113: Chilled Water System supply: BMS control screen (Up), MEP drawing (Down)



The actions recommended are as follows:

- i. Verify physically if the valves are installed as per the original design. If yes, then work with JCI to integrate them into the BMS control with the option to enable or disable them and to set their control if enabled according to the Delta T setpoint. As Delta T is lower than the setpoint, recycling return chilled water will occur in adequate % .
- ii. If physically the blending valves are not present as per the design, then consider installing them and implementing the strategy as per the previous point.
- iii. In both above cases, VFDs of the primary pumps will need to be readjusted in operation and PID loop.

The below table provides the key indicators of this EEM with the note that the implementation cost considered a full installation of new valves. Obviously, if it happens that these valves are already installed, then the CAPEX to connect and control them via the BMS is much lower.

Energy savings are from both the reductions of chilled water energy (due to the recycling effect) and primary pumps' electricity consumption (due to higher usage of VFD speed reduction)

EEM - 12		
Operation of the Blending Mixing Valve on Chilled Water Line		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT COOLING	CO ₂
KWh	RTH	Kg
39,307	90,000	40,706
0.9%	3.2%	1.3%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$10,624
Annual Share of Cost Savings [%]		0.6%
Implementation Cost [\$]		\$20,000
Simple Payback [Years]		1.9
Net Present Value [\$]		\$64,062
Internal Rate of Return [%]		53%

6.13. EEM – 13: [CAPEX] Recovery Wheels for FAHUs

6.13.1. Rationale

FSB has a number of Fresh Air Handling Units that provide 100% Outdoor Air to specific areas without any heat recovery in the process from the exhausted air. The below figure provides two such examples whereby the supply and exhaust fans are totally separated. An Energy Recovery wheel will allow recuperating substantial energy from the exhausted air and reducing the thermal energy (cooling and heating) of the treatment of the Outside Air as seen in the right picture.

It is to be noted that the residences have installed such units (Energy Recovery Units) and the present EEM would be to check the viability of such an action on some of the hotel FAHUs.

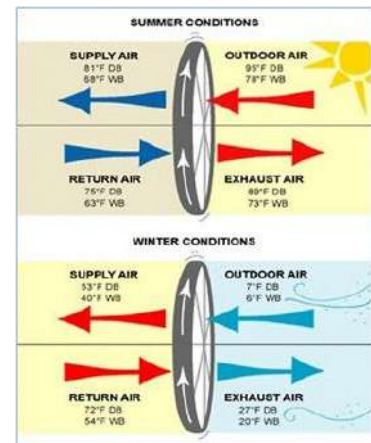
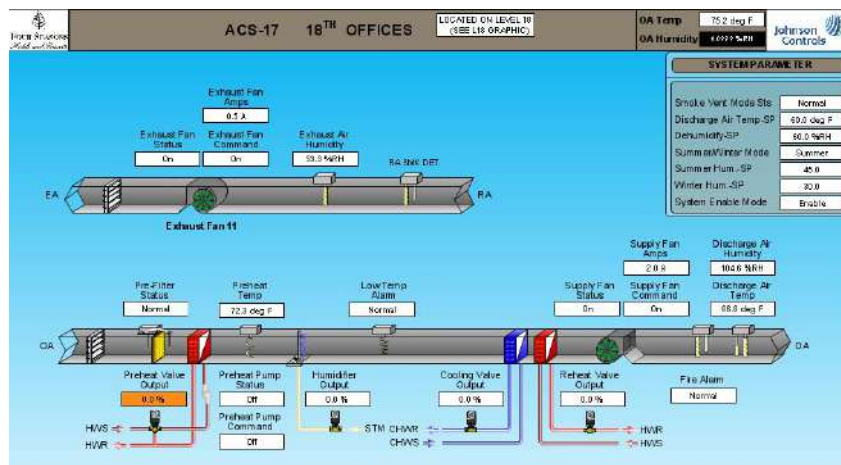
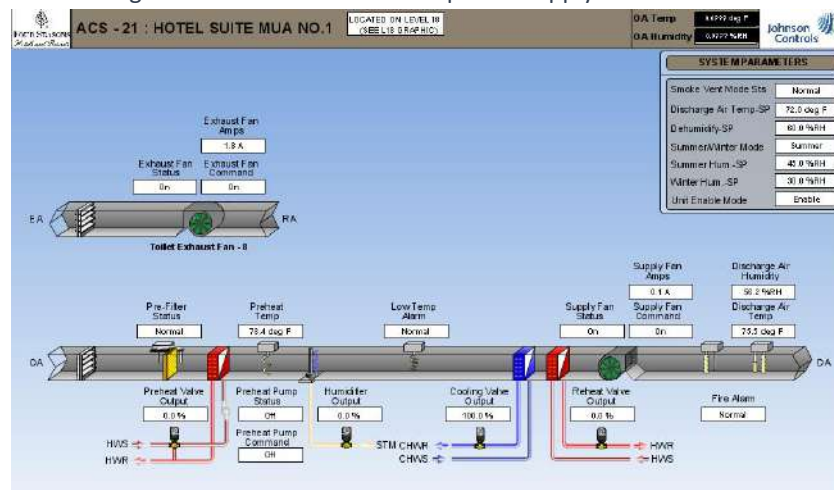


Figure 114: Fresh AHUs with separate supply and exhaust



6.13.2. Recommended action and energy efficiency

The recommended actions are as follows:

- i. Review all Fresh Air Handling Units physical installations and their ducting with the primary targets being those who have separate exhaust – like the 2 cases in the previous page.
- ii. Check the possibility to modify the ducting of the intake and exhaust air in a way to install an Energy Recovery Wheel connecting them together
- iii. Undertake a pricing of the execution according to the modifications needed.

The below pictures provide one such example AHU 23 which serves guest rooms and which has the TE 6 exhaust fan that is just 10m away with the possibility of connecting both intake and exhausted ducts.

Figure 115: AHU 23 (Left); TE 6 (Right)



The savings to be achieved would be a minimum of 25% of the thermal energy used by each targeted AHU (cooling and heating). It is impossible to calculate the appropriate indicators of the present EEM as it would depend on the physical viability of such a project and the related execution cost .

Clearly an enthalpy/energy wheel is a great opportunity to optimize the thermal energy consumption in the key FAHUs in FSB and should be investigated in the forthcoming time.

6.14. EEM – 14: [CAPEX] Solar Water Heaters Installation ■

6.14.1. Rationale

The roof of FSB has ample space for the installation of solar thermal panels. In fact, with around 3230m² available on the upper roof, a large number of panels can be installed for domestic hot water production, which will reduce the hot water energy demand from the District Heating supply.

Figure 116: FSB Roof



6.14.2. Efficiency Action and Energy Savings

A Solar Thermal system works by harnessing the sun's energy and converting it into heat, that is then transferred to the building's hot water system.

Solar Thermal panels are used in conjunction with a boiler, collector or immersion heater. The solar collector utilizes the sun's rays to heat a transfer fluid which is a mixture of water and glycol (if temperatures may freeze). The heated water from the collectors is pumped to a heat exchanger inside a water cylinder. The heat from the exchanger will then heat the water inside the cylinder. After the liquid releases its heat, the water flows back to the collectors for reheating. A controller will ensure the fluid circulates to the collector when there is sufficient heat available. Solar Thermal technology is proven, reliable and low maintenance.

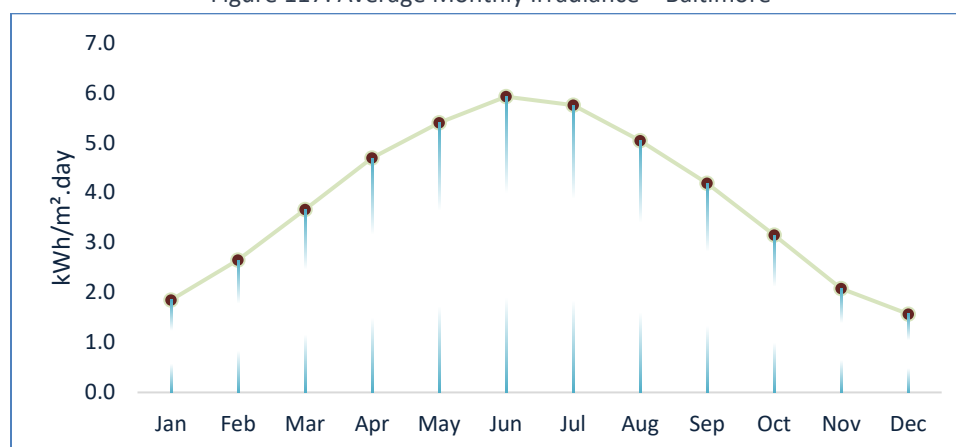
Key components of the systems are:

- **Collectors:** Solar Thermal collectors capture the sun's energy by heating a fluid contained within reinforced glass pipes.
- **Pump & Controller:** The pump ensures the transfer fluid is circulated between the collectors and the water cylinder efficiently. A system control panel regulates the pump, providing information on the system's performance, and highlighting any issues, should they develop.

- **Water Cylinder:** The heat captured by the solar thermal collectors is pumped to a coil in the water cylinder. A solar thermal water cylinder has a dedicated coil that allows the heat from the collectors to be transferred as efficiently as possible.

In FSB, the system would be integrated to the current calorifiers infrastructure and will act as a preheating hot water production with the district heating making up the gap to cover the thermal energy needs in any given day. The main parameters for the solar thermal installation are the solar irradiance, surface available and energy prices. The solar irradiance in Baltimore, as seen in the below figure, is considered very convenient for solar related energy production, and the roof is directly exposed to the sun, which means that shading won't have an effect on the system's performance and efficiency.

Figure 117: Average Monthly Irradiance – Baltimore



A solar thermal system should be installed on the hotel's roof to tackle the hot water consumption used for domestic hot water production. The set of tables related to the savings calculation are found in the below table.

Table 23: Solar Panels Key Figures

Total Area [sq. m]	2,100	Specific Heat [KJ/Kg.K]	4,186
Active Panel Area [sq. m]	3	Water [L or Kg]	30,000
Actual # of Panels	700	Energy [MJ]	3,767,400
Preliminary Percentage Savings	389%	Energy [KWh]	1,046,500
Collector Efficiency / Yield	50%	Solar Plant Size [KW]	253

Table 24: Solar Thermal Energy Figures from Installing Solar Panels

Month	Average Irradiation	Transposition Factor	Monthly Solar Irradiation	Collected Energy	Exported Energy
Unit	KWh/m ²	-	KWh/m ²	KWh	KWh
Jan	1.85	1.41	80.86	169,813	84,907
Feb	2.65	1.31	97.20	204,124	102,062
Mar	3.66	1.17	132.75	278,771	139,386
Apr	4.69	1.03	144.92	304,334	152,167
May	5.40	0.96	160.70	337,478	168,739
Jun	5.92	0.91	161.62	339,394	169,697
Jul	5.75	0.93	165.77	348,122	174,061
Aug	5.04	1.01	157.80	331,385	165,693
Sep	4.19	1.13	142.04	298,286	149,143
Oct	3.15	1.29	125.97	264,534	132,267
Nov	2.08	1.43	89.23	187,387	93,694
Dec	1.57	1.52	73.98	155,355	77,677
Total	-	-	-	3,218,984	1,609,492
Maximum	5.9 (Jun)	1.5 (Dec)	165.8 (Jul)	348,122 (Jul)	174,061 (Jul)
Average	3.8	1.2	127.7	268,249	134,124
Minimum	1.6 (Dec)	0.9 (Jun)	74.0 (Dec)	155,355 (Dec)	77,677 (Dec)

The recommended panels are the flat panel types to be installed with an adequate angle of maximum 20 to 30°.

Figure 118: Flat Panel solar thermal panels



The following table summarizes the savings provided from the solar panels taking into consideration the allocated area, solar irradiation, and the lower efficiency due to the shading of the trees and the non-optimal tilt.

Table 25: SWH Savings

Month	Existing Consumption District Heating		Heat Production Solar	District Heating Savings	Savings Ratio
Unit	mmBTU	MJ	MJ	mmBTU	%
Jan	1,397	1,179,131	305,664	290	26%
Feb	1,209	1,020,450	367,424	348	36%
Mar	1,339	1,130,176	501,788	476	44%
Apr	1,239	1,045,772	547,801	519	52%
May	1,222	1,031,423	607,461	576	59%
Jun	1,126	950,394	610,908	579	64%
Jul	1,164	982,468	626,620	594	64%
Aug	1,164	982,468	596,493	565	61%
Sep	1,126	950,394	536,915	509	56%
Oct	1,280	1,080,377	476,161	451	44%
Nov	1,295	1,093,038	337,297	320	31%
Dec	1,397	1,179,131	279,638	265	24%
Total	14,958	12,625,222	5,794,171	5,492	46%
Maximum	1,397 (Jan)	1,179,131 (Jan)	626,620 (Jul)	594 (Jul)	64% (Jun)
Average	1,247	1,052,102	482,848	458	47%
Minimum	1,126 (Jun)	950,394 (Jun)	279,638 (Dec)	265 (Dec)	24% (Dec)

The key performance indicators of this measure are found in the table below. The project includes the solar thermal panels, piping, related pumps and control as well as a set of buffer tanks.

EEM - 14		
Solar Water Heaters Installation		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT HEATING	CO ₂
KWh	mmBTU	Kg
	5,492	313,405
0.0%	22.0%	10.3%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$45,729
Annual Share of Cost Savings [%]		2.7%
Implementation Cost [\$]		\$280,000
Simple Payback [Years]		6.1
Net Present Value [\$]		\$81,838
Internal Rate of Return [%]		16%

6.15. EEM – 15: [CAPEX] Photovoltaic Electricity Generation ■

6.15.1. Rationale

An alternative to the solar thermal system installation and the usage of the roof space for the installation of renewable energy would be the development of a PV (Photovoltaic) power plant that would be integrated with the main feeder of the property. Savings will come from the voided KWh to be purchased from the utility at 0.089 \$/KWh. The recommended solution will be an online PV Power Plant without batteries whereby produced energy is used instantaneously by the property by integrating it to the present main electrical panels.

6.15.2. Efficiency Action and Energy Savings

The key plant's parameters from allocated space, number of panels and overall plant size is shown in the following table.

Table 26: PV Plant Parameters

Panel Yield	20.0%	Inverter losses (6% to 15%)	8.0%
W _p per Panel	450	Temperature losses (5% to 15%)	6.0%
Number of Panels	767	DC cables losses (1% - 3%)	1.0%
Allocated Area [m ²]	2,300	AC cables losses (1% - 3%)	1.0%
Losses Factor	78%	Shadings (0% - 40%)	2.5%
PV Plant Size [KW _p]	345	Losses weak irradiation (3% - 7%)	3.0%
Effective PV Load [KW]	69	Losses due to dust, snow... (2%)	0.5%
Energy [KWh]	502,184	Other Losses	0.0%

The savings calculation is based on the available solar irradiance in Baltimore along the number of panels installed which would give the potential electricity produced (taking into consideration the efficiency of the panels).

The below tables provide first the end monthly electricity produced by the described PV plant, and then the cost savings based on the electricity tariff of 0.089 \$/KWh. The annual cost savings is seen to top the 28.3 %.

Table 27: PV Electricity Production Calculation

Month	Monthly Solar Radiation	Captured Energy	Monthly PV Energy Produced	
Unit	kWh/m ² .day	kWh/m ² .day	kWh/m ²	KWh
Jan	1.85	0.37	11.5	20,577
Feb	2.65	0.53	14.8	26,623
Mar	3.66	0.73	22.7	40,709
Apr	4.69	0.94	28.1	50,483
May	5.40	1.08	33.5	60,063
Jun	5.92	1.18	35.5	63,723
Jul	5.75	1.15	35.7	63,956
Aug	5.04	1.01	31.2	56,059

Month	Monthly Solar Radiation	Captured Energy	Monthly PV Energy Produced	
Unit	kWh/m ² .day	kWh/m ² .day	KWh/m ²	KWh
Sep	4.19	0.84	25.1	45,101
Oct	3.15	0.63	19.5	35,037
Nov	2.08	0.42	12.5	22,389
Dec	1.57	0.31	9.7	17,463
Total	-	-	-	502,184
Maximum	5.92 (Jun)	1.18 (Jun)	35.65 (Jul)	63,956 (Jul)
Average	3.83	0.77	23.33	41,849
Minimum	1.57 (Dec)	0.31 (Dec)	9.73 (Dec)	17,463 (Dec)

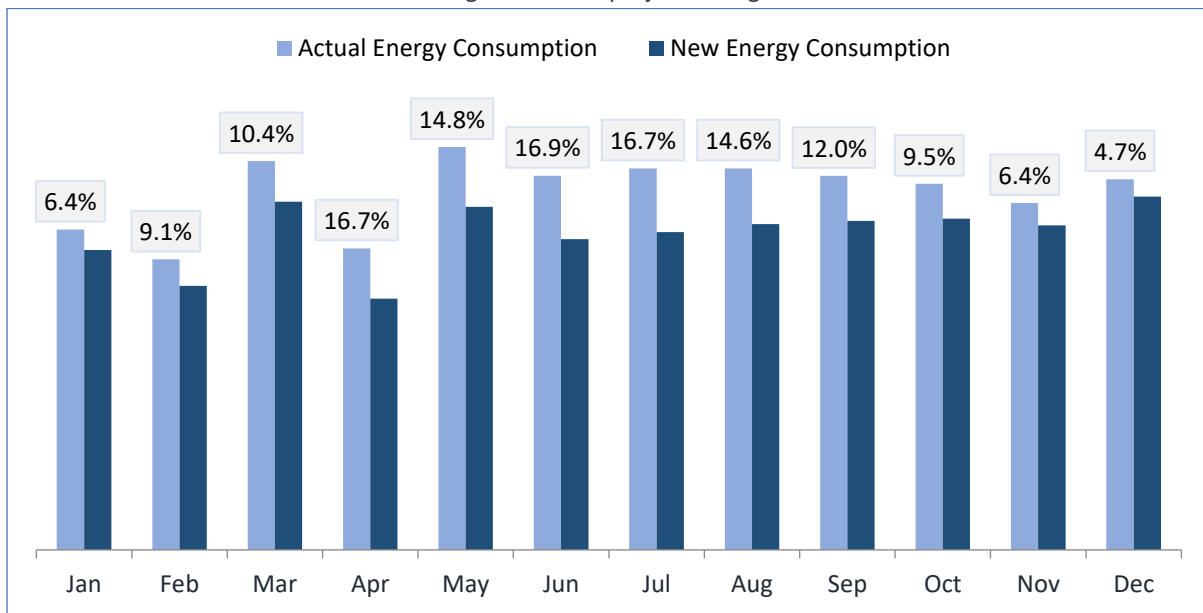
Table 28: PV Energy Savings

Month	Actual Energy Consumption	PV Energy Produced	New Energy Consumption	Energy Cost Savings	
Unit	kWh	kWh	kWh	US\$	%
Jan	321,756	20,577	301,179	1,831	6.4%
Feb	291,819	26,623	265,196	2,369	9.1%
Mar	390,629	40,709	349,920	3,623	10.4%
Apr	302,987	50,483	252,504	4,493	16.7%
May	404,833	60,063	344,770	5,346	14.8%
Jun	375,971	63,723	312,248	5,671	16.9%
Jul	383,263	63,956	319,307	5,692	16.7%
Aug	383,262	56,059	327,203	4,989	14.6%
Sep	375,772	45,101	330,671	4,014	12.0%
Oct	367,878	35,037	332,841	3,118	9.5%
Nov	348,482	22,389	326,093	1,993	6.4%
Dec	372,403	17,463	354,940	1,554	4.7%
Total	4,319,055	502,184	3,816,872	44,694	11.6%
Maximum	404,833 (May)	63,956 (Jul)	354,940 (Dec)	5,692 (Jul)	16.9% (Jun)
Average	359,921	41,849	318,073	3,725	11.5%
Minimum	291,819 (Feb)	17,463 (Dec)	252,504 (Apr)	1,554 (Dec)	4.7% (Dec)

The overall PV project includes the following parts/components:

- PV modules each of 450Wp of Efficiency > 17% at STC, total system power of 288KWp.
- Inverters & Related Junction Boxes.
- Steel Structure and roofing system to support all panels
- Cabling and adequate accessories and cable trays.
- Provision of a room to install the inverters.

Figure 119: PV project savings



The below table provides the key indicators of the PV project with a payback period of 6.9 years and a positive NPV which means that this Renewable Energy option is a financially feasible one.

EEM - 15		
Photovoltaic Electricity Generation		
ENERGY SAVINGS		
ELECTRICITY	DISTRICT COOLING	CO ₂
KWh	RTH	Kg
502,184	0	186,611
11.6%	0.0%	7.7%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$44,694
Annual Share of Cost Savings [%]		2.6%
Implementation Cost [\$]		\$310,500
Simple Payback [Years]		6.9
Net Present Value [\$]		\$43,154
Internal Rate of Return [%]		14%

6.16. EEM – 16: [CAPEX] Re-Commissioning and Water balancing : Chilled and Hot Water Loops

6.16.1. Rationale

During the upcoming renovation of FSB, it is highly recommended to have a comprehensive re-commissioning of the whole infrastructure including a Test and Air/Water Balance across all branches and premises.

The service shall include the following scope:

- Re-commission all sensors and valves in the HVAC system
- TAB (Testing Adjusting and Balancing) across all entities
- Re-Commissioning of the chilled and hot water infrastructures including all valves, actuators, sensors,...
- Review of AHUs and parasitic loads operation.

6.16.2. Action Plan for TAB and Re-commissioning

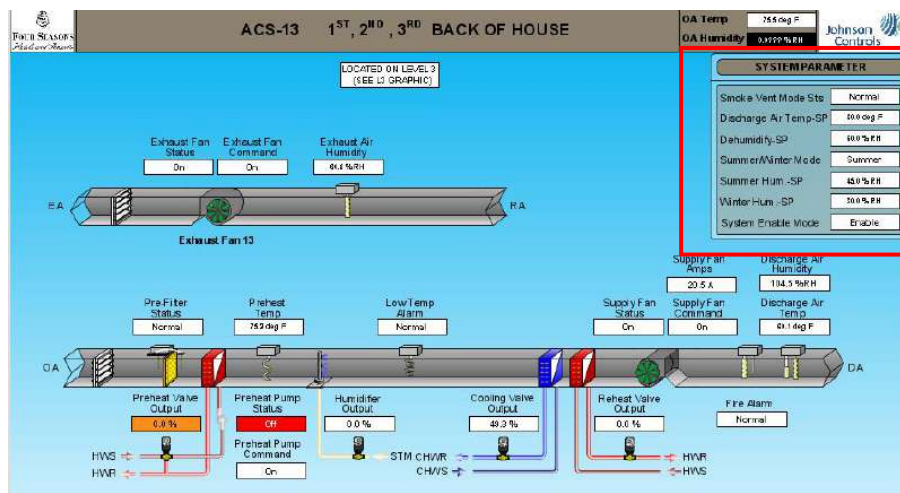
The actions to be completed are detailed in the following sections

i. Re-commissioning of the HVAC systems' components

With approximately 10 years of operation, it is crucial to have a proper re-commissioning of all the key branches, equipment end uses of the HVAC system of FSB. This would include the overall components from actuators to valves, sensors, cooling and heating coils etc...

This has to be done top to bottom, starting from the District Cooling and District Heating Transfer Stations and then going meticulously across all the entities' individual infrastructure,

The BMS review during the field audit showcased various sensors that were not properly operating as per the below example.



ii. TAB: Testing Adjusting and Balancing:

TAB ensures correct water balancing for the chilled and hot water systems. This procedure is accomplished by checking installations for conformity to design, measuring and having required chilled water to meet design specifications, and recording/reporting results. The table summarizes the TAB procedure for a chilled water and would be the same for the hot water network.

Table 29: Water Balancing Procedure

Preparation Procedure	Test and Balance Procedure
1. Air balance should be completed before water balance begins	1. Set chilled water pumps to proper flow
2. Open all valves to full open position. Close coil bypass stop valves. Set mixing valves to full coil flow.	2. Adjust chilled water flow through chillers
3. Verify that the responsible parties have removed and cleaned all strainers.	3. Check leaving water and entering water temperatures through chillers. Reset to correct design temperatures.
4. Examine water in system and determine if water has been treated and cleaned.	4. Check water temperatures at inlet side of cooling coils. Note any rise or drop of temperatures from source.
5. Check chilled water pumps rotation.	5. Proceed to balance each cooling coil.
6. Check expansion tanks to determine they are not air bound and the system is completely full of water.	6. Upon completion of low readings and adjustments at coils, mark settings and record data.
7. Check all vents at high points of chilled water system and determine all are operating freely.	7. After adjustments to coil are made, recheck the settings at the chilled water pumps and chillers and readjust if required.
8. Set all temperature controls so all coils are calling for full cooling.	8. With full flow through the bypass, set the pressure drop across the bypass valve to match coil flow pressure drop. This prevents unbalanced flow conditions when coils are on bypass. Same procedure must be applied on chillers to adjust chillers' bypass valves.
9. Check operation of automatic bypass valves.	9. Record and check pressure drop on each cooling coil and across bypass valves.
10. Check and set temperatures of chillers to design requirements.	10. Chilled water pumps operating suction and discharge pressures and final THD

iii. AHUs and parasitic loads in all entities

AHUs: Air handling systems and parasitic loads should not be overlooked in the pursuit for energy optimization strategies.

Ensuring that air handling systems are operating efficiently can have a major effect on chilled and hot water plants operation and proper delta T. Air handlers with chilled and hot water coils are designed with a specific ΔT at which they must operate for maximum efficiency. It is not uncommon to find air handling systems operating at much lower ΔT s than intended. This can occur when an operator manually adjusts the supply air temperature too quickly to resolve an immediate comfort problem. When the set point is fixed too low, the cooling valve opens to 100 percent flow to the cooling coil, but the new set point can never be achieved. Even a modest change to a supply air set point of a few degrees can double the amount of flow through the coil. The result of this high flow can decrease the ΔT at the plant.

Supply air set points that are operating properly based on design are seldom the source of comfort issues. Whenever a comfort issue is encountered, first check other areas such as terminal box operation, clogged filters/coils, duct integrity, and actual fan operation before lowering the supply air temperature. Properly functioning airside systems will benefit the efficiency of the chiller plant operation.

Parasitic Loads: In addition to the issues raised above, other challenges can arise that affect the ability to achieve design ΔT . For example, uncontrolled, non-HVAC (parasitic) chilled and hot water loads can also contribute to a ΔT in the chilled and hot water plants. For example, water cooled refrigeration racks and ice machines served by the chilled water system for their condenser loop. These loads do not require water temperature as low as the chilled water system, as they can effectively work off of higher water temperatures. The key is sufficient water flow, which is critical to the proper operation of the racks and ice machines. With this uncontrolled process, the chilled water system furnishes a specific amount of chilled water without regard to the operation of the racks and ice machines. Since all of the machines are on the same chilled water loop, they receive the same amount of chilled water regardless of whether they are powered on or off. The solution is to decouple the parasitic loads so a local pump serves the loads from the chilled water return line. This modification will actually increase the ΔT and requires no increase in the chilled water plant pumping capacity. The end result is better control of the plant with enhanced energy efficiency.

iii, Total Potential Savings

The present EEM covers the whole FSB chilled water, hot water, and HVAC infrastructures and accordingly, savings are going to be at the levels of both a reduction of chilled water usage (from all end uses) and hot water usage due to the reduced cooling/heating demand.

The estimated savings as well as the Key indicators of this EEM are found in the below table with the main savings estimated to top the 5% of the total chilled water usage and 5% of the total hot water usage.

EEM - 16		
Re-Commissioning and Water balancing : Chilled and Hot Water Loops		
ENERGY SAVINGS		
DISTRICT HEATING	DISTRICT COOLING	CO ₂
mmBTU	RTH	Kg
1,249	139,069	111,582
5.0%	5.0%	2.9%
ECONOMIC PERFORMANCE		
Annual Cost Savings [\$]		\$21,407
Annual Share of Cost Savings [%]		1.3%
Implementation Cost [\$]		\$35,000
Simple Payback [Years]		1.6
Net Present Value [\$]		\$134,384
Internal Rate of Return [%]		61%

6. Energy Efficiency Summary Figures

The following table provides the summary of the Efficiency Measures with key indicators including CAPEX, % Savings from Total Utilities Cost and Payback period.

As can be seen, if all EEMs are implemented, the total annual cost savings would be **\$329.851** at a CAPEX of **\$421,000** and a payback of **1.3 Years** (including the LC/NC EEMs which had savings of **\$230,639** – out of which \$146,260 are from the district cooling/heating capacity subscription reduction)

Table 30: EEMs Summary

CODE	EEM Name	ANNUAL SAVINGS		COST	PAYBACK
-	-	\$	KGs CO ₂	\$	Years
EEM - 01 [LCNC]	Reduce Cooling and Heating subscription Capacity	\$146,260 (8.7%)	0	\$0	-
EEM - 02 [LCNC]	Lighting Control	\$975 (0.1%)	4,069	\$0	-
EEM - 03 [CAPEX]	Motion Detectors Installation	\$2,411 (0.1%)	10,068	\$3,000	1.2
EEM - 04 [CAPEX]	Lighting Retrofit	\$2,723 (0.2%)	11,369	\$4,000	1.5
EEM - 05 [CAPEX]	Improve Thermal Insulation of the Hot Water Network	\$3,123 (0.2%)	21,401	\$10,000	3.2
EEM - 06 [LCNC]	HVAC Control	\$39,171 (2.3%)	174,289	\$0	-
EEM - 07 [LCNC]	Hot Water Primary Pumps Operation	\$28,610 (1.7%)	147,300	\$0	-
EEM - 08 [CAPEX]	Installation of VFDs on Fans and Pumps	\$11,822 (0.7%)	49,362	\$28,000	2.4
EEM - 09 [CAPEX]	Pool Cover - Indoor Pool residences	\$2,407 (0.1%)	16,163	\$5,000	2.1
EEM - 10 [LCNC]	Antifreeze control management for outdoor pool	\$12,803 (0.8%)	85,127	\$6,000	0.5
EEM - 11 [LCNC]	Building Envelope - Heat gain mitigation for the first 21 floors	\$2,820 (0.2%)	8,700	\$0	-
EEM - 12 [CAPEX]	Operation of the Blending Mixing Valve on Chilled Water Line	\$10,624 (0.6%)	40,706	\$20,000	1.9
EEM - 13 [CAPEX]	Recovery Wheels for FAHUs	\$0 (0.0%)	0	\$0	0.0
EEM - 14 [CAPEX]	Solar Water Heaters Installation (*)	\$45,729 (2.7%)	313,405	\$280,000	6.1
EEM - 15 [CAPEX]	Photovoltaic Electricity Generation	\$44,694 (2.6%)	186,611	\$310,500	6.9
EEM - 16 [CAPEX]	Re-Commissioning and Water balancing : Chilled and Hot Water Loops	\$21,407 (1.3%)	111,582	\$35,000	1.6
	TOTAL (*)	\$329,851 (19.6%)	866,748	\$421,500	1.3

(*) The total figures exclude EEM 14 as EEM 15 was considered instead

The below table provides the annual saving for each utility. Electricity savings are seen to be **1,189,386 KWh** per year representing **28%** of the actual electricity consumption (including EEM 16/PV Power Plant), while chilled water Consumption will be reduced by **403,167** representing **14%** of the actual chilled water consumption, and the Hot Water consumption will decrease by **10,801 mmBTU** representing **43%** of the actual consumption (with the application of EEM 14/ Solar Water Heaters).

Table 31: EEMs Summary

CODE	EEM Name	ELECTRICITY	DISTRICT COOLING	DISTRICT HEATING
-	-	KWh	RTH	mmBTU
EEM - 01	Reduce Cooling and Heating subscription Capacity	0 (0.0%)	0 (0.0%)	0 (0.0%)
EEM - 02	Lighting Control	10,950 (0.3%)	0 (0.0%)	0 (0.0%)
EEM - 03	Motion Detectors Installation	27,094 (0.6%)	0 (0.0%)	0 (0.0%)
EEM - 04	Lighting Retrofit	30,594 (0.7%)	0 (0.0%)	0 (0.0%)
EEM - 05	Improve Thermal Insulation of the Hot Water Network	0 (0.0%)	0 (0.0%)	375 (1.5%)
EEM - 06	HVAC Control	224,368 (5.2%)	144,097 (5.2%)	744 (3.0%)
EEM - 07	Hot Water Primary Pumps Operation	204,651 (4.7%)	0 (0.0%)	1,249 (5.0%)
EEM - 08	Installation of VFDs on Fans and Pumps	132,836 (3.1%)	0 (0.0%)	0 (0.0%)
EEM - 09	Pool Cover - Indoor Pool residences	1,401 (0.0%)	0 (0.0%)	274 (1.1%)
EEM - 10	Antifreeze control management for outdoor pool	11,000 (0.3%)	0 (0.0%)	1,420 (5.7%)
EEM - 11	Building Envelope - Heat gain mitigation for the first 21 floors	5,000 (0.1%)	30,000 (1.1%)	0 (0.0%)
EEM - 12	Operation of the Blending Mixing Valve on Chilled Water Line	39,307 (0.9%)	90,000 (3.2%)	0 (0.0%)
EEM - 13	Recovery Wheels for FAHUs	0 (0.0%)	0 (0.0%)	0 (0.0%)
EEM - 14	Solar Water Heaters Installation	0 (0.0%)	0 (0.0%)	5,492 (22.0%)
EEM - 15	Photovoltaic Electricity Generation	502,184 (11.6%)	0 (0.0%)	0 (0.0%)
EEM - 16	Re-Commissioning and Water balancing: Chilled and Hot Water Loops	0 (0.0%)	139,069 (5.0%)	1,249 (5.0%)
	TOTAL	1,189,386 28%	403,167 14%	10,801 43%

END OF REPORT